

Information Theory Part I

Alex Alemi

2021-03-23

Part 1

Part 2

Part 3

Problem Set

Solutions

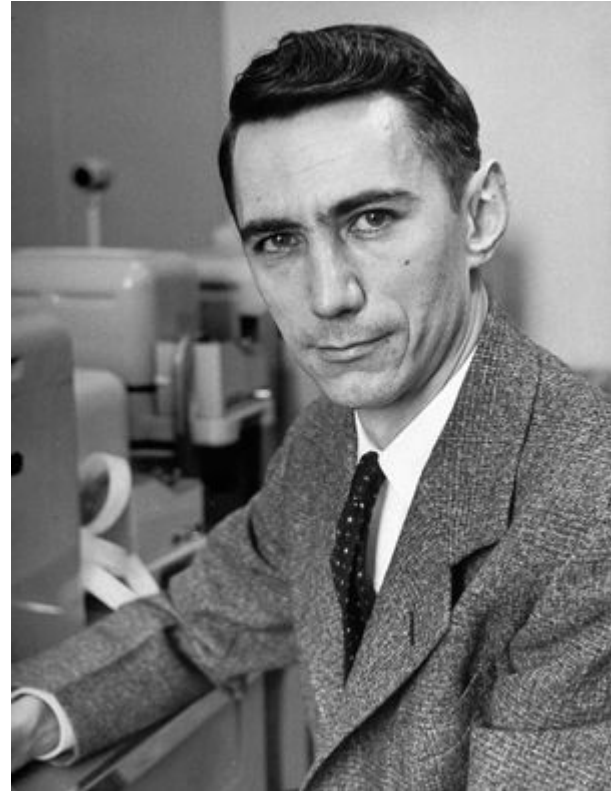
Information Theory is
about communication

Claude Shannon

✧ The Mathematical Theory of
Communication - 1948

Information theory primarily concerned
with communication.

Have it to thank for basically all things
digital.



'48 Shannon publishes *A Mathematical Theory of Communication*

'72 Discrete Cosine Transform developed

'51 Kullback-Leibler Divergence

'62 Low density parity codes developed

'77 Lempel-Ziv

'89 zip

'06 Asymmetric numeral systems zstandard

'94

Burrows-Wheeler
bzip2

1870s Boltzmann's H-theorem and Gibbs entropy

'60 Reed-Solomon codes

'76 Arithmetic coding

'47 Hamming Codes

'93 Turbo codes

'08 polar codes

'67 Viterbi algorithm

'51 Huffman encoding

Timeline of Information Theory

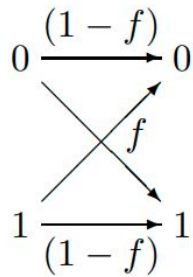


Figure 1.5. A binary data sequence of length 10 000 transmitted over a binary symmetric channel with noise level $f = 0.1$. [Dilbert image Copyright©1997 United Feature Syndicate, Inc., used with permission.]

From MacKay's Information Theory, Inference and Learning Algorithms

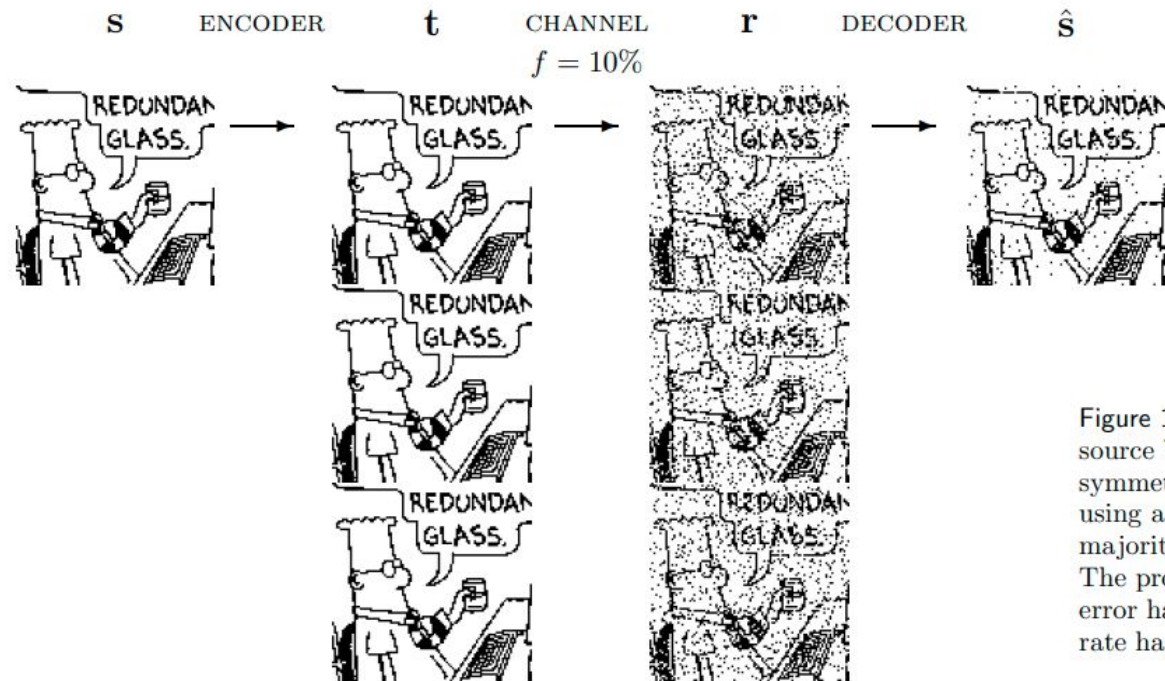


Figure 1.11. Transmitting 10 000 source bits over a binary symmetric channel with $f = 10\%$ using a repetition code and the majority vote decoding algorithm. The probability of decoded bit error has fallen to about 3%; the rate has fallen to $1/3$.

From MacKay's Information Theory, Inference and Learning Algorithms

Repetition Codes

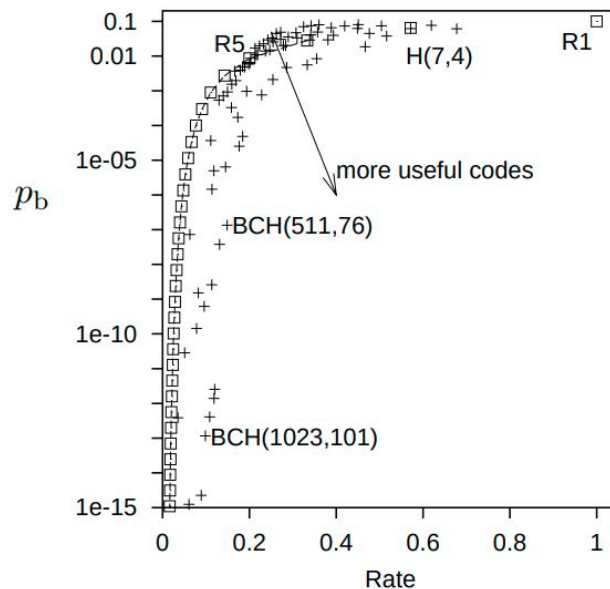
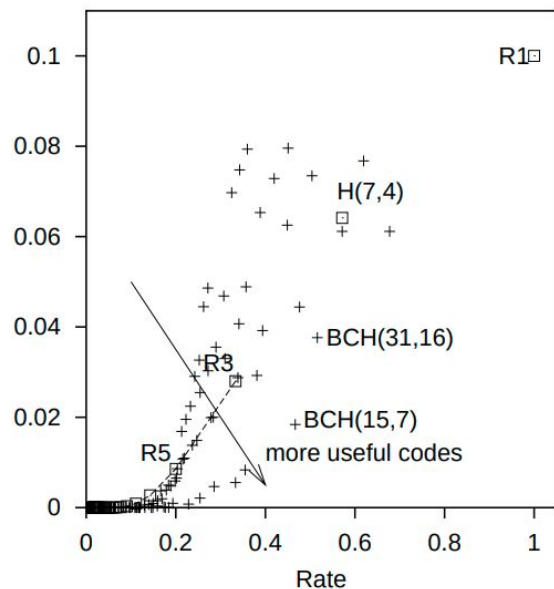


Figure 1.18. Error probability p_b versus rate R for repetition codes, the (7,4) Hamming code and BCH codes with blocklengths up to 1023 over a binary symmetric channel with $f = 0.1$. The righthand figure shows p_b on a logarithmic scale.

From MacKay's Information Theory, Inference and Learning Algorithms

The Noisy Channel Coding Theorem

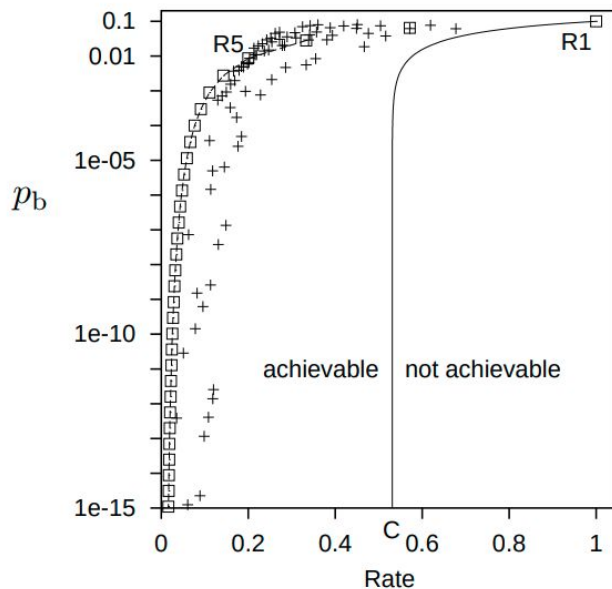
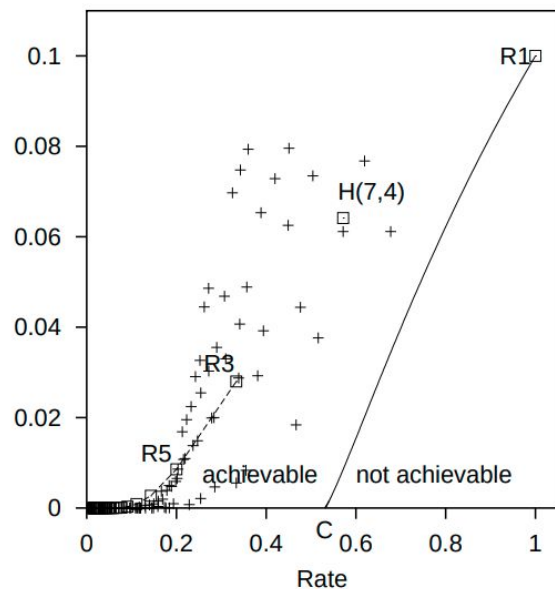
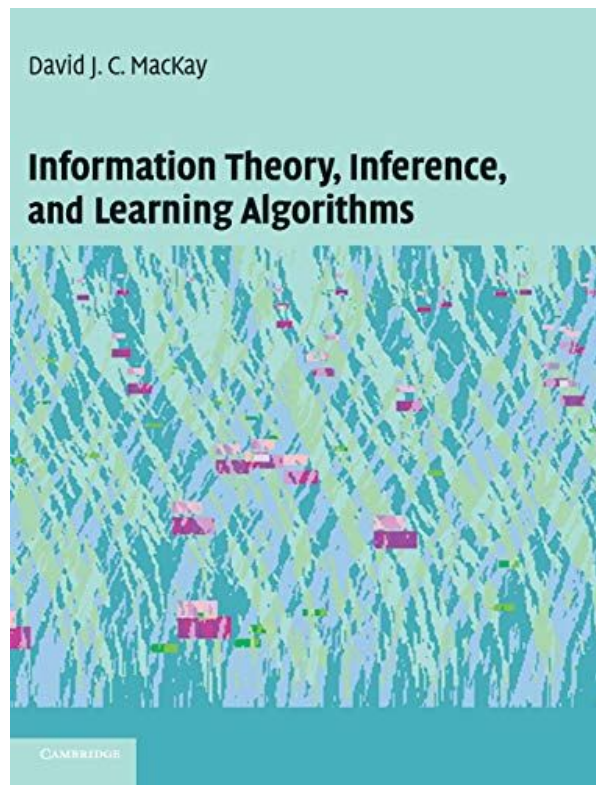
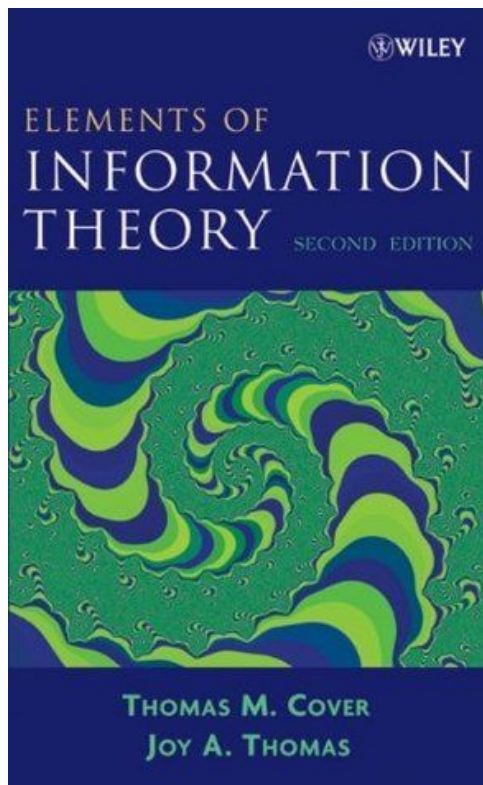


Figure 1.19. Shannon's noisy-channel coding theorem. The solid curve shows the Shannon limit on achievable values of (R, p_b) for the binary symmetric channel with $f = 0.1$. Rates up to $R = C$ are achievable with arbitrarily small p_b . The points show the performance of some textbook codes, as in figure 1.18.

The equation defining the Shannon limit (the solid curve) is $R = C/(1 - H_2(p_b))$, where C and H_2 are defined in equation (1.35).

Where to get Help



Frank Nielsen
@FrnkNlsn

Some books on information theory...



3:58 AM · Mar 21, 2021 · Twitter Web App

Information Theory is
about measurements

Cast of Characters

weight of evidence measures the importance of data for distinguishing hypotheses

relative entropy measures a "distance" between two distributions

entropy measures a "complexity" of a distribution

mutual information measures the "complexity" of the interaction between two random variables.

Cast of Characters

weight of evidence $w[x; p, q] \equiv \log \frac{p(x)}{q(x)}$

relative entropy $\mathcal{S}[p; q] \equiv \int dx p(x) \log \frac{p(x)}{q(x)}$

entropy $\mathcal{S}[p] \equiv - \int dx p(x) \log p(x)$

mutual information $\mathcal{I}(X; Y) = \int dx dy p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$

Weight of Evidence



Bayes Rule

$$\Pr(A, B) = \Pr(A) \Pr(B|A) = \Pr(B) \Pr(A|B)$$

$$\Pr(B|A) = \frac{\Pr(A|B) \Pr(B)}{\Pr(A)}$$

Bayes Rule = Joint probability sliced two ways + algebra.

Bayes Rule

$$\Pr(H|D) = \frac{\Pr(D|H) \Pr(H)}{\Pr(D)}$$

What if H is hypothesis, D is data?

Bayes Rule

The probability a hypothesis is true
given the data

The probability of the data given
the hypothesis

The probability of
the hypothesis

$$\Pr(H|D) = \frac{\Pr(D|H) \Pr(H)}{\Pr(D)}$$

The probability of
the data

Bayes Rule

$$\text{Pr}(H|D) = \frac{\text{Pr}(D|H) \text{Pr}(H)}{\text{Pr}(D)}$$

Posterior Probability

Likelihood

Prior Probability

Marginal Likelihood

Bayes Factor

Posterior Odds

Bayes Factor

Prior Odds

$$\frac{\Pr(H_1|D)}{\Pr(H_2|D)} = \frac{\Pr(D|H_1)}{\Pr(D|H_2)} \frac{\Pr(H_1)}{\Pr(H_2)}$$

Bayes Factor

Posterior Log Odds

Weight of Evidence

Prior Log Odds

$$\log \frac{\Pr(H_1|D)}{\Pr(H_2|D)} = \log \frac{\Pr(D|H_1)}{\Pr(D|H_2)} + \log \frac{\Pr(H_1)}{\Pr(H_2)}$$

Posterior log odds are prior log odds + weight of evidence

Odds

$$\text{Odds}(A) = \frac{\text{Pr}(A)}{\text{Pr}(\neg A)} = \frac{\text{Pr}(A)}{1 - \text{Pr}(A)}$$

$$\text{Pr}(A) = \frac{1}{2} \rightarrow \text{Odds}(A) = \frac{\frac{1}{2}}{\frac{1}{2}} = 1 = 1 : 1$$

$$\text{Pr}(A) = \frac{3}{4} \rightarrow \text{Odds}(A) = \frac{\frac{3}{4}}{\frac{1}{4}} = 3 = 3 : 1$$



"Information"

Posterior Log Odds

Weight of Evidence

Prior Log Odds

$$\log \frac{\Pr(H_1|D)}{\Pr(H_2|D)} = \log \frac{\Pr(D|H_1)}{\Pr(D|H_2)} + \log \frac{\Pr(H_1)}{\Pr(H_2)}$$

before the observation. This difference, which can be positive or negative, may be considered as the information resulting from the observation $X = x$, and we define the logarithm of the likelihood ratio, $\log [f_1(x)/f_2(x)]$, as the information in $X = x$ for discrimination in favor of H_1 against H_2 . [Cf. Good (1950, p. 63), who describes it also as the *weight of evidence* for H_1 given x .] The mean information for discrimination in favor of

Hypotheses

$$\Pr(X = x | P) \rightarrow p(x)$$

$$\Pr(X = x | Q) \rightarrow q(x)$$

The probability that some random variable X takes on a value x given that some model is true

Probability Densities

$$\Pr(x \leq X \leq x + dx \mid P) \rightarrow p(x) dx$$

The probability that some continuous random variable X takes on a value between x and $x + dx$ given that some model is true

Weight of Evidence

$$\log \frac{p(x)}{q(x)}$$

Units

$\log_2$: bit, shannon

$\log_e$: nat

$\log_{10}$: dit, ban, hartley

$10 \log_{10}$: deciban

$$2^{10} \sim e^7 \sim 10^3$$

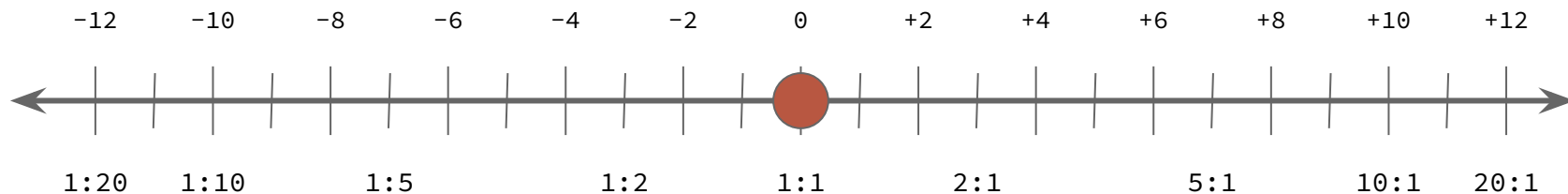
$$1 \text{ bit} = \frac{1}{\log 2} \text{ nat} = 1.44 \text{ nat}$$

$$\lg x \sim \ln x + \log_{10} x$$

Decibans

db	0	1	2	3	4	5	6	7	8	9	10	11	12	13
bit	0	$\frac{1}{3}$	$\frac{2}{3}$	1	$1\frac{1}{3}$	$1\frac{2}{3}$	2	$2\frac{1}{3}$	$2\frac{2}{3}$	3	$3\frac{1}{3}$	$3\frac{2}{3}$	4	$4\frac{1}{3}$
nat	0.00	0.23	0.46	0.69	0.92	1.15	1.38	1.61	1.84	2.07	2.30	2.53	2.76	2.99
odds	1:1	5:4	π :2	2:1	5:2	π :1	4:1	5:1	2π :1	8:1	10:1	4π :1	16:1	20:1
p	50%	56%	61%	67%	72%	76%	80%	83%	86%	89%	91%	93%	94%	95%

Belief-O-Meter



Examples: Loaded Dice

Hypothesis 1: Die is loaded and shows 1/3 of the time.

Hypothesis 2: Die is fair.

$$\text{1/3} : 10 \log_{10} \frac{\frac{1}{3}}{\frac{1}{6}} = 3 \text{ db}$$

$$\text{1} \mid \text{2} \mid \text{3} \mid \text{4} \mid \text{5} : 10 \log_{10} \frac{\frac{2}{3}}{\frac{5}{6}} = 10 \log_{10} \frac{4}{5} = -1 \text{ db}$$

Multiple Observations

$$\log \frac{p(x_1, x_2, \dots)}{q(x_1, x_2, \dots)} = \log \frac{p(x_1)p(x_2) \cdots}{q(x_1)q(x_2) \cdots} = \log \frac{p(x_1)}{q(x_1)} + \log \frac{p(x_2)}{q(x_2)} + \cdots$$

Independent observations decompose additively given the logarithmic form.

Example: Loaded Dice

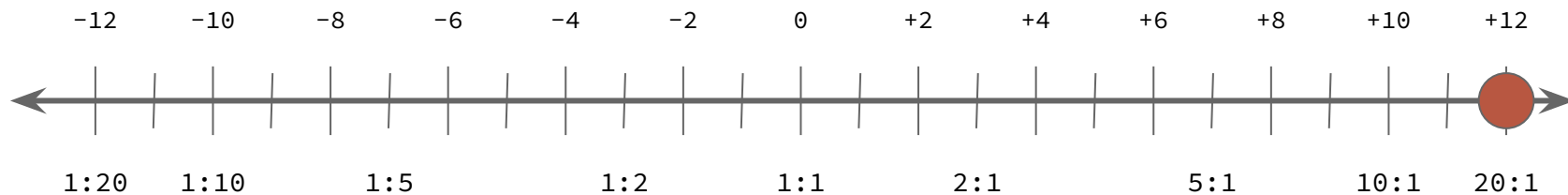


10  and 10 non-.

3 db for each , -1 for each non-,

we end up with 20 db in favor.

Belief-O-Meter



Examples: Coin

Hypothesis 1: Coin is doubled headed.  / 

Hypothesis 2: Coin is fair.  / 

$$\text{light gray circle} : 10 \log_{10} \frac{1}{\frac{1}{2}} = 3 \text{ db}$$

$$\text{dark gray circle} : 10 \log_{10} \frac{0}{\frac{1}{2}} = -\infty \text{ db}$$

Examples: Births

Hypothesis 1: Human births are 51% male ♂ - 49% female ♀

Hypothesis 2: Human births are 50% male ♂ - 50% female ♀

Laplace looked at Parisian birth records and found that between 1745 and 1770, 251,527 males and 241,945 females were born.

$$251527 \cdot 10 \log_{10} \frac{0.51}{0.50} + 241945 \cdot 10 \log_{10} \frac{0.49}{0.50} = 404 \text{ db}$$

this fraction is even a little too great, and, as it is of excessive smallness, we can regard as certain as any other moral truth, that the difference observed in Paris between the births of boys and those of girls is due to a greater possibility in the births of boys. We

Examples: Covid

Hypothesis 1: You have Covid.

Hypothesis 2: You don't have Covid.

$$+ : 10 \log_{10} \frac{\Pr(+|\text{Covid})}{\Pr(+|\neg\text{Covid})} = 10 \log_{10} \frac{0.938}{0.044} = 13.3 \text{ db}$$

$$- : 10 \log_{10} \frac{\Pr(-|\text{Covid})}{\Pr(-|\neg\text{Covid})} = 10 \log_{10} \frac{0.062}{0.956} = -11.9 \text{ db}$$

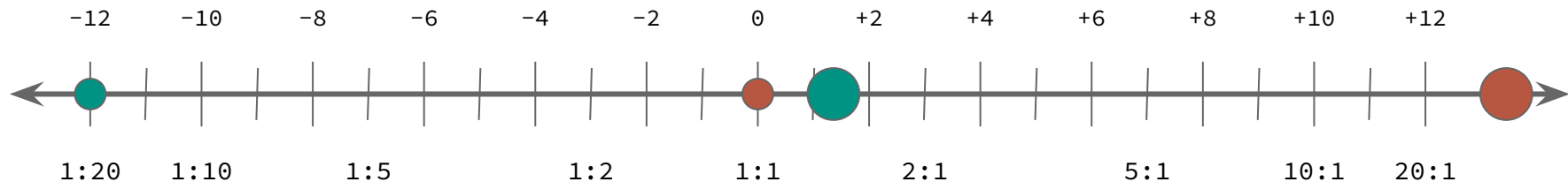
Combine with Prior

Posterior Log Odds

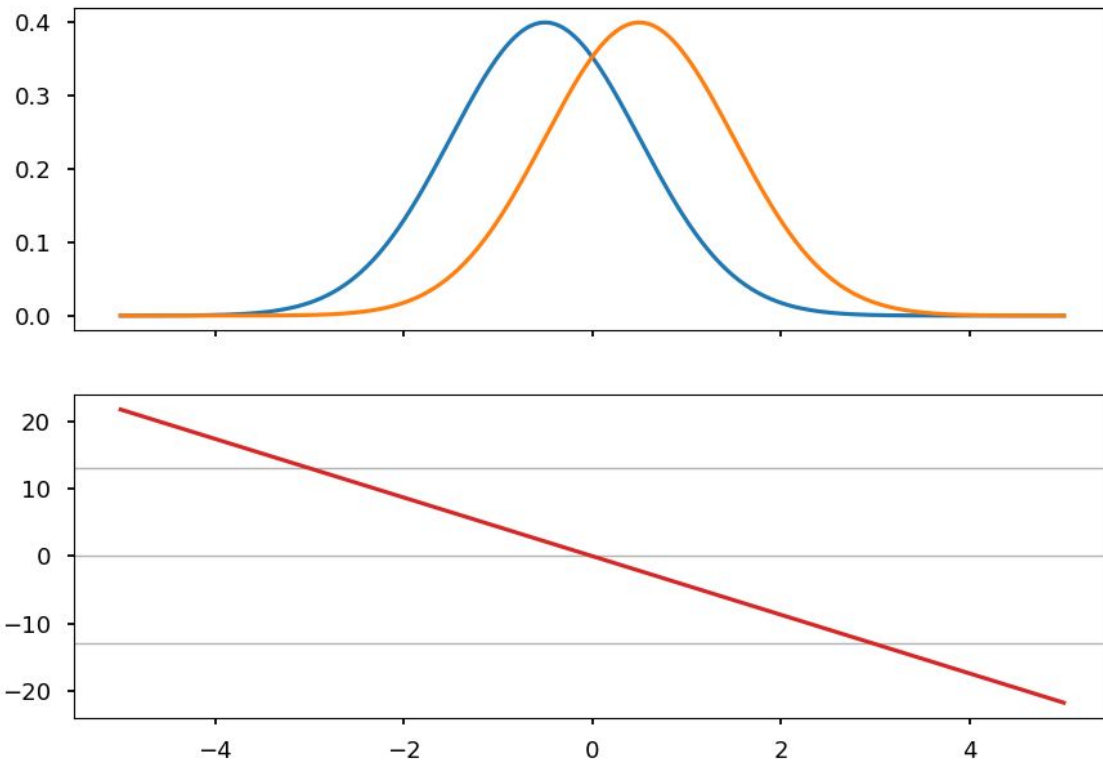
Weight of Evidence

Prior Log Odds

$$\log \frac{\Pr(H_1|D)}{\Pr(H_2|D)} = \log \frac{\Pr(D|H_1)}{\Pr(D|H_2)} + \log \frac{\Pr(H_1)}{\Pr(H_2)}$$



Examples: Gaussians



Examples: M&Ms

Hypothesis 1: M&M Bag is from NJ.

Hypothesis 2: M&M Bag is from TN.

						
Hackettstown, NJ	12.5%	12.5%	12.5%	12.5%	25%	25%
Cleveland, TN	12.4%	13.1%	13.5%	19.8%	20.5%	20.7%
db	0.035	-0.20	-0.33	-2.0	0.86	0.82

Should we expect to be able
to tell where our M&M bag
came from?

Relative Entropy (KL Divergence)



Relative Entropy is
expected weight of
evidence.

Expected Weight of Evidence

$$\mathbb{E} \left[\log \frac{p(x)}{q(x)} \right]$$

But expected with respect to what?

Relative Entropy

$$\mathcal{S}[p; q] = \mathbb{E}_{x \sim p(x)} \left[\log \frac{p(x)}{q(x)} \right]$$

Relative Entropy

$$\mathcal{S}[p; q] = \sum_i p(x_i) \log \frac{p(x_i)}{q(x_i)}$$

Relative Entropy

$$\mathcal{S}[p; q] = \int \mathrm{d}x \, p(x) \log \frac{p(x)}{q(x)}$$

Relative Entropy

$$\mathcal{S}[p; q] = \left\langle \log \frac{p(x)}{q(x)} \right\rangle_p$$

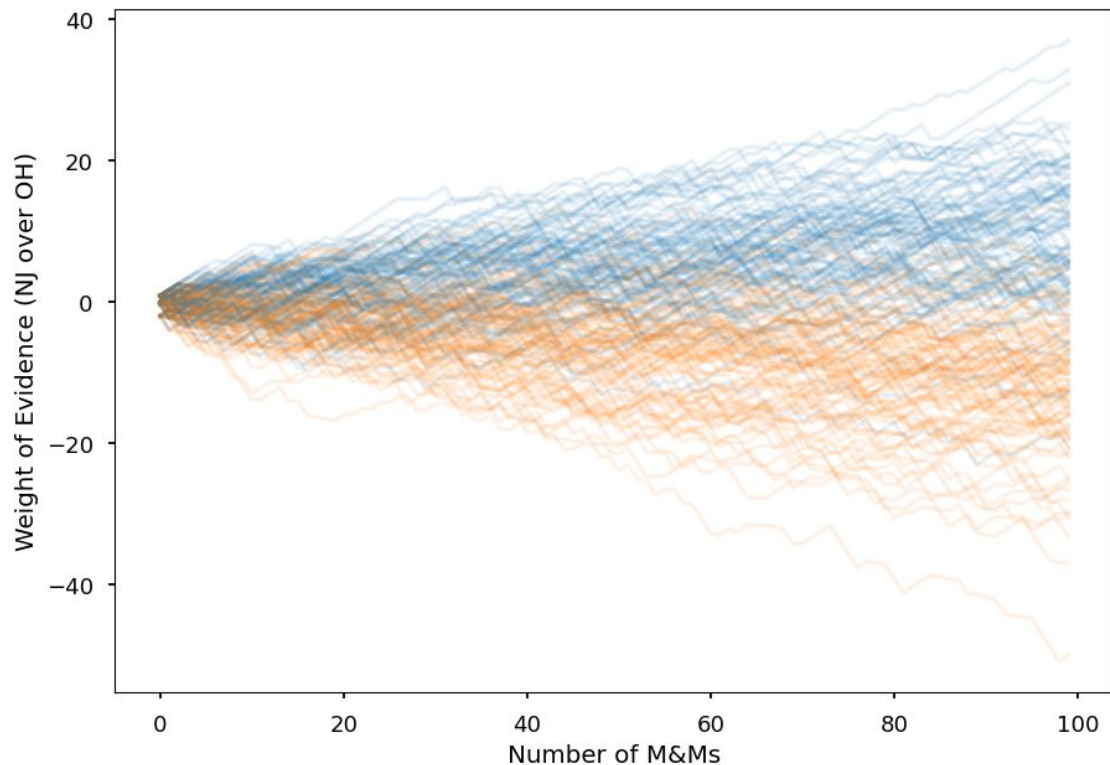
Examples: M&Ms

$$\mathbb{E}_{\text{NJ}} : 0.108 \text{ db} \quad \mathbb{E}_{\text{TN}} : -0.117 \text{ db}$$

$$\mathcal{S}[\text{NJ}; \text{TN}] = 0.108 \text{ db} \quad \mathcal{S}[\text{TN}; \text{NJ}] = 0.117 \text{ db}$$

						
Hackettstown, NJ	12.5%	12.5%	12.5%	12.5%	25%	25%
Cleveland, TN	12.4%	13.1%	13.5%	19.8%	20.5%	20.7%
db	0.035	-0.20	-0.33	-2.0	0.86	0.82

Examples: M&Ms



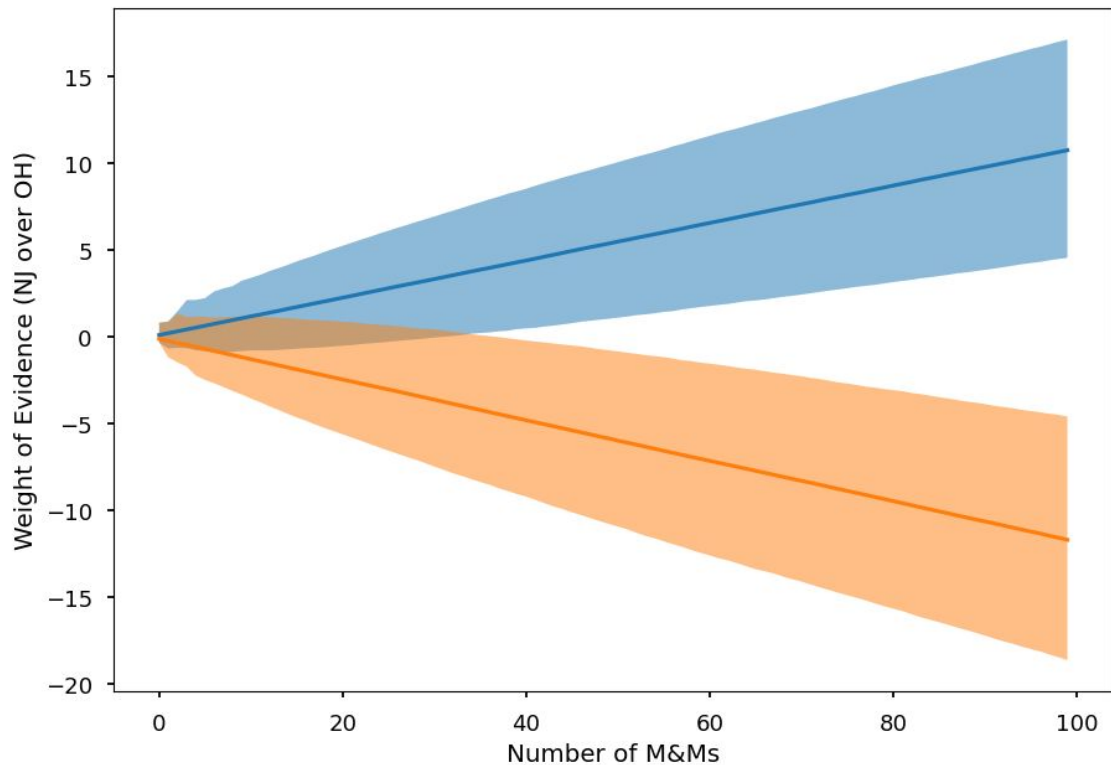
$$\mathbb{E}_{\text{NJ}} : 0.108 \text{ db}$$

$$\mathbb{E}_{\text{TN}} : -0.117 \text{ db}$$

$$\mathcal{S}[\text{NJ}; \text{TN}] = 0.108 \text{ db}$$

$$\mathcal{S}[\text{TN}; \text{NJ}] = 0.117 \text{ db}$$

Examples: M&Ms



$$\mathbb{E}_{\text{NJ}} : 0.108 \text{ db}$$

$$\mathbb{E}_{\text{TN}} : -0.117 \text{ db}$$

$$\mathcal{S}[\text{NJ}; \text{TN}] = 0.108 \text{ db}$$

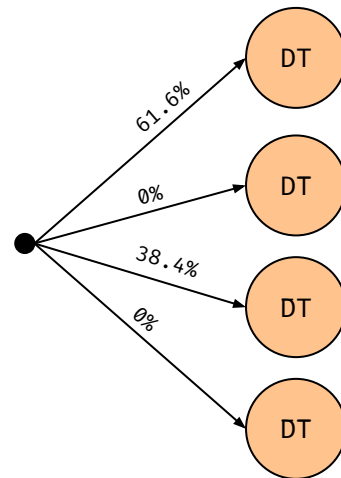
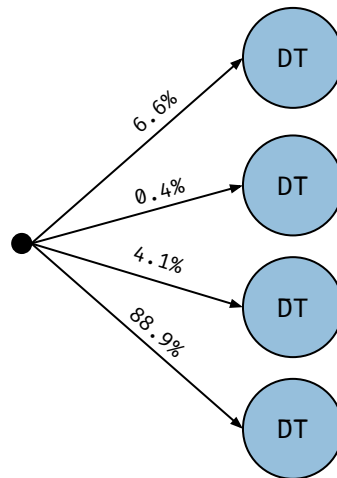
$$\mathcal{S}[\text{TN}; \text{NJ}] = 0.117 \text{ db}$$

Relative Entropy is
asymmetric in its
arguments.

Updating Beliefs

Imagine the COVID example again, let's say you started with some belief distribution over outcomes (blue distribution) and then updated to some new beliefs (orange distribution).

How much did your beliefs change?



Break

Q: How would you measure the magnitude of your updated beliefs? What properties does that magnitude have?

Relative Entropy is a
measure of the magnitude
in your updated beliefs

Continuous

$$\mathcal{S}[p; q] = \left\langle \log \frac{p(x)}{q(x)} \right\rangle_p$$

~~$$\mathcal{S}[\text{NJ}; \text{TN}] = 0.108 \text{ db}$$~~

$$\mathcal{S}[\text{NJ}; \text{TN}] = 0.109 \text{ db}$$

~~$$\mathcal{S}[\text{TN}; \text{NJ}] = 0.117 \text{ db}$$~~

$$\mathcal{S}[\text{TN}; \text{NJ}] = 0.118 \text{ db}$$

						
Hackettstown, NJ	12.5%	12.5%	12.5%	12.5%	25% 24%	25% 26%
Cleveland, TN	12.4%	13.1%	13.5%	19.8%	20.5%	20.7%
db	0.035	-0.20	-0.33	-2.0	0.86	0.82

Relative Entropy is
reparameterization
independent

Permutation Invariant

$$\mathcal{S}[p; q] = \sum_i p(x_i) \log \frac{p(x_i)}{q(x_i)}$$

$$\mathcal{S}[\text{NJ}; \text{OH}] = 0.108 \text{ db} \quad \mathcal{S}[\text{OH}; \text{NJ}] = 0.117 \text{ db}$$

						
Hackettstown, NJ	12.5%	25%	12.5%	12.5%	12.5%	25%
Cleveland, TN	19.8%	20.7%	12.4%	13.5%	13.1%	20.5%
db	-2.0	0.82	0.035	-0.33	-0.20	0.86

Reparameterization Invariant $\mathcal{S}[p; q] = \left\langle \log \frac{p(x)}{q(x)} \right\rangle_p$

$$\mathcal{S}[p(x, y, z); q(x, y, z)] = \mathcal{S}[p(r, \theta, \phi); q(r, \theta, \phi)]$$

Our belief update magnitude should be invariant to change of coordinates.

Reparameterization Invariant $\mathcal{S}[p; q] = \left\langle \log \frac{p(x)}{q(x)} \right\rangle_p$

Consider a remapping of our coordinate system. The Relative Entropy is invariant to such a mapping.

$$\begin{aligned}\mathcal{S}[p; q] &= \mathcal{S}[p(x); q(x)] \\ &= \int dx \, p(x) \log \frac{p(x)}{q(x)} \\ &= \int dy \, p(y) \log \left(\frac{p(y) \left| \frac{dy}{dx} \right|}{q(y) \left| \frac{dy}{dx} \right|} \right) \\ &= \mathcal{S}[p(y); q(y)]\end{aligned}$$

$y = f(x)$
 $p(x) dx = p(y) dy$

Relative Entropy is
non-negative

Non-Negative

Very useful property!

$$\mathcal{S}[p; q] \geq 0$$

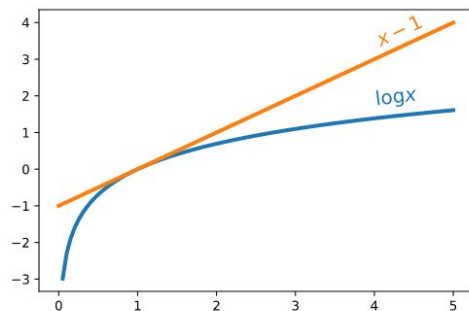
Intuitively, viewing Relative Entropy as expected weight of evidence, this is saying that the expected weight of evidence in favor of the truth is always positive.

More data is always a good thing when we are trying to learn the truth.

Non-Negative

Relative
Entropy is
always
non-negative!

$$\begin{aligned}\mathcal{S}[p; q] &= \int dx p(x) \log \frac{p(x)}{q(x)} \\ &= - \int dx p(x) \log \frac{q(x)}{p(x)} \\ &\geq - \int dx p(x) \left(\frac{q(x)}{p(x)} - 1 \right) \\ &= - \int dx q(x) + \int dx p(x) \\ &= -1 + 1 = 0\end{aligned}$$



$$\log x \leq x - 1$$

$$\mathcal{S}[p; q] = 0 \implies p = q$$

Edges

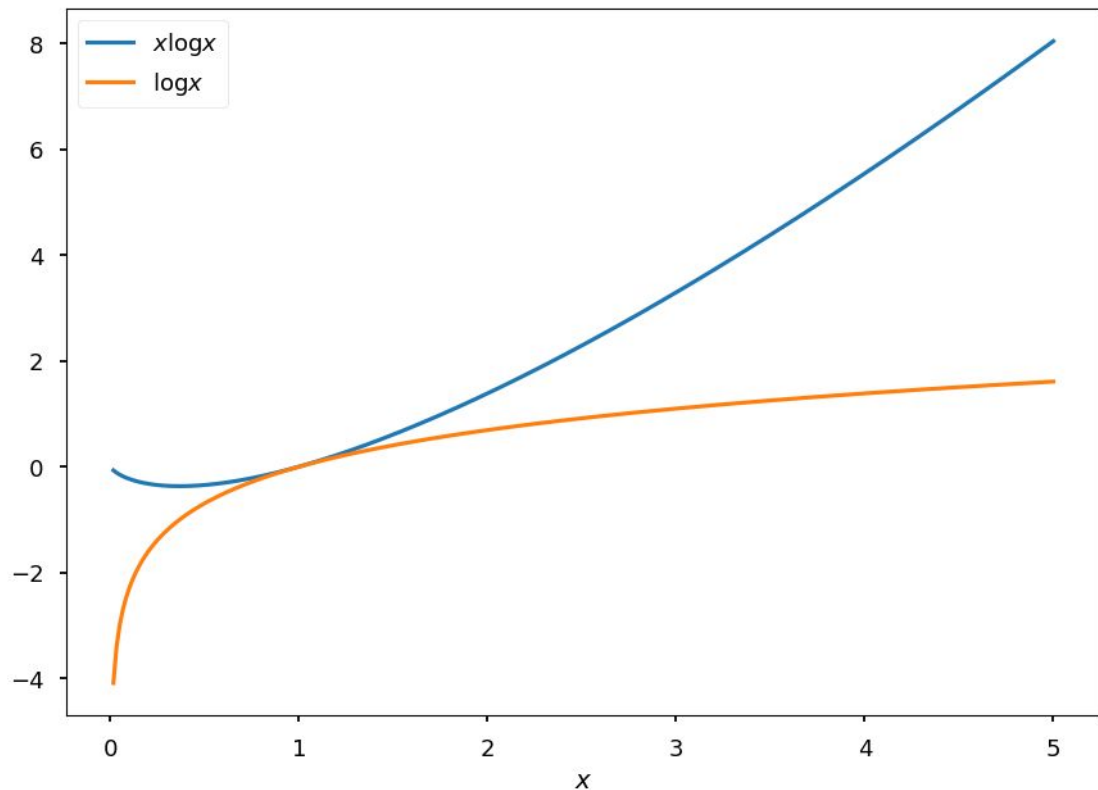
$$\mathcal{S}[p; q] = \left\langle \log \frac{p(x)}{q(x)} \right\rangle_p$$

$$\lim_{x \rightarrow 0} x \log x = 0$$

$$\lim_{x \rightarrow 0} \log x = -\infty$$

p can be zero where q is not zero, and this doesn't count towards the KL.

q cannot be zero where p is *not* zero.



Monotonic

Consider updating our beliefs from a uniform distribution on N elements to one on $N' < N$ elements.

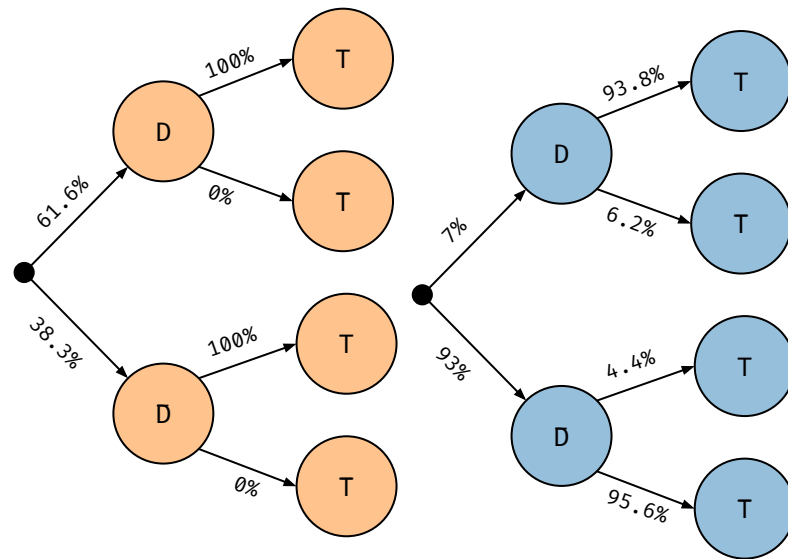
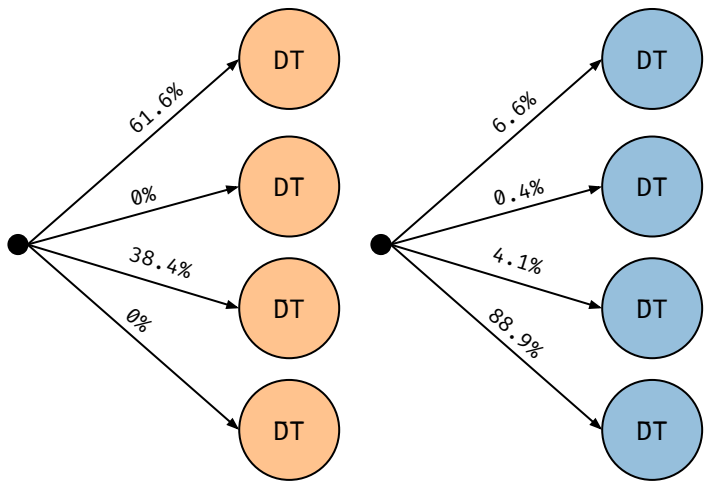
$$\mathcal{S}[p; q] = \sum_k \frac{1}{N'} \log \frac{\frac{1}{N'}}{\frac{1}{N}} = \log \frac{N}{N'}$$

Cutting a space in half is 1 bit of KL.

In general Relative Entropy is the number of optimal yes-no questions needed to move from q to p .

Chain Rule

We could decompose a situation into multiple steps. Want our measurement of information gain to be the same in either.



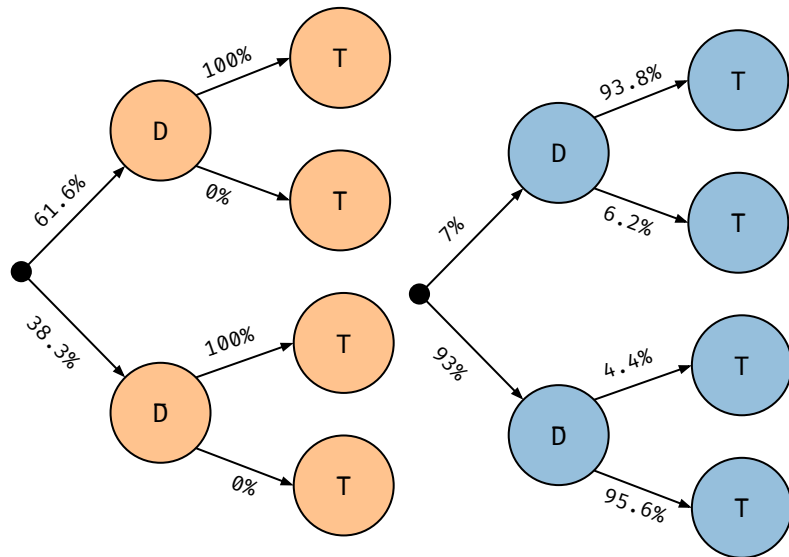
Chain Rule

$$\mathcal{S}[p(T, D); q(T, D)] = 9.7 \text{ db}$$

$$\mathcal{S}[p(D); q(D)] = 4.3 \text{ db}$$

$$\mathcal{S}[p(T|D=1); q(T|D=1)] = 0.28 \text{ db}$$

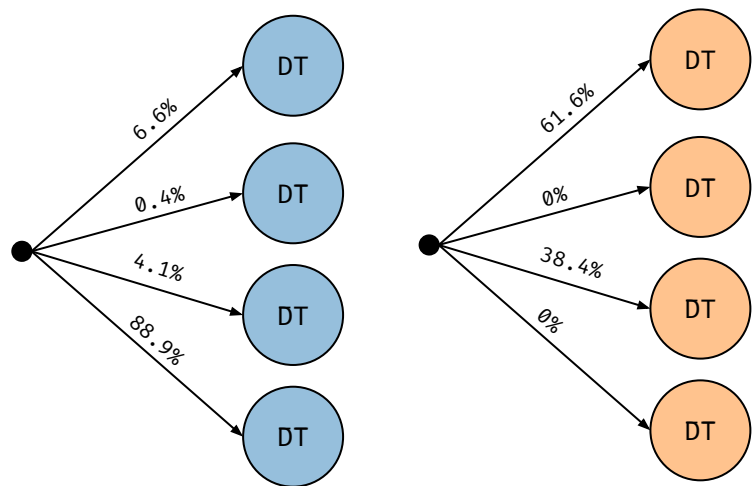
$$\mathcal{S}[p(T|D=0); q(T|D=0)] = 13.6 \text{ db}$$



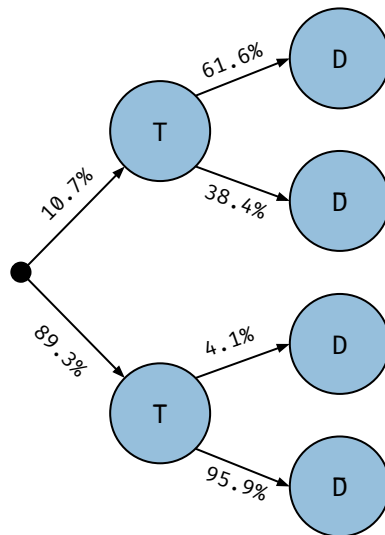
$$4.3 + 0.616 \cdot 0.28 + 0.383 \cdot 13.6 = 9.7$$

Bayes Rules

Bayes Rule corresponds to a minimal update!

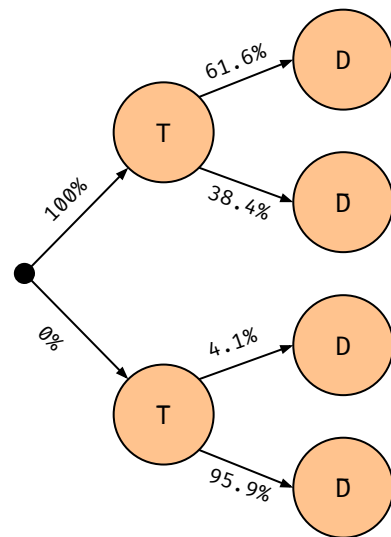


$$\mathcal{S}[p(T, D); q(T, D)] = 9.7 \text{ db}$$



$$\mathcal{S}[p(T); q(T)] = 9.7 \text{ db}$$

$$\mathcal{S}[p(D|T); q(D|T)] = 0.0 \text{ db}$$



Chain Rule

$$\begin{aligned}\mathcal{S}[p(x, y); q(x, y)] &= \int dx dy p(x, y) \log \frac{p(x, y)}{q(x, y)} \\ &= \int dx dy p(x)p(y|x) \log \frac{p(x)p(y|x)}{q(x)q(y|x)} \\ &= \int dx p(x) \log \frac{p(x)}{q(x)} + \int dx p(x) \int dy p(y|x) \log \frac{p(y|x)}{q(y|x)} \\ &= \mathcal{S}[p(x); q(x)] + \mathbb{E}_{p(x)}[\mathcal{S}[p(y|x); q(y|x)]]\end{aligned}$$

A New Theorem of Information Theory

Arthur Hobson¹

Received September 9, 1969

Consider a random experiment whose possible outcomes are $z_1, z_2, \dots, z_n$. Let the prior probabilities be $p_1^0, \dots, p_n^0$, and let the posterior probabilities be $p_1, \dots, p_n$. It is shown that, subject to certain prescribed and intuitively reasonable conditions, the expression $I = k \sum p_i \ln(p_i/p_i^0)$, where k is a positive constant, is the unique expression for the information contained in a message which alters the probabilities from the p_i^0 to the p_i .

KEY WORDS: information theory; mathematics; probability theory; quantitative measure of uncertainty; communication theory; entropy; foundations of statistical mechanics.

Hobson

Relative Entropy is unique given some reasonable axiomatic requirements.

We want our measure of information gain to be:

1. *Continuous*. Small changes in the distributions should correspond to small changes in the information gain.
2. *Reparameterization Independent*. Changing our coordinate system shouldn't have an effect on our measurement.
3. *Non-negative*. Should always be positive.
4. *Monotonic*. In a natural way, restricting to fewer and fewer things should be a larger and larger change.
5. Satisfy a natural *additive chain rule*. If we decompose our probability into steps, we should get the same answer.

The Relative Entropy is the only function to satisfy all of these requirements.

Relative Entropy is the
Kullback-Leibler
Divergence, a "distance"
between distributions

Number of Samples to Distinguish

Relative Entropy is like a "distance" between distributions.

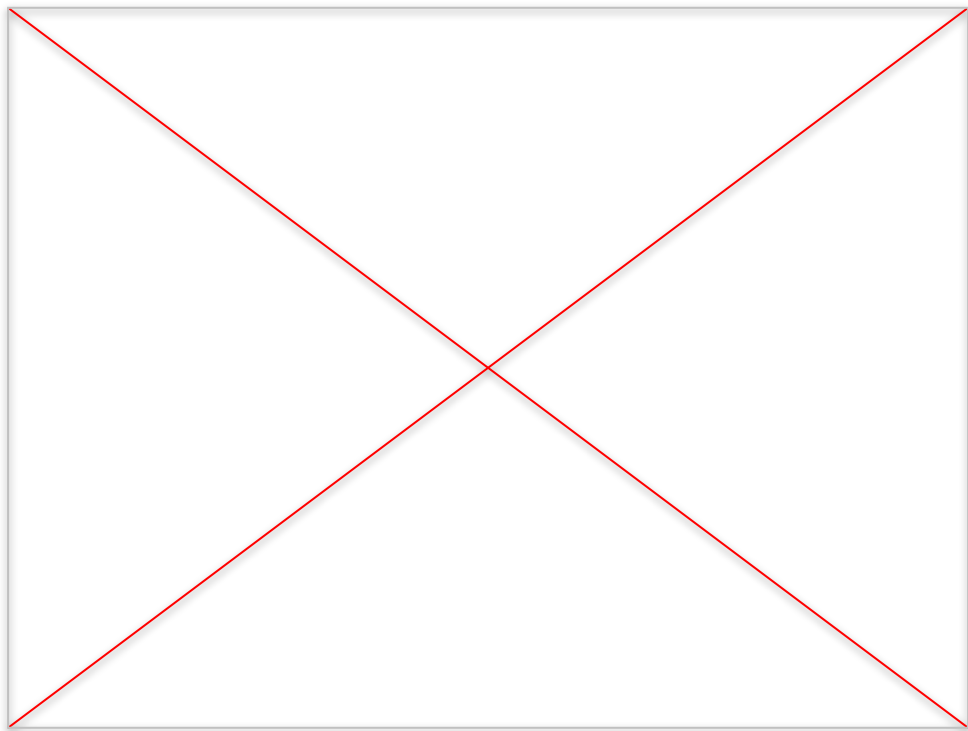
As the expected weight of evidence, it's how easy it would be to distinguish p from q if p were true.

$$N_{\text{eff}} \sim \frac{3 \text{ nat}}{\mathcal{S}[p; q]}$$

$$N_{\text{eff}} \sim 15\,000$$

$$\text{Births: } \mathcal{S} = 0.51 \cdot 10 \log_{10} \frac{0.51}{0.50} + 0.49 \cdot 10 \log_{10} \frac{0.49}{0.50} = 870 \times 10^{-9} \text{ db}$$

KL Divergence is asymmetric



https://twitter.com/ari_seff/status/1303741288911638530

A Principle For Learning

Make it hard to tell the difference between the real world and your model.

$$\min \mathcal{S}[\text{real}; \text{model}]$$

$$\min_q \int dx p(x) \log \frac{p(x)}{q(x)}$$

$$\min_q \int dx p(x) \log \frac{p(x)}{q(x)} = \int dx p(x) \log p(x) - \min_q \int dx p(x) \log q(x)$$

-Data Entropy

Cross Entropy

Maximum Likelihood

We don't have access to the true distribution of data, only samples.

$$\hat{p}(x) = \frac{1}{N} \sum_i \delta(x - x_i)$$

$$-\int dx \hat{p}(x) \log q(x) = -\int dx \frac{1}{N} \sum_i \delta(x - x_i) \log q(x) = -\frac{1}{N} \sum_i \log q(x_i)$$

Empirical Cross
Entropy

$$\min_q \mathcal{S}[\hat{p}; q]$$

Relative Entropy to a Uniform Distribution

The uniform distribution is special, the Relative entropy with respect to the uniform distribution is special.

$$\mathcal{S}[p; u] = \sum_i p(x_i) \log \frac{p(x_i)}{u(x_i)} = \sum_i p(x_i) \log p(x_i) + \log N$$

Entropy is a measure of how un-uniform a distribution is.

$$\mathcal{S}[p; u] = \log N - \mathcal{H}[p]$$

How easy it is to distinguish it from a uniform distribution.

$$\mathcal{H}[p] = - \sum_i p(x_i) \log p(x_i)$$

Mutual Information

$$\mathcal{I}[X; Y] = \mathcal{S}[p(x, y); p(x)p(y)] = \int dx dy p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$$

Mutual information is the relative entropy between a joint distribution and the product of its marginals.

It tests how easily we can discern whether there is a relationship between the two variables, rather than them being independent.

Summary

weight of evidence $w[x; p, q] \equiv \log \frac{p(x)}{q(x)}$

relative entropy $\mathcal{S}[p; q] \equiv \int dx p(x) \log \frac{p(x)}{q(x)}$

entropy $\mathcal{S}[p] \equiv - \int dx p(x) \log p(x)$

mutual information $\mathcal{I}(X; Y) = \int dx dy p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$

FIN

(THE END)

*Please check out the Problem Set [here](#)
Solutions [here](#)*

Information Theory Part II

[Alex Alemi](#)

2021-03-30

[Part 1](#)

[Part 2](#)

[Part 3](#)

[Problem Set](#)

[Solutions](#)

Errata

Cleveland, TN not Cleveland, OH

[\[quartz article\]](#)

[\[SAS article\]](#)

COLOR THEORY

A statistician got curious about M&M colors and went on an endearingly geeky quest for answers



By **Corinne Purtill**
Reporter

March 15, 2017

This article is more than 2 years old.

Correction: An earlier version of this story misstated the location of one of the M&Ms factories. The candies are produced in Cleveland, Tennessee, not Cleveland, Ohio.



Analysis



M&M Analysis

Hypothesis 1: M&M Bag is from NJ.

Hypothesis 2: M&M Bag is from TN.

—93.8 db

						
Hackettstown, NJ	12.5%	12.5%	12.5%	12.5%	25%	25%
Cleveland, TN	12.4%	13.1%	13.5%	19.8%	20.5%	20.7%
db	0.035	-0.20	-0.33	-2.0	0.86	0.82
My Bag	87	69	93	124	136	96

M&M Analysis

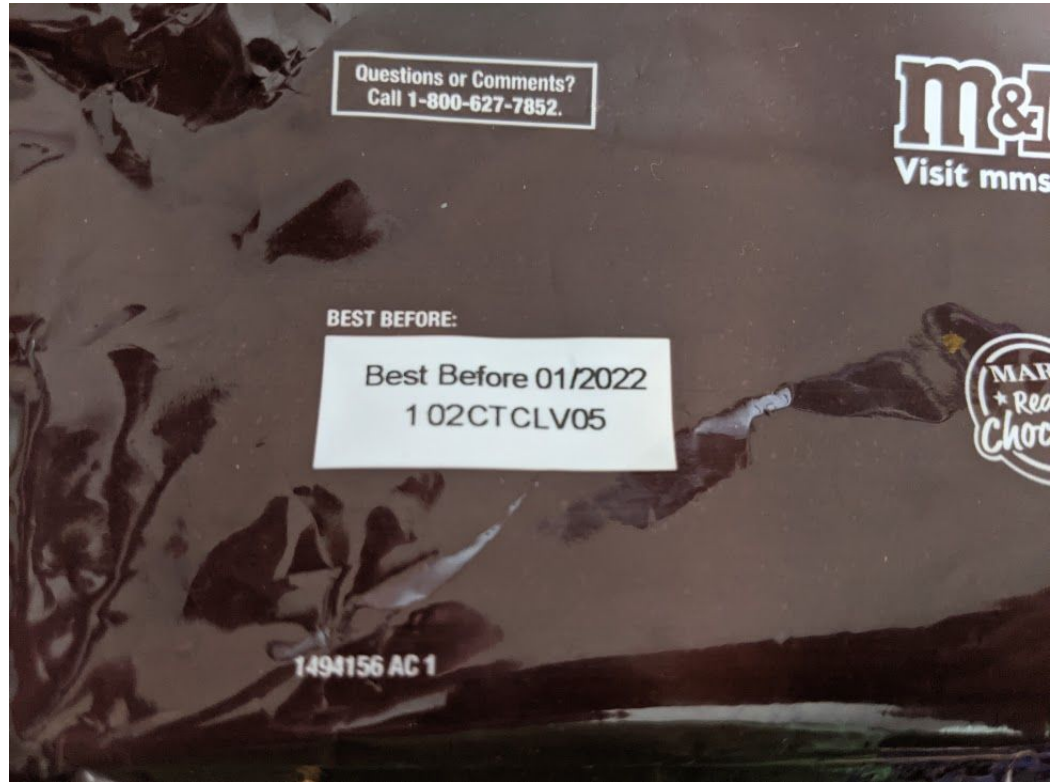
Hypothesis 1: M&M Bag is from NJ.

Hypothesis 2: M&M Bag is from TN.

—93.8 db

						
Hackettstown, NJ	12.5%	12.5%	12.5%	12.5%	25%	25%
Cleveland, TN	12.4%	13.1%	13.5%	19.8%	20.5%	20.7%
db	0.035	-0.20	-0.33	-2.0	0.86	0.82
My Bag	14.4%	11.4%	15.4%	20.5%	22.5%	15.8%

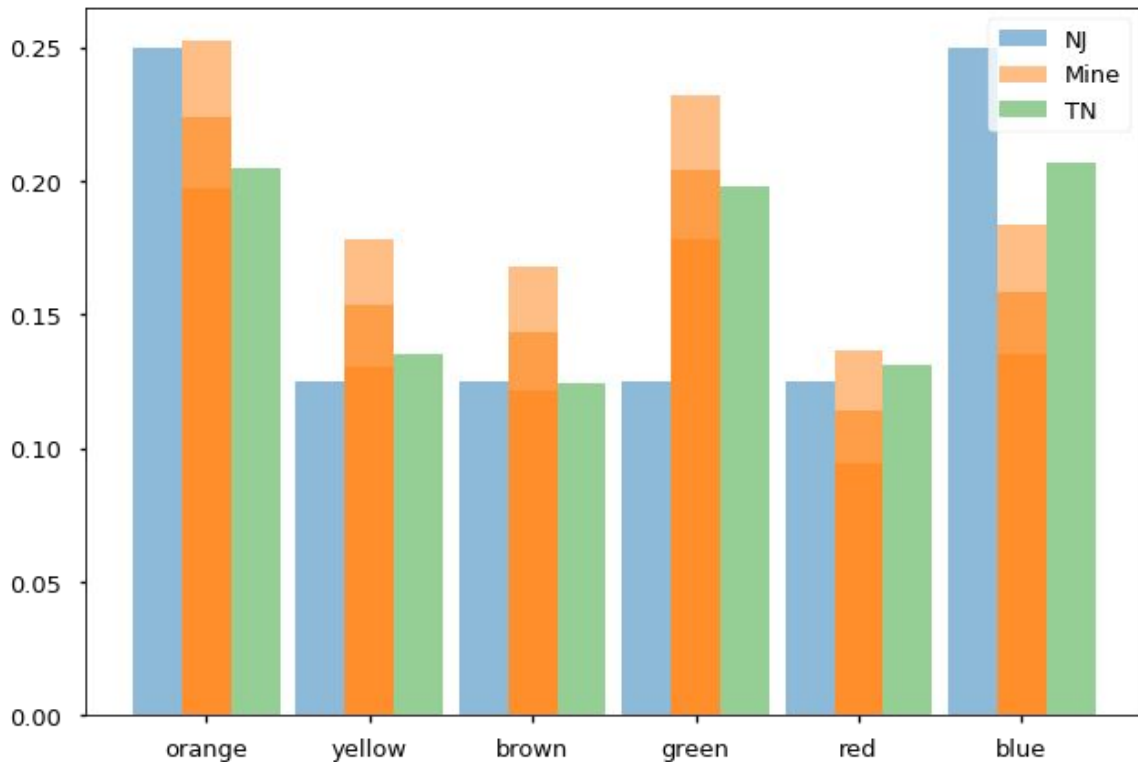
Smoking Gun



M&M Analysis

Deviation as large as the one
observed from NJ happened 0 out
of a million trials,

For TN it was 1 out of 48, ~2%



Unhelpful



MARS WRIGLEY NORTH AMERICA <contact@usc.mars.com>

Sat, Mar 27, 11:47 AM (2 days ago)

to me ▾

Hi Alex,

Thank you for contacting Mars Wrigley about M&M'S®.

We'd really like to help you, unfortunately, the information that you were asking for is strictly confidential.

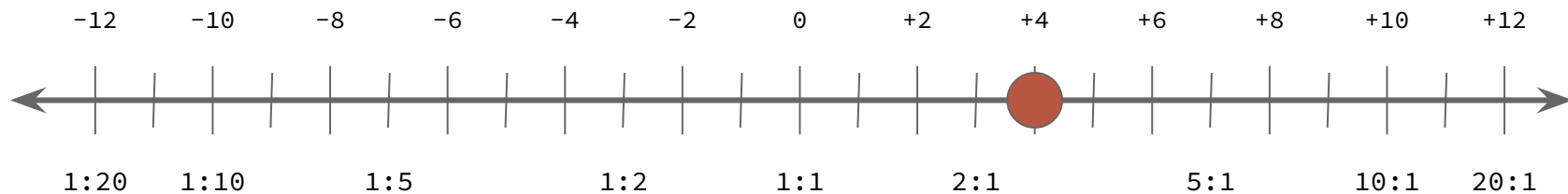
Thank you for understanding.

Regards,

Angellowe

Your Friend at Mars Wrigley

Two Ends

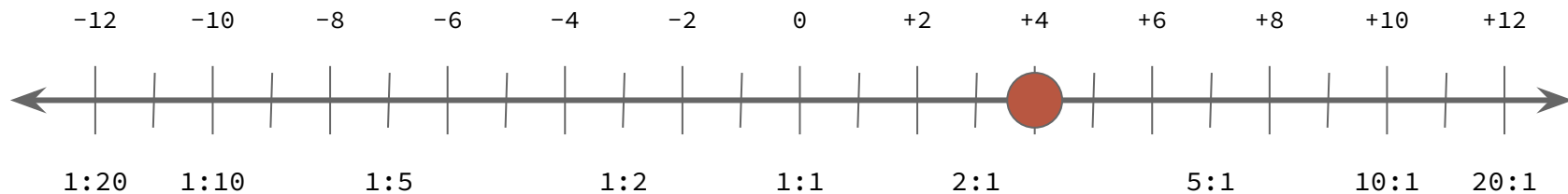


Your Bag of
M&Ms came
from TN

$$\log \frac{p(x)}{q(x)}$$

Your Bag of
M&Ms came
from NJ

Two Ends

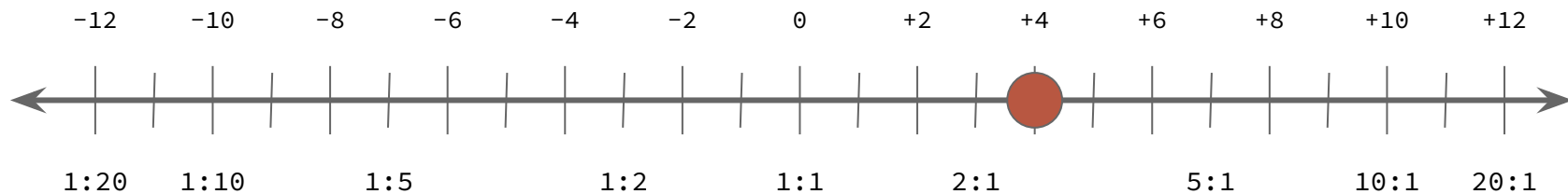


It'll snow
tomorrow

It'll rain
tomorrow

$$\log \frac{p(x)}{q(x)}$$

Two Ends



You don't win
the lottery

You win the
lottery

$$\log \frac{p(x)}{1 - p(x)}$$

Recap

weight of evidence

$$\log \frac{p(x)}{q(x)}$$

relative entropy

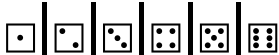
$$\mathcal{S}[p; q] = \left\langle \log \frac{p(x)}{q(x)} \right\rangle_p$$

expected weight of evidence | magnitude of belief update

Multiple Hypotheses

Exhaustive Set

Hypothesis 1: The dice landed 

Hypothesis 2: The dice landed 

$$P(H_2|D) = 1 = P(H_2)$$

Posterior Log Odds

Weight of Evidence

Prior Log Odds

$$\log \frac{\Pr(H_1|D)}{\Pr(H_2|D)} = \log \frac{\Pr(D|H_1)}{\Pr(D|H_2)} + \log \frac{\Pr(H_1)}{\Pr(H_2)}$$

Exhaustive Set

Hypothesis 1: The dice landed 

Hypothesis 2: The dice landed      

$$P(H_2|D) = 1 = P(H_2)$$

Posterior Log Odds

Weight of Evidence

Prior Log Odds

$$\log \frac{\Pr(H_1|D)}{\cancel{\Pr(H_2|D)}} = \log \frac{\Pr(D|H_1)}{\Pr(D|H_2)} + \log \frac{\cancel{\Pr(H_1)}}{\Pr(H_2)}$$

Information in Data for outcome

Hypothesis 1: The dice landed 

Hypothesis 2: The dice landed      

$$P(H_2|D) = 1 = P(H_2)$$

$$\text{w.o.e.} = \log \frac{P(D|H_1)}{P(H_1)}$$

Data: the dice is showing 

$$P(D|H_1) = 1$$

Information in Data for outcome

Hypothesis 1: The dice landed 

Hypothesis 2: The dice landed      

$$P(H_2|D) = 1 = P(H_2)$$

$$\text{w.o.e.} = \log \frac{\cancel{P(D|H_1)}}{P(H_1)}$$

Data: the dice is showing 

$$P(D|H_1) = 1$$

Information in Discerning Data

Hypothesis 1: The dice landed 

Hypothesis 2: The dice landed |||||

$$\text{w.o.e} = \log \frac{1}{P(H_1)}$$

Information in the discerning observation of a hypothesis amongst a set of exhaustive alternatives

Observations of a Random Variable

Hypothesis 1: $X = x$

Hypothesis 2: $x \in \mathcal{X}$

$$\log \frac{1}{p(x)}$$

Information in the observation of a particular value of a random variable.

"Surprise"

$$-\log p(x)$$

Small probability -> Large Surprise

Large Probability -> Small Surprise

"Surprise"

$$\log \frac{1}{p(x)}$$

Small probability -> Large Surprise

Large Probability -> Small Surprise

Entropy



Entropy is Expected "Surprise"

$$\mathcal{H}[X] = \sum_i p(x_i) \log \frac{1}{p(x_i)}$$

$$\mathcal{H}[X] = - \sum_i p(x_i) \log p(x_i)$$

Entropy is the expected "information content" of an observation of a random variable.

Shannon's Axioms of Entropy

Entropy is the unique measure of "surprise" satisfying some simple axioms.
(from Shannon)

1. *Continuous*. Small changes in the probabilities should correspond to small changes in the entropy.
2. *Monotonic*. For uniform distributions, entropy should be an increasing function of N .
3. *Satisfy a natural chain rule*. If a choice is decomposed into a series of choices, the entropy would be the weighted sum of the pieces.

The Entropy is the only function to satisfy all of these requirements.

Relative Entropy to a Uniform Distribution

The uniform distribution is special, the Relative entropy with respect to the uniform distribution is special.

$$\mathcal{S}[p; u] = \sum_i p(x_i) \log \frac{p(x_i)}{u(x_i)} = \sum_i p(x_i) \log p(x_i) + \log N$$

Entropy is a measure of how un-uniform a distribution is.

$$\mathcal{S}[p; u] = \log N - \mathcal{H}[p]$$

How easy it is to distinguish it from a uniform distribution.

$$\mathcal{H}[p] = - \sum_i p(x_i) \log p(x_i)$$

Entropy and Cross Entropy

$$\int dx p(x) \log \frac{p(x)}{q(x)} = \left[- \int dx p(x) \log q(x) \right] - \left[- \int dx p(x) \log p(x) \right]$$

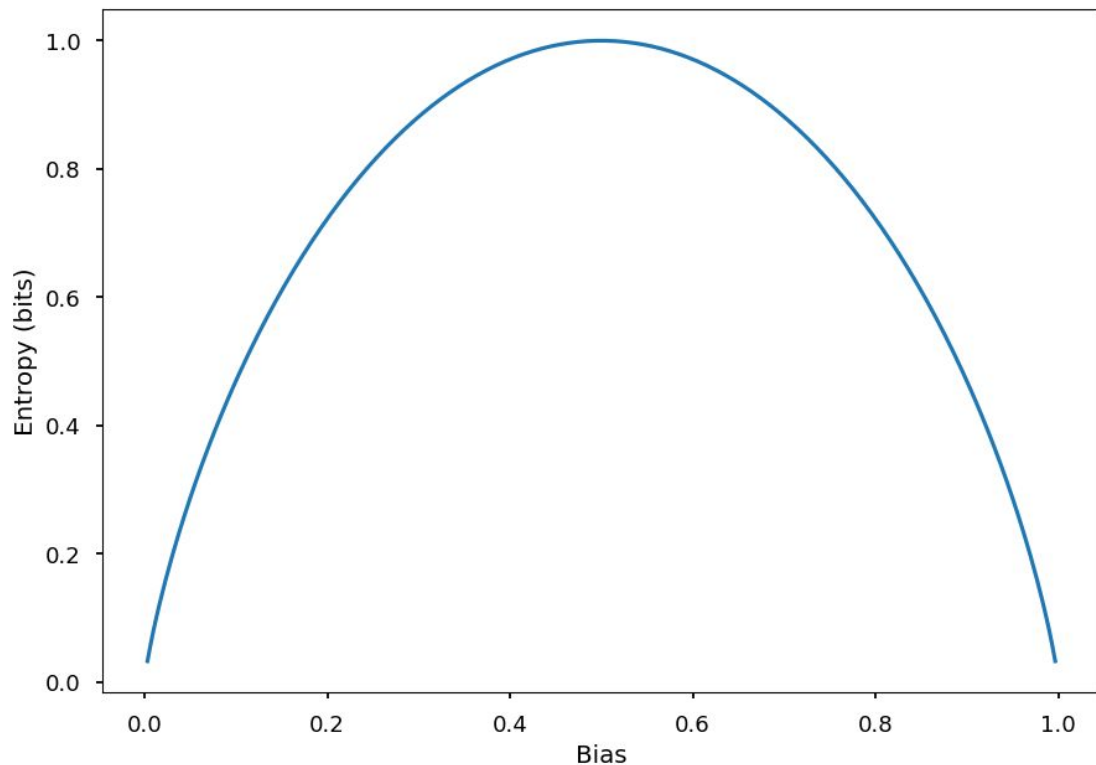
$$\mathcal{S}[p; q] = \mathcal{H}[p; q] - \mathcal{H}[p]$$

Relative Entropy is Cross Entropy minus Entropy

Entropy is a measure of
ignorance

Entropy of a Biased Coin

For a biased coin, entropy is maximal when the coin is fair, and is 1 bit for a single coin.



Example: Uniform Distributions

$$\mathcal{H}[X] = - \sum_i \frac{1}{4} \log_2 \frac{1}{4} = 2 \text{ bits}$$

$$\mathcal{H}[X] = \log_2 N$$

Number of yes-no questions to identify an element.

$$2^{20} \sim 10^6$$

20 questions => 1 in a million

$$\log_2(7.713 \times 10^9) = 32.8 \text{ bits}$$

Entropy is the Data Compression Limit

In order to represent a sample from a discrete distribution, the entropy tells how many bits it takes to represent a sample from the distribution, on average.

Example: Cards

diamonds
clubs
diamonds
diamonds
spades
spades
hearts
clubs
diamonds
clubs
...

$$\frac{5 \cdot 8}{4} + \frac{5 \cdot 8}{4} + \frac{6 \cdot 8}{4} + \frac{8 \cdot 8}{4} = 48 \text{ bits}$$

Example: Cards

◇
♣
◇
◇
♠
♠
♥
♣
◇
♣
...

$$\frac{3 \cdot 8}{4} + \frac{3 \cdot 8}{4} + \frac{3 \cdot 8}{4} + \frac{3 \cdot 8}{4} = 24 \text{ bits}$$

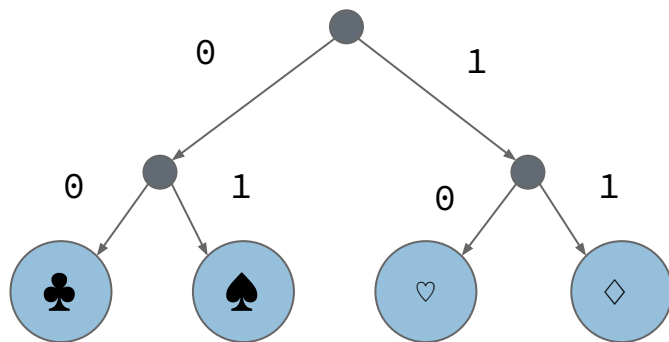
Example: Cards

d
c
d
d
s
s
h
c
d
c
...

$$\frac{8}{4} + \frac{8}{4} + \frac{8}{4} + \frac{8}{4} = 8 \text{ bits}$$

Example: Cards

11
00
11
11
01
01
10
00
11
00
...



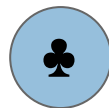
$$\frac{2}{4} + \frac{2}{4} + \frac{2}{4} + \frac{2}{4} = 2 \text{ bits}$$

This is the best we can do.

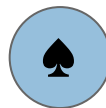
$$\mathcal{H}[X] = 2 \text{ bits}$$

Example: Biased Cards

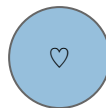
11
00
11
11
01
01
10
00
11
00
...



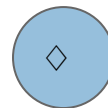
$1/2$



$1/4$



$1/8$



$1/8$

$$\frac{2}{2} + \frac{2}{4} + \frac{2}{8} + \frac{2}{8} = 2 \text{ bits}$$

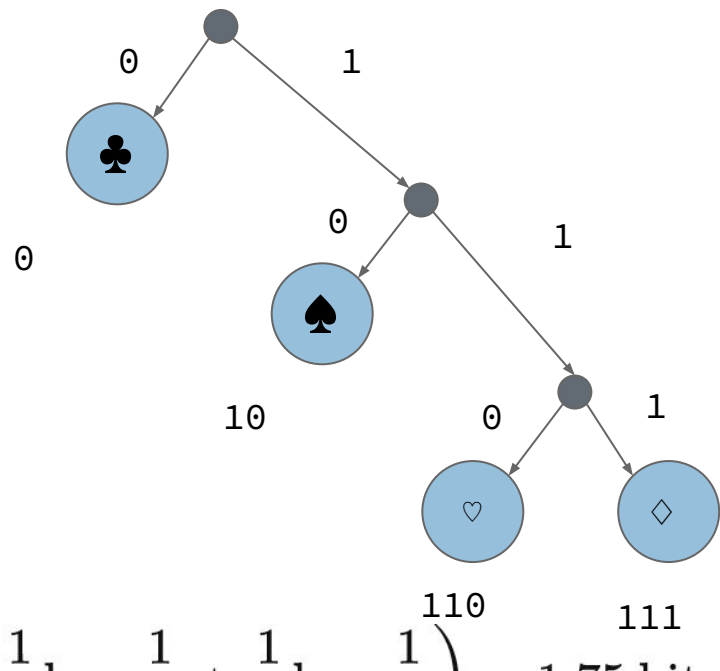
$$\mathcal{H}[X] = -\left(\frac{1}{2}\log_2 \frac{1}{2} + \frac{1}{3}\log_2 \frac{1}{4} + \frac{1}{8}\log_2 \frac{1}{8} + \frac{1}{8}\log_2 \frac{1}{8}\right) = 1.75 \text{ bits}$$

Example: Biased Cards

111
0
111
111
10
10
110
0
111
...

$$\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{3}{8} = 1.75 \text{ bits}$$

$$\mathcal{H}[X] = -\left(\frac{1}{2}\log_2 \frac{1}{2} + \frac{1}{3}\log_2 \frac{1}{4} + \frac{1}{8}\log_2 \frac{1}{8} + \frac{1}{8}\log_2 \frac{1}{8}\right) = 1.75 \text{ bits}$$

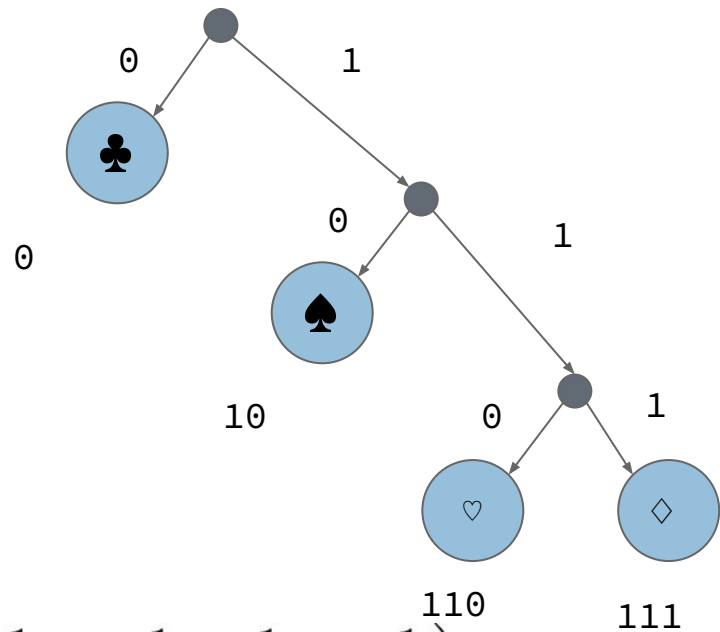


Example: Biased Cards

1110111111101011001110

$$\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{3}{8} = 1.75 \text{ bits}$$

$$\mathcal{H}[X] = -\left(\frac{1}{2}\log_2 \frac{1}{2} + \frac{1}{3}\log_2 \frac{1}{4} + \frac{1}{8}\log_2 \frac{1}{8} + \frac{1}{8}\log_2 \frac{1}{8}\right) = 1.75 \text{ bits}$$

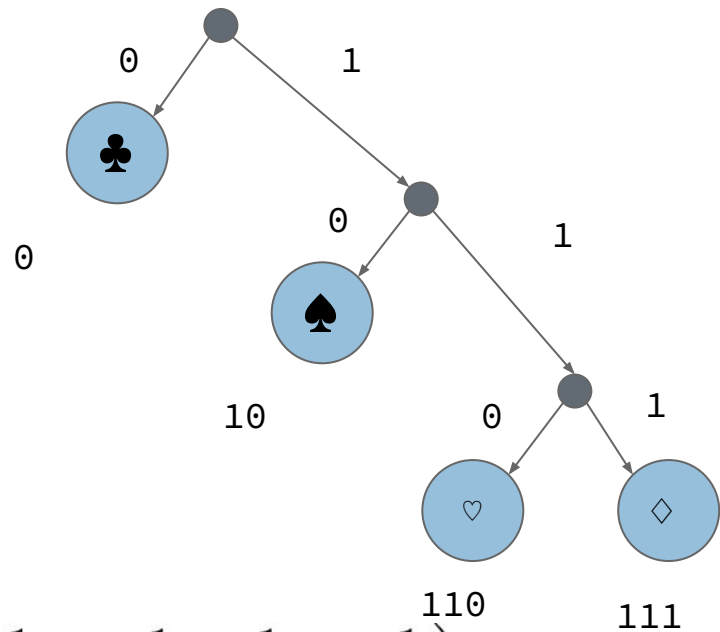


Sampling Biased Cards

1110111111101011001110

$$\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{3}{8} = 1.75 \text{ bits}$$

$$\mathcal{H}[X] = -\left(\frac{1}{2}\log_2 \frac{1}{2} + \frac{1}{3}\log_2 \frac{1}{4} + \frac{1}{8}\log_2 \frac{1}{8} + \frac{1}{8}\log_2 \frac{1}{8}\right) = 1.75 \text{ bits}$$

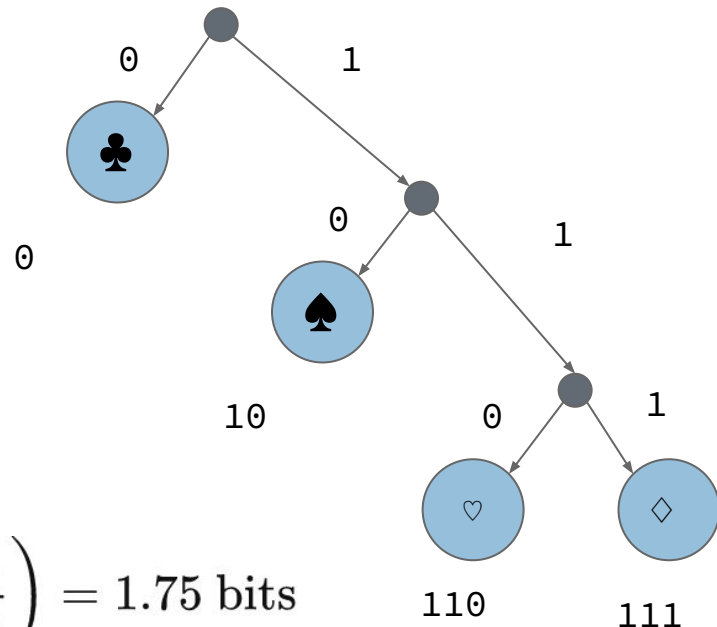


Entropy is the Number of Coin Flips to Sample

In order to generate a sample from a discrete distribution, on average it takes as many fair coin flips as the entropy of the distribution.

$$\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{3}{8} = 1.75 \text{ bits}$$

$$\mathcal{H}[X] = -\left(\frac{1}{2}\log_2 \frac{1}{2} + \frac{1}{3}\log_2 \frac{1}{4} + \frac{1}{8}\log_2 \frac{1}{8} + \frac{1}{8}\log_2 \frac{1}{8}\right) = 1.75 \text{ bits}$$



Relative Entropy is Excess Cost

Relative entropy is how much more expensive it is to represent a sample from p if you use q to encode rather than p .

$$\mathbb{E}_p \left[\log \frac{1}{q(x)} \right] = \mathcal{H}[p] + \mathcal{S}[p; q]$$

Asymptotic Equipartition Theorem

Weak Law of Large Numbers

Averages tend to means

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_i X_i \rightarrow \mu$$

Weak Law of Large Numbers

Joint likelihoods tend to entropies.

$$\lim_{N \rightarrow \infty} -\frac{1}{N} \log p(X_1, X_2, \dots, X_N) \rightarrow \mathcal{H}[X]$$

Typical Sets

Sequences with likelihoods $\sim$ entropy

$$2^{-N(\mathcal{H}(X)+\epsilon)} \leq p(x_1, \dots, x_N) \leq 2^{-N(\mathcal{H}(X)-\epsilon)}$$

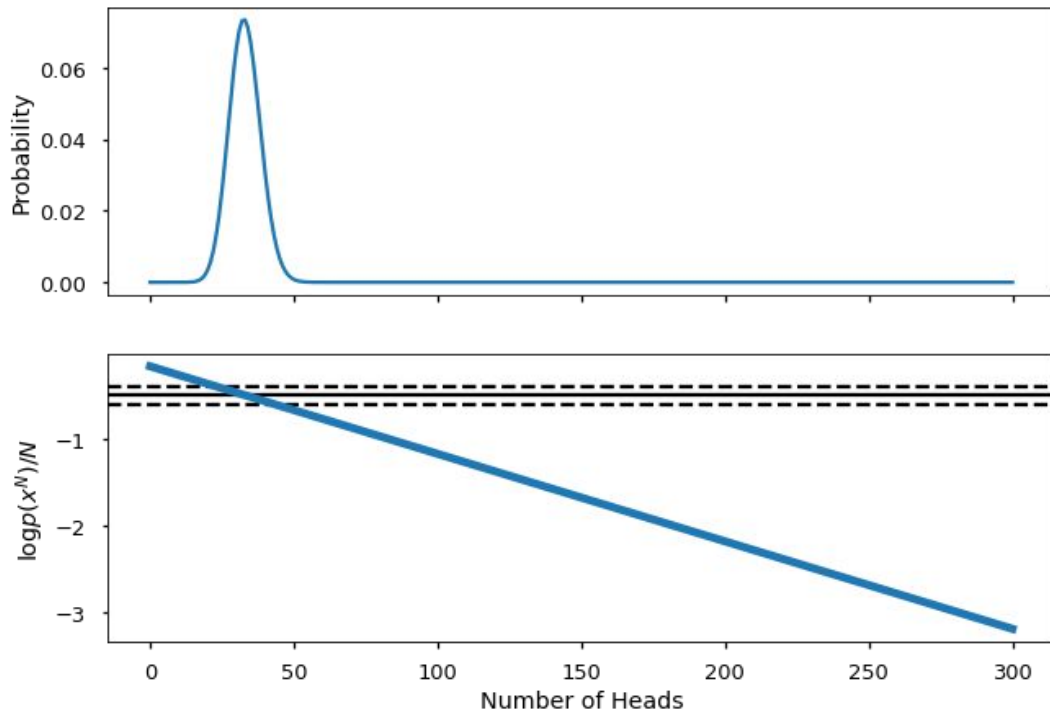
Asymptotic Equipartition Property

- The typical set has probability near 1
- Elements of the typical set are nearly equiprobable
- The number of elements in the typical set is nearly $2^{N\mathcal{H}} = (2^{\mathcal{H}})^N$

Biased Coin

- Coin with $p=0.11$
- 300 flips

$$\mathcal{H} = 0.5 \text{ bits}$$



Sketch

- Separate all possible sequences into two groups, typical and atypical. Prepend 0 if typical, 1 if atypical.
- Index each of the typical examples
- Index each of the atypical examples.

$$P(\text{typical}) \log_2 |\text{typical}| + (1 - P(\text{typical})) \log_2 |\mathcal{X}| + 1 \\ \sim N(\mathcal{H} + \epsilon)$$

Entropy is ~ how many
things there are

Continuous entropy is silly.

Continuous Entropy

$$\mathcal{H}[p] = - \int dx \, p(x) \log p(x)$$

$$[p(x)] = [X]^{-1}$$

$$\int [dx] [p(x)] \log [p(x)]$$

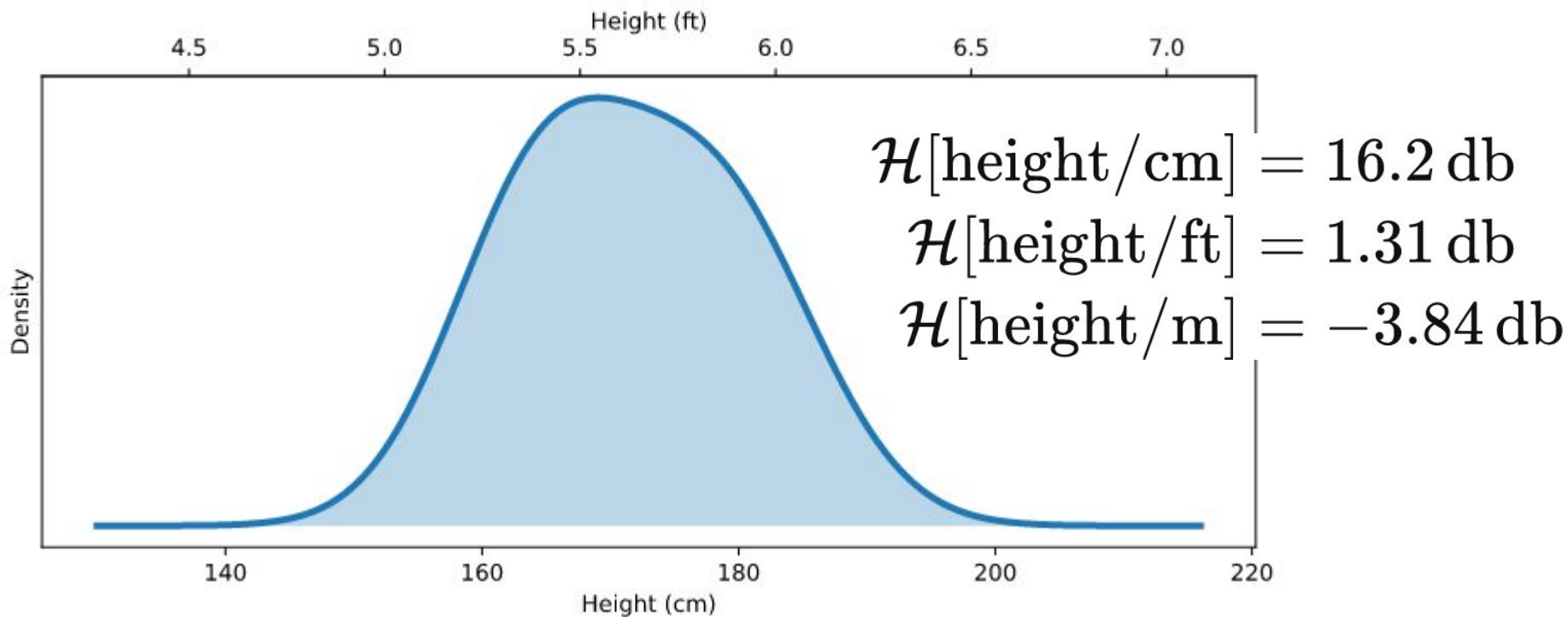
$$\int [X] [X]^{-1} \log [X]^{-1}$$

?

$$y = \alpha x$$

$$\mathcal{H}[X] = - \int dx \, p(x) \log p(x) = - \int dy \, p(y) \log \alpha p(y) = \mathcal{H}[Y] + \log \alpha$$

Continuous Entropy



Cautionary Tale: Images

$$[-1, 1] \quad [0, 1] \quad [0, 256]$$

$$\log(2) \sim 0.69 \text{ nat/dim}$$

$$\log(256) \sim 5.5 \text{ nat/dim}$$

Mutual Information



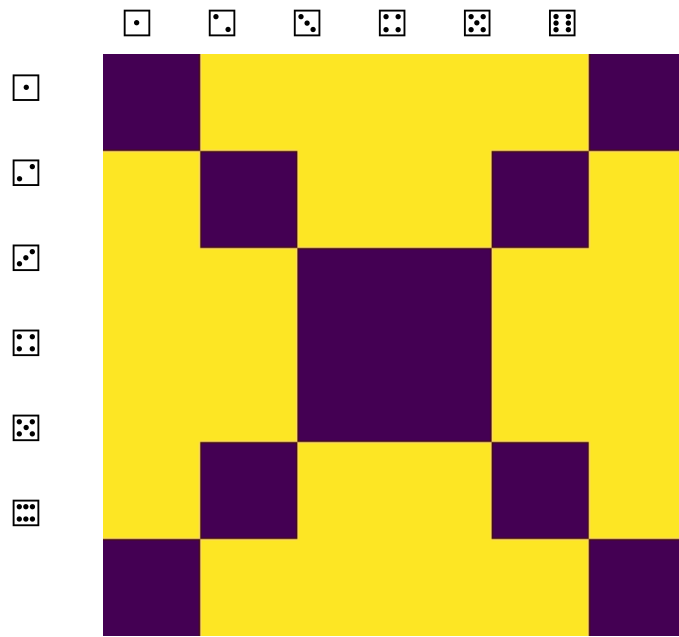
Front and Top of Die

If I tell you the front face of a die,
how much does that reveal about
the top?

$$10 \log_{10}(6) = 7.8 \text{ db}$$

$$10 \log_{10}(4) = 6.0 \text{ db}$$

$$I(X; Y) = \mathcal{H}(X) - \mathcal{H}(X|Y)$$



Mutual Information is a reduction in Entropy

$$\mathcal{I}[X; Y] = \mathcal{H}[X] - \mathcal{H}[X|Y]$$

Bits you learn
about X after
observing Y

Bits you need
to represent
an X sample

Bits you need
to represent
an X sample, if
you know Y

Mutual Information

$$\mathcal{I}[X; Y] = \mathcal{S}[p(x, y); p(x)p(y)] = \int dx dy p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$$

Mutual information is the relative entropy between a joint distribution and the product of its marginals.

It tests how easily we can discern whether there is a relationship between the two variables, rather than them being independent.

Mutual Information

$$\begin{aligned}\mathcal{I}[X; Y] &= \int dx dy p(x, y) \log \frac{p(x, y)}{p(x)p(y)} = \mathcal{S}[p(x, y); p(x)p(y)] \\ &= \int dx p(x) dy p(y|x) \log \frac{p(y|x)}{p(y)} = \mathbb{E}_{p(x)}[\mathcal{S}[p(y|x); p(y)]] \\ &= \int dy p(y) dx p(x|y) \log \frac{p(x|y)}{p(x)} = \mathbb{E}_{p(y)}[\mathcal{S}[p(x|y); p(x)]]\end{aligned}$$

Mutual Information is symmetric. It's also the relative entropy between the conditional distribution after observing a sample and the marginal.

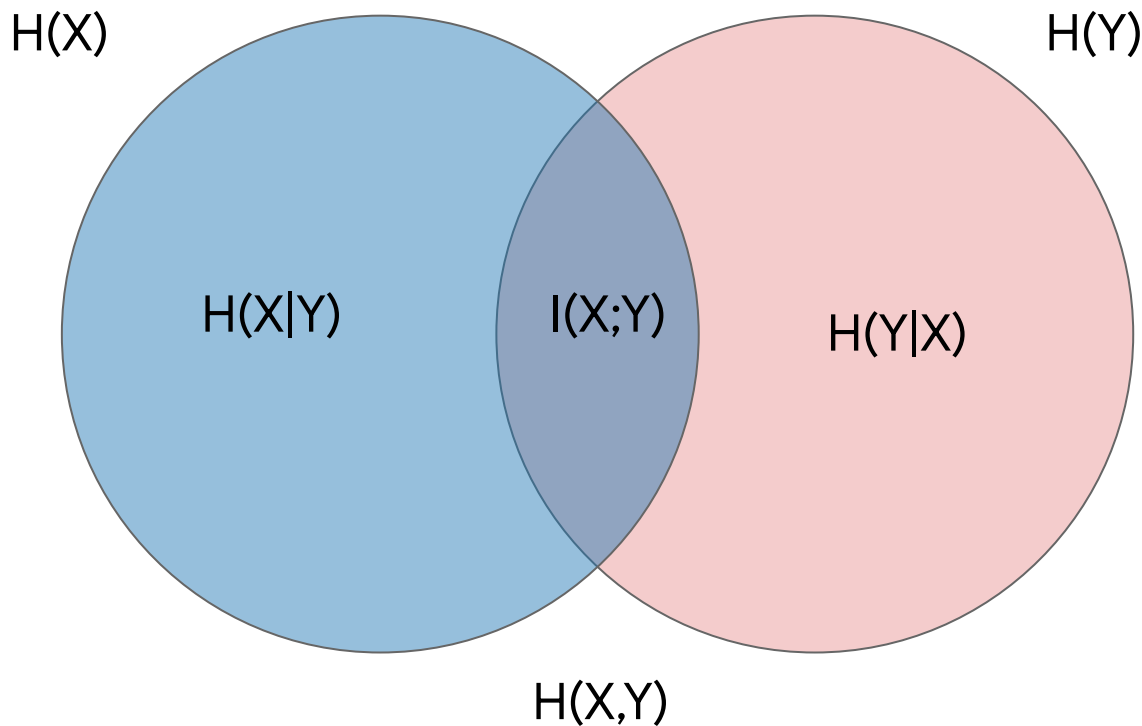
Mutual Information is Positive

$$\mathcal{I}[X; Y] = \mathcal{S}[p(x, y); p(x)p(y)] \geq 0$$

Mutual information is positive because relative entropy is positive.

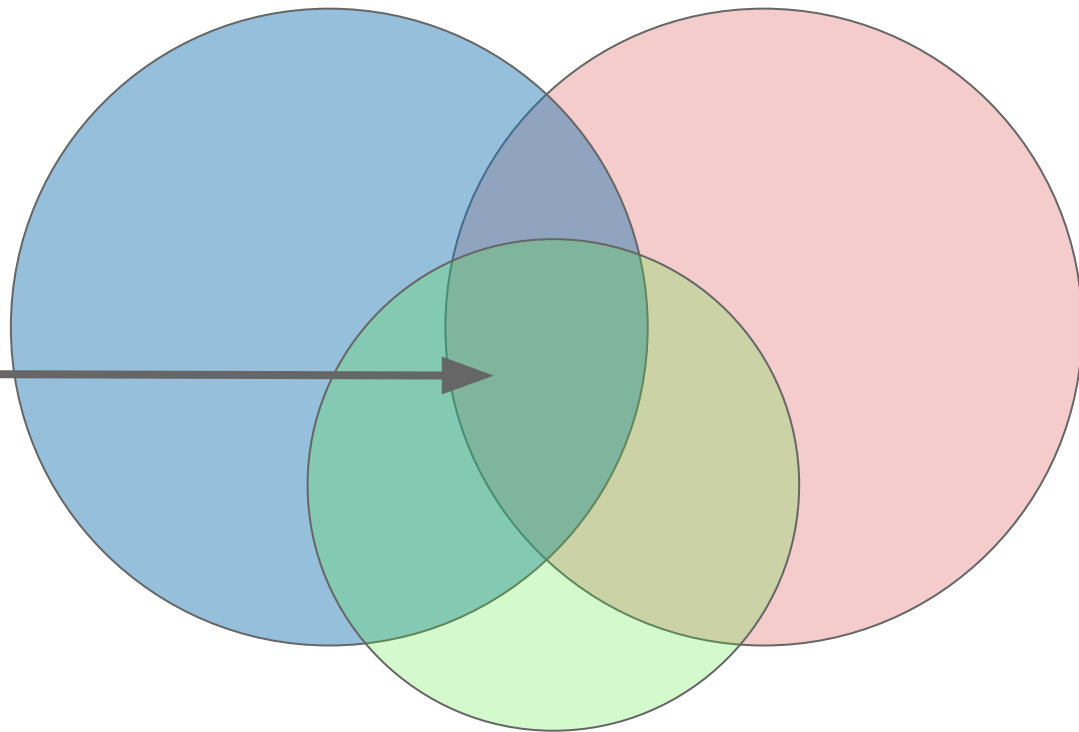
Venn Diagrams

$$\begin{aligned}\mathcal{I}[X;Y] &= \mathcal{H}[X] - \mathcal{H}[Y|X] \\ &= \mathcal{H}[Y] - \mathcal{H}[Y|X] \\ &= \mathcal{H}[X] + \mathcal{H}[Y] - \mathcal{H}[X,Y] \\ &= \mathcal{H}[X,Y] - \mathcal{H}[X|Y] - \mathcal{H}[Y|X]\end{aligned}$$



Caution

Diagrams suggest all quantities positive, but the intersection of three random variables is not necessarily positive!

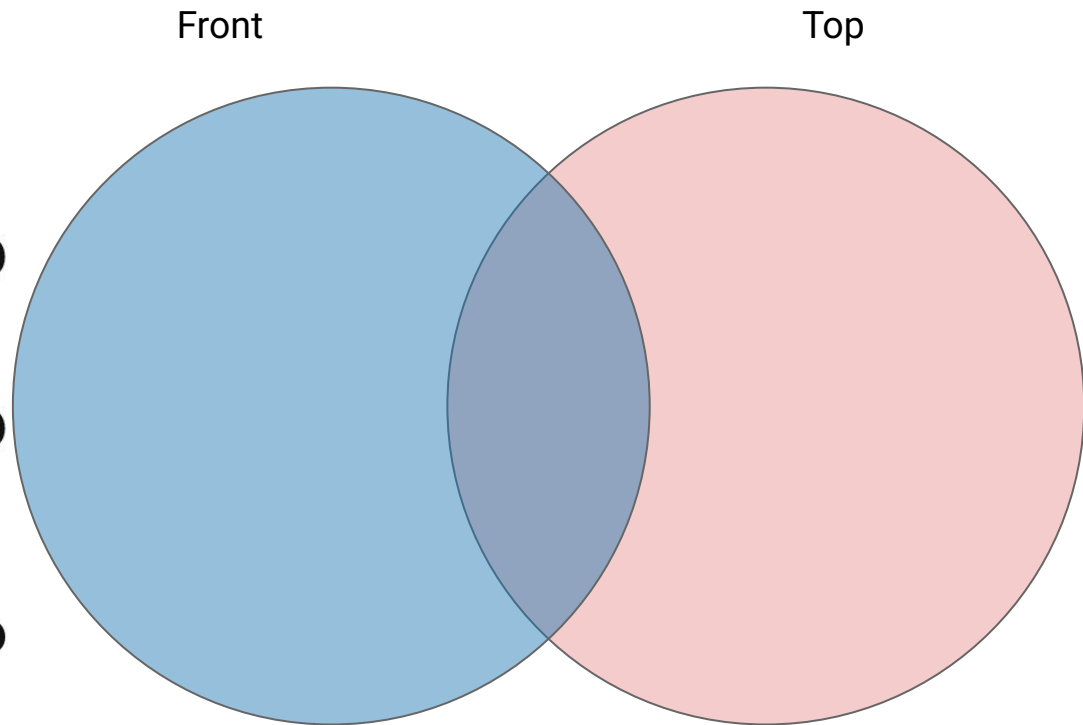


Example: Die

$$10 \log_{10}(6) = 7.8 \text{ db}$$

$$10 \log_{10}(4) = 6.0 \text{ db}$$

$$10 \log_{10} \left(\frac{6 \cdot 6}{4 \cdot 6} \right) \sim 1.8 \text{ db}$$



Cast of Characters

weight of evidence measures the importance of data for distinguishing hypotheses

relative entropy measures a "distance" between two distributions

entropy measures a "complexity" of a distribution

mutual information measures the "complexity" of the interaction between two random variables.

Summary

weight of evidence $w[x; p, q] \equiv \log \frac{p(x)}{q(x)}$

relative entropy $\mathcal{S}[p; q] \equiv \int dx p(x) \log \frac{p(x)}{q(x)}$

entropy $\mathcal{S}[p] \equiv - \int dx p(x) \log p(x)$

mutual information $\mathcal{I}(X; Y) = \int dx dy p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$

Relative Entropy as a Principle for Learning

A Principle For Learning

Make it hard to tell the difference between the real world and your model.

$$\min \mathcal{S}[\text{real}; \text{model}]$$

$$\min_q \int dx p(x) \log \frac{p(x)}{q(x)}$$

$$\min_q \int dx p(x) \log \frac{p(x)}{q(x)} = \int dx p(x) \log p(x) - \min_q \int dx p(x) \log q(x)$$

-Data Entropy

Cross Entropy

Maximum Likelihood

We don't have access to the true distribution of data, only samples.

$$\hat{p}(x) = \frac{1}{N} \sum_i \delta(x - x_i)$$

$$-\int dx \hat{p}(x) \log q(x) = -\int dx \frac{1}{N} \sum_i \delta(x - x_i) \log q(x) = -\frac{1}{N} \sum_i \log q(x_i)$$

Empirical Cross
Entropy

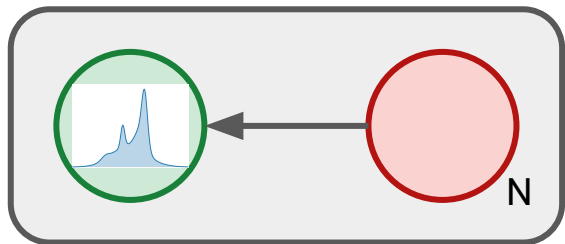
$$\min_q \mathcal{S}[\hat{p}; q]$$

Variational Autoencoders

World P

world we have

$$p(x, z) = p(z|x)p(x)$$



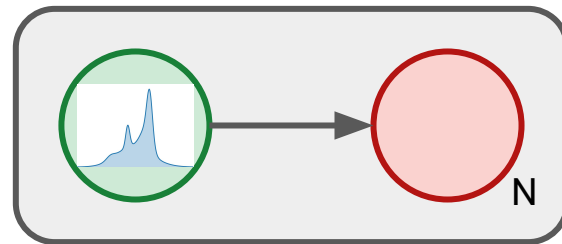
Rep

Images

World Q

world we want

$$q(z, x) = q(z)q(x|z)$$



Rep

Images

VAE

$$\begin{aligned}\text{KL}[p; q] &= \mathbb{E}_{p(x, z)} \left[\log \frac{p(z, x)}{q(z, x)} \right] \\&= \left\langle \log \frac{p(x)p(z|x)}{q(x|z)q(z)} \right\rangle_p \\&= \langle \log p(x) \rangle + \left\langle -\log q(x|z) + \log \frac{p(z|x)}{q(z)} \right\rangle \\&= \mathbb{H}_p[x] + \left\langle -\log q(x|z) + \beta \log \frac{p(z|x)}{q(z)} \right\rangle\end{aligned}$$

FIN

(THE END)

*Please check out the Problem Set [here](#)
Solutions [here](#)*

Information Theory Part III

[Alex Alemi](#)

2021-03-30

[Part 1](#)

[Part 2](#)

[Part 3](#)

[Problem Set](#)

[Solutions](#)

Skipped Example

PII – Population

$$10 \log_{10} \frac{1}{\text{US Pop}} = -85 \text{ db}$$

PII – Birthday

$$10 \log_{10} \frac{1}{\text{US Pop}} = -85 \text{ db}$$

$$10 \log_{10}(365) = 26.6 \text{ db}$$

PII - Age

$$10 \log_{10} \frac{1}{\text{US Pop}} = -85 \text{ db}$$

$$10 \log_{10}(365) = 26.6 \text{ db}$$

$$10 \log_{10}(\text{age}) = 19 \text{ db}$$

$$10 \log_{10}(63) = 18 \text{ db}$$

PII – Zip Code

$$10 \log_{10} \frac{1}{\text{US Pop}} = -85 \text{ db}$$

$$10 \log_{10}(365) = 26.6 \text{ db}$$

$$10 \log_{10}(\text{age}) = 19 \text{ db}$$

$$10 \log_{10}(\text{zip}) = 41.6 \text{ db}$$

$$10 \log_{10}(10^5) = 50 \text{ db}$$

$$10 \log_{10}(43 \cdot 10^3) = 46 \text{ db}$$

PII – Gender

$$10 \log_{10} \frac{1}{\text{US Pop}} = -85 \text{ db}$$

$$10 \log_{10}(365) = 26.6 \text{ db}$$

$$10 \log_{10}(\text{age}) = 19 \text{ db}$$

$$10 \log_{10}(\text{zip}) = 41.6 \text{ db}$$

$$10 \log_{10}(2) = 3 \text{ db}$$

PII ~ Birthday + Age + Zip Code + Gender

$$10 \log_{10}(\text{US Pop}) = 85 \text{ db}$$

$$10 \log_{10}(365) = 26.6 \text{ db}$$

$$10 \log_{10}(\text{age}) = 19 \text{ db}$$

$$10 \log_{10}(\text{zip}) = 41.6 \text{ db}$$

$$10 \log_{10}(2) = 3 \text{ db}$$

5 db

*87% of American are
uniquely identified by
DOB + Age + Zip Code
+ Gender*

*Simple Demographics Often Identify People
Uniquely - Sweeney 2000*

Continuous and Deterministic

For continuous random variables, invertible transformations contain infinite information.

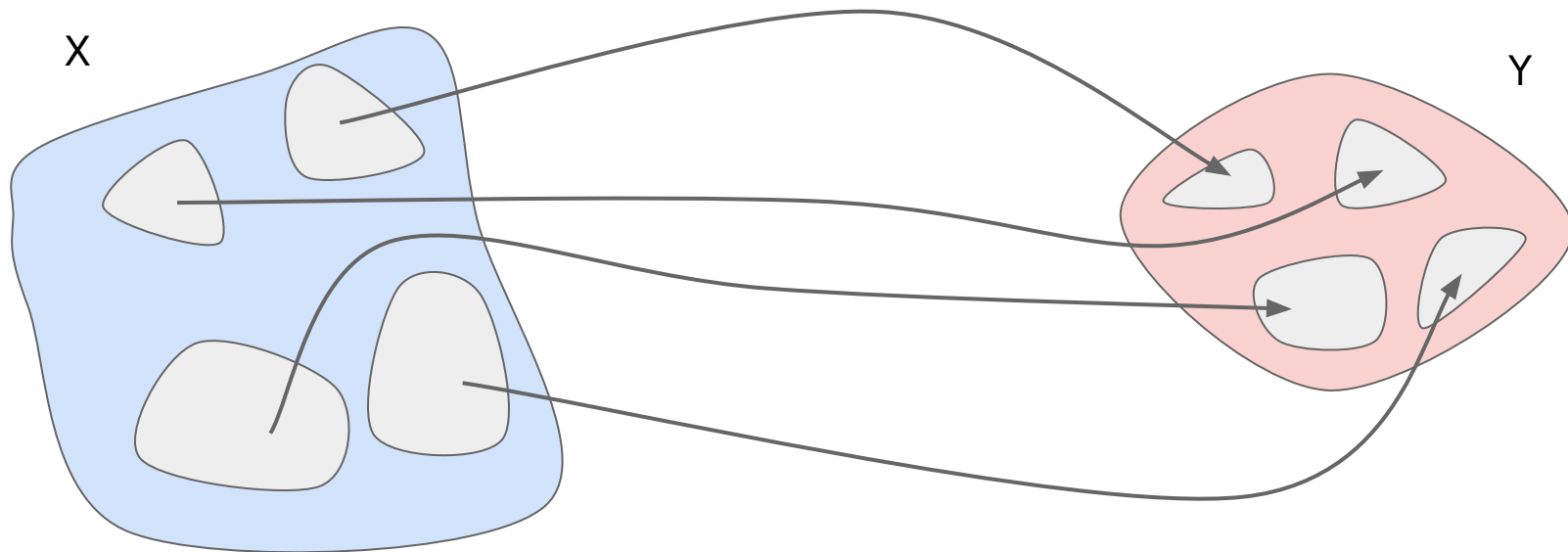
$$\mathcal{I}[X; g(X)] = \infty \text{ bit}$$

Deterministic $\neq$ Invertible.

$$\mathcal{I}[X; \text{sgn}(X)] = 1 \text{ bit}$$

Number of Tubes

Joint AEP $\rightarrow$ Mutual information is the number of "tubes" between X and Y

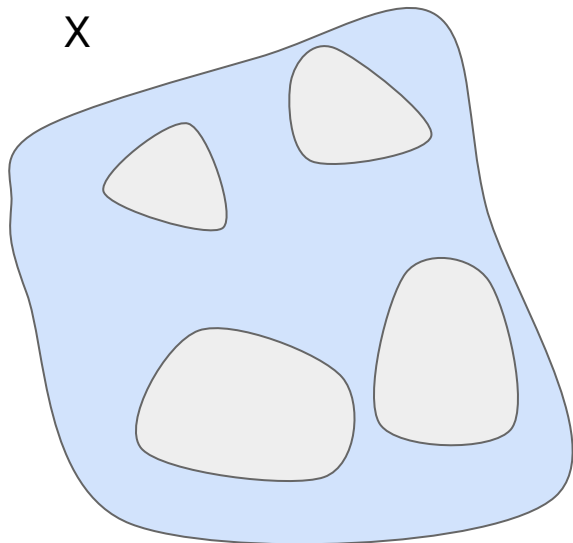


Number of Tubes

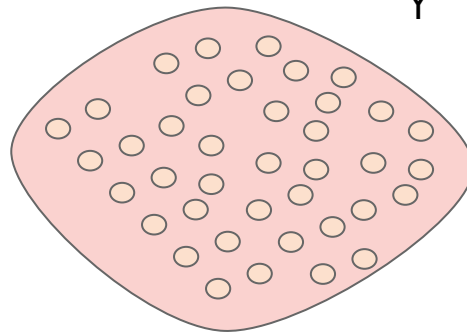
Joint AEP $\rightarrow$ Mutual information is the number of "tubes" between X and Y

$$2^{N\mathcal{H}[Y]}$$

X



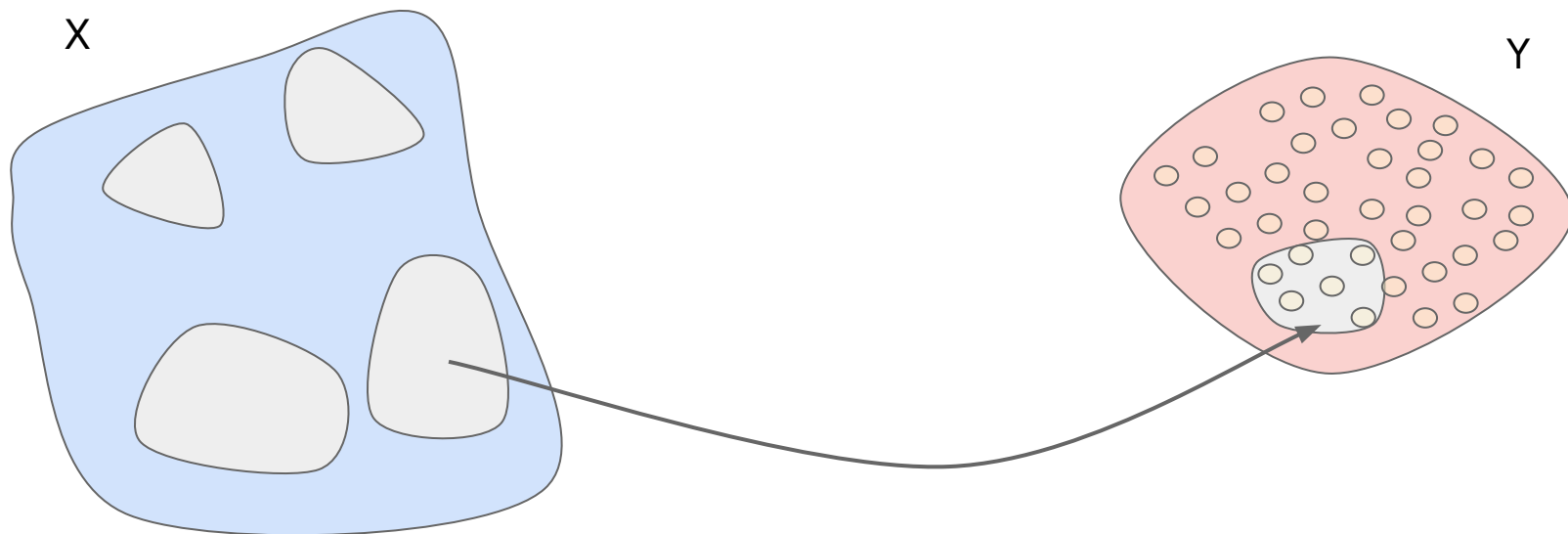
Y



Number of Tubes

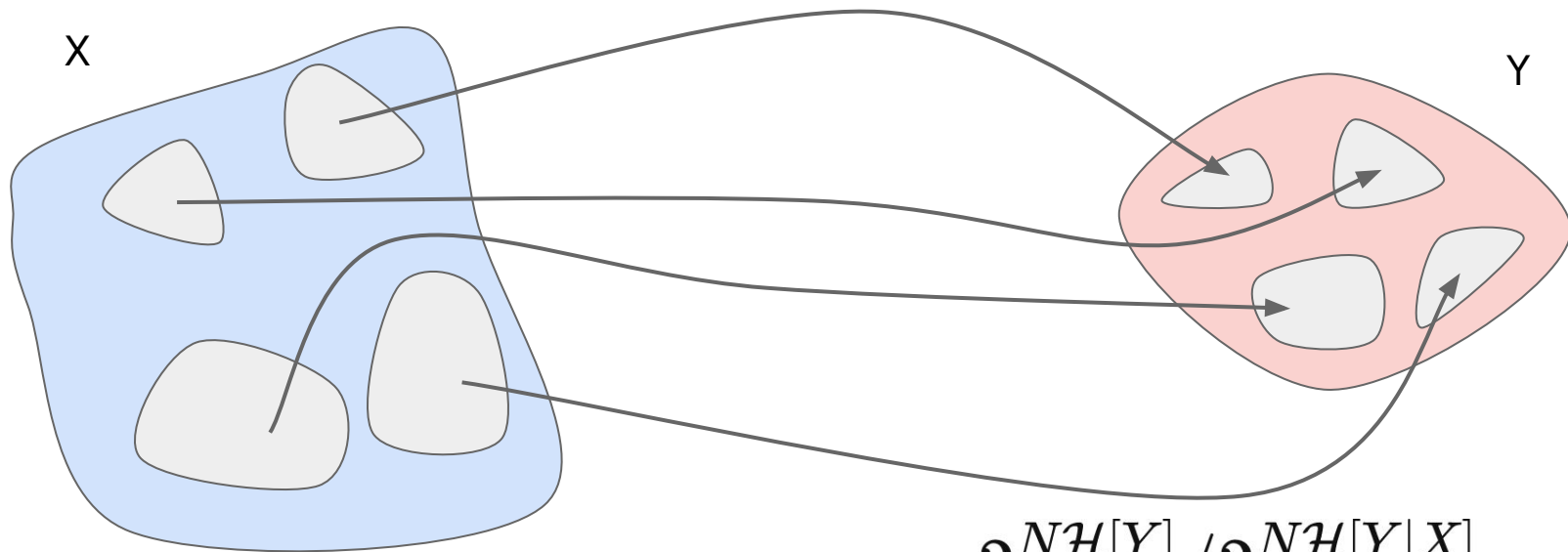
Joint AEP $\rightarrow$ Mutual information is the number of "tubes" between X and Y

$$2^{N\mathcal{H}[Y|X=x]}$$



Number of Tubes

Joint AEP $\rightarrow$ Mutual information is the number of "tubes" between X and Y



$$2^{N\mathcal{H}[Y]} / 2^{N\mathcal{H}[Y|X]} = 2^{N\mathcal{I}[X;Y]}$$

Processing only destroys
information

Data Processing Inequality

$$X \rightarrow Z \rightarrow Y$$

$$\mathcal{I}[X; Y] \leq \mathcal{I}[X; Z]$$

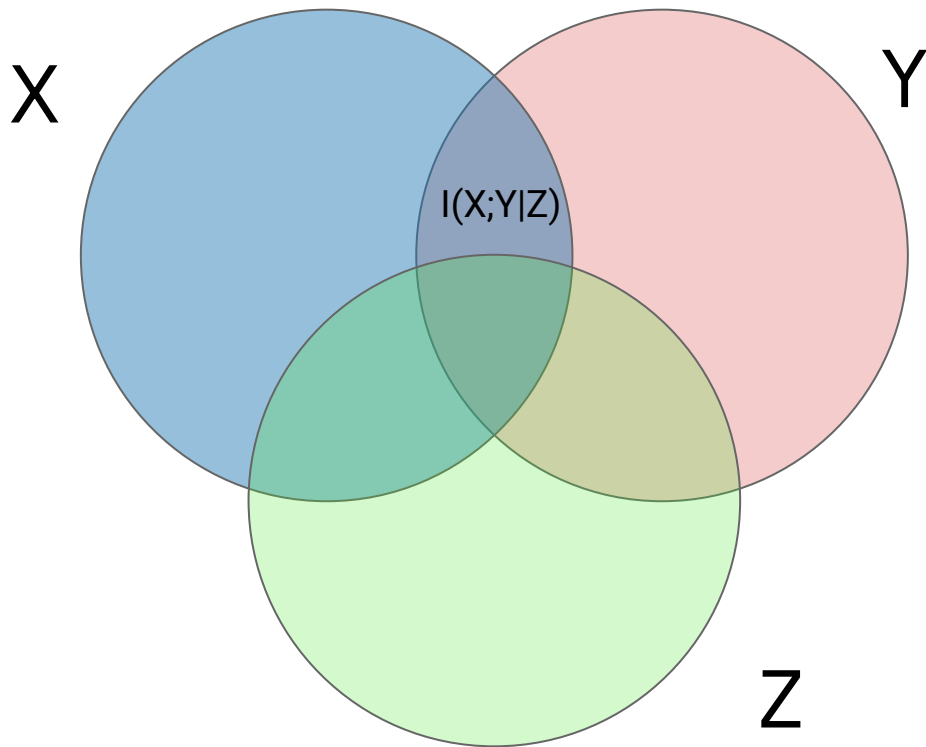
Information cannot be created, only destroyed.

Data Processing Inequality

$$X \rightarrow Z \rightarrow Y$$

$$\mathcal{I}[X; Y|Z] = 0$$

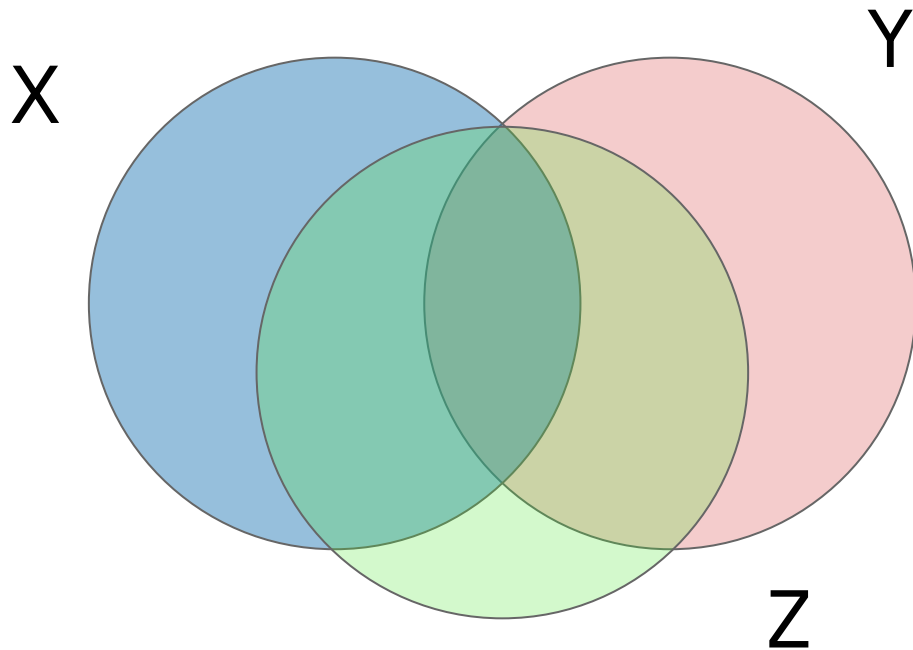
$$\mathcal{I}[X; Y] \leq \mathcal{I}[X; Z]$$



Data Processing Inequality

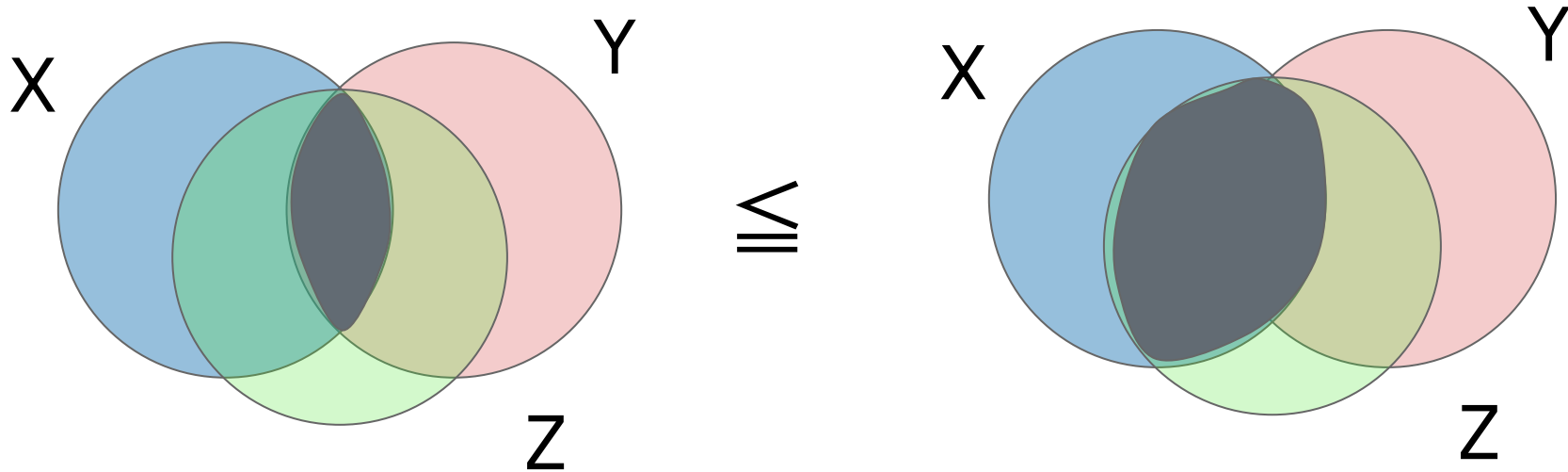
$$X \rightarrow Z \rightarrow Y$$

$$\mathcal{I}[X; Y] \leq \mathcal{I}[X; Z]$$



Data Processing Inequality

$$X \rightarrow Z \rightarrow Y \quad \mathcal{I}[X; Y] \leq \mathcal{I}[X; Z]$$



Data Processing Inequality

What happens if we "process" our distribution?

$$x \rightarrow \boxed{t(y|x)} \rightarrow y$$

$$p(y) = \int dx \, t(y|x) p(x) \quad q(y) = \int dx \, t(y|x) q(x)$$

$$\mathcal{S}[p(x); q(x)] \geq \mathcal{S}[p(y); q(y)]$$

Any processing makes it harder to distinguish between the two distributions.

Monotonicity

Marginalization is a kind of processing.

$$(x, y) \rightarrow \boxed{} \rightarrow y$$

$$\mathcal{S}[p(x, y); q(x, y)] \geq \mathcal{S}[p(y); q(y)]$$

If you only look at a subset of variables, its harder to distinguish between two distributions.

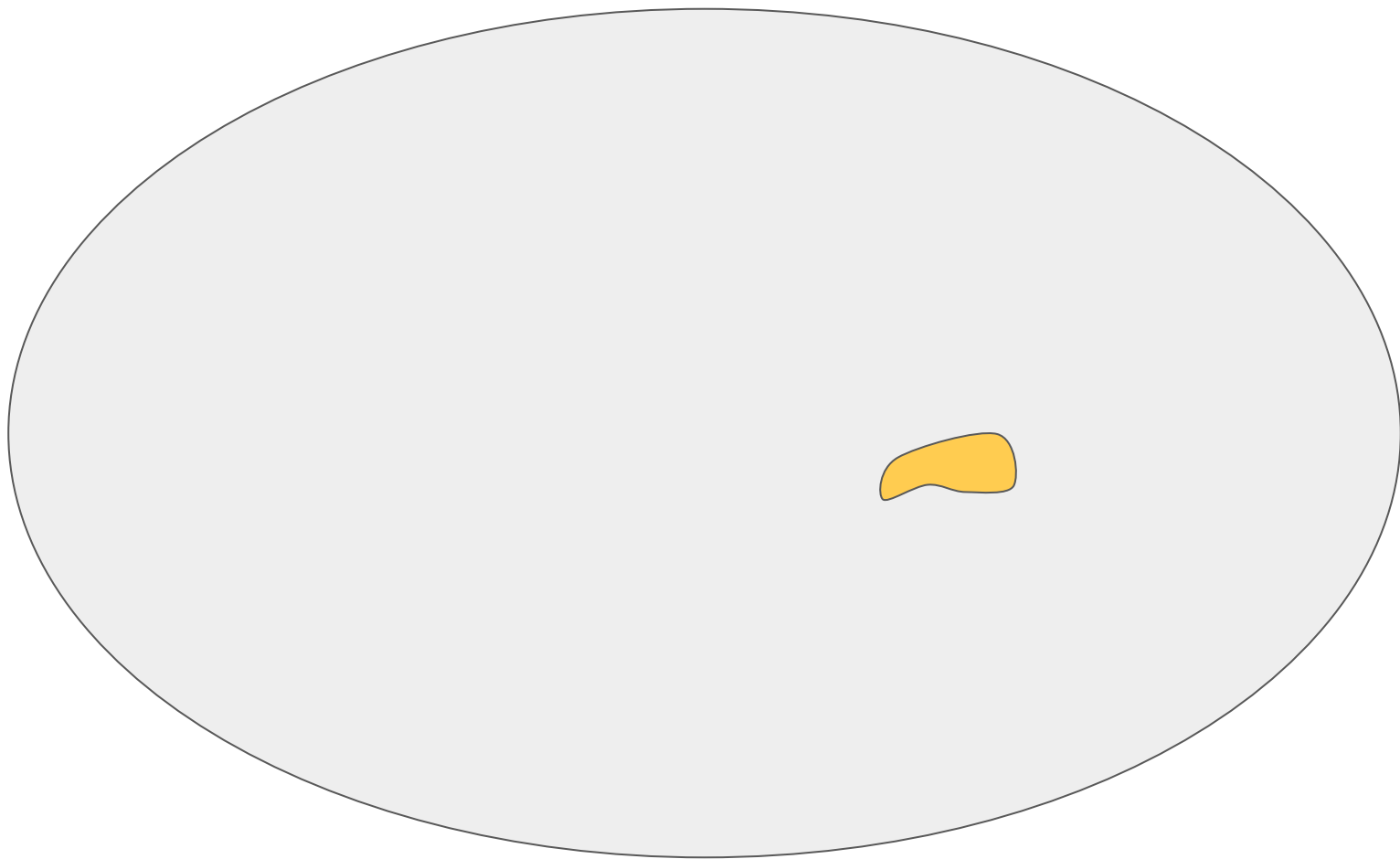
Signal and Noise

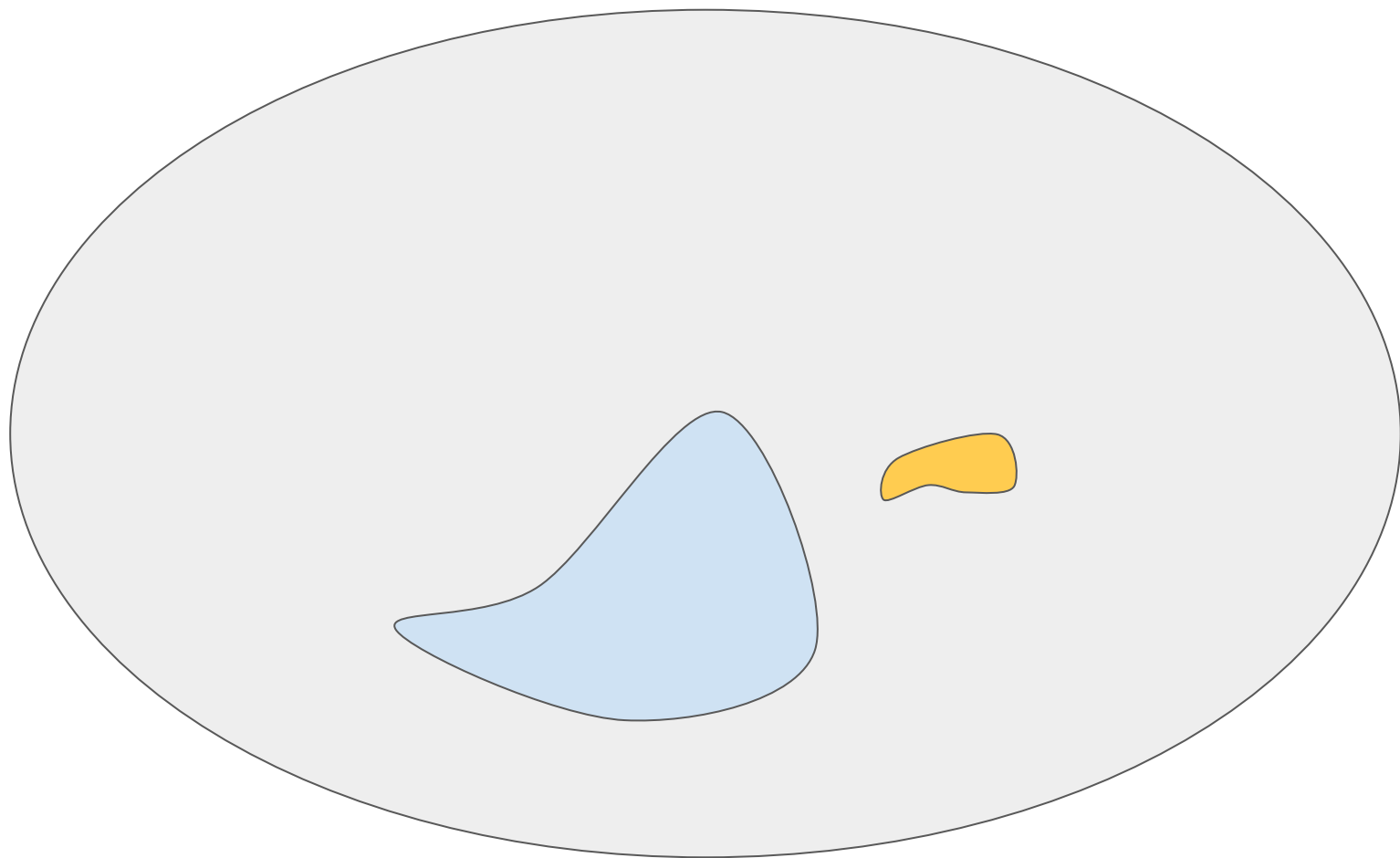
$$S = k \cdot \log W$$

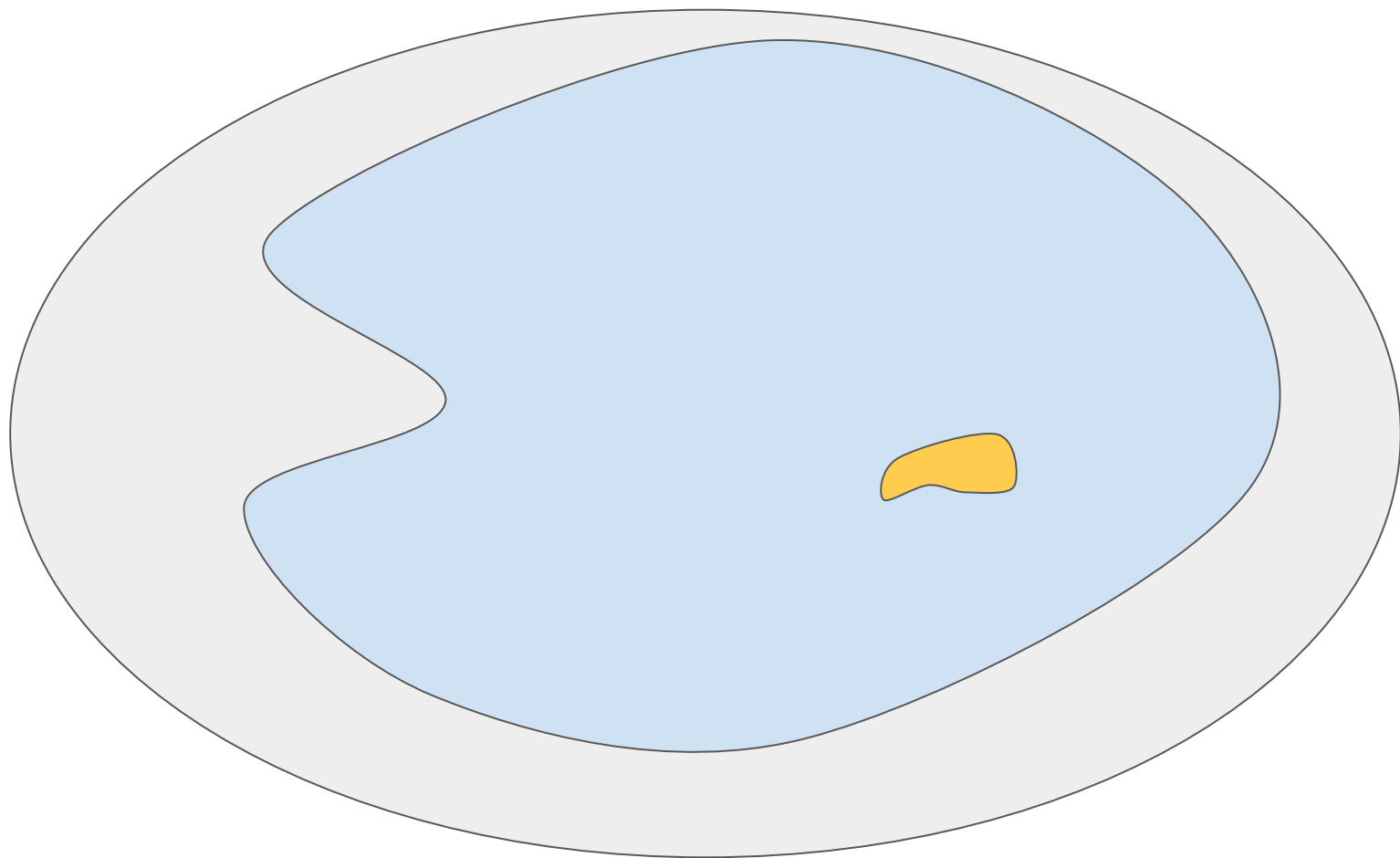
4585693498231234
7490532893947833
5905810199809343
9046308731891504
6945241060702756
5791755416289583
1722653391013743
3364338109261211
6752724308343748
3055138227637198
9681284143710092
5505916112347391
5251326864194069
7123281077722824
1716032982092282
6418603617365559

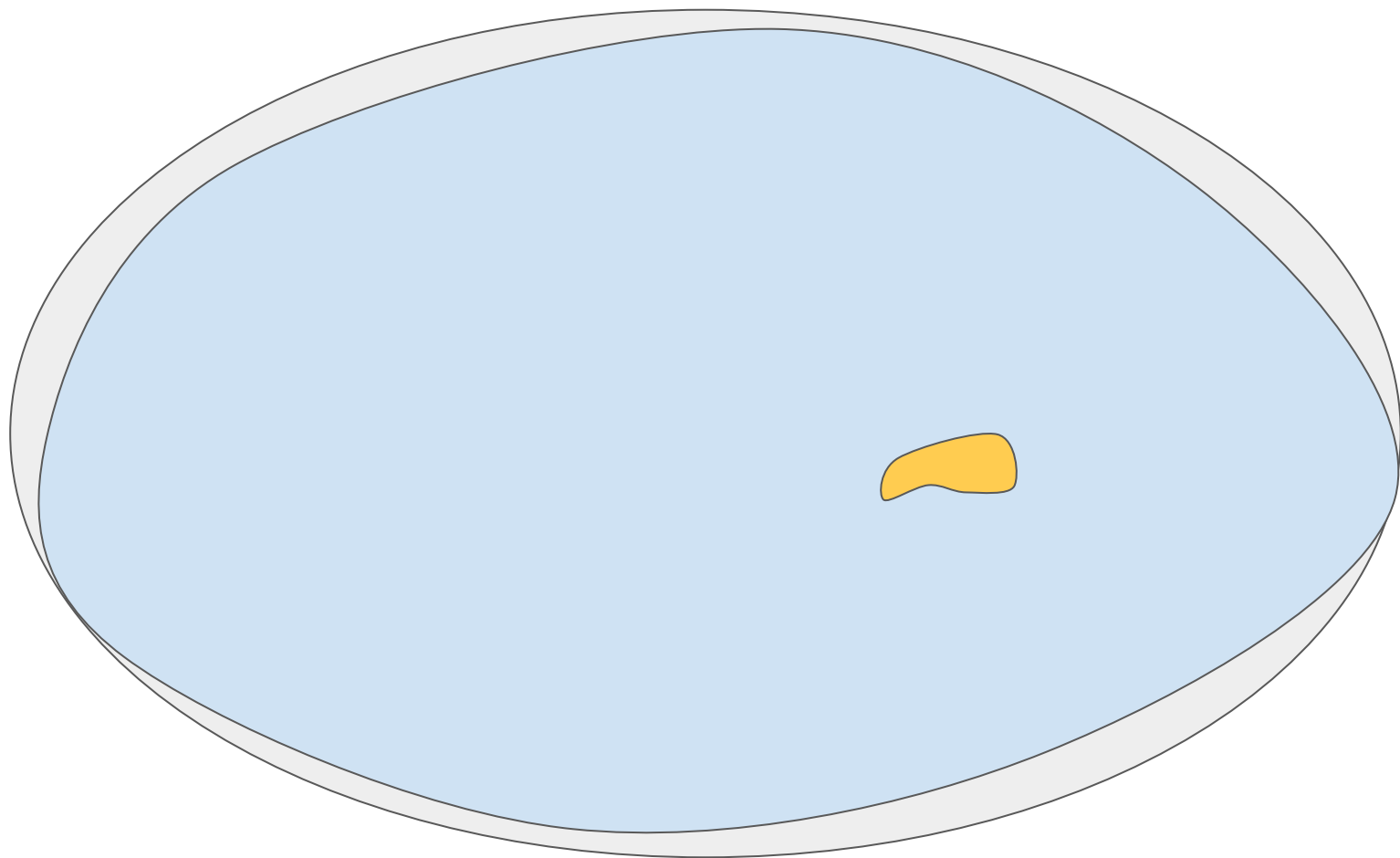
303 bits total

3.32 bits of label information









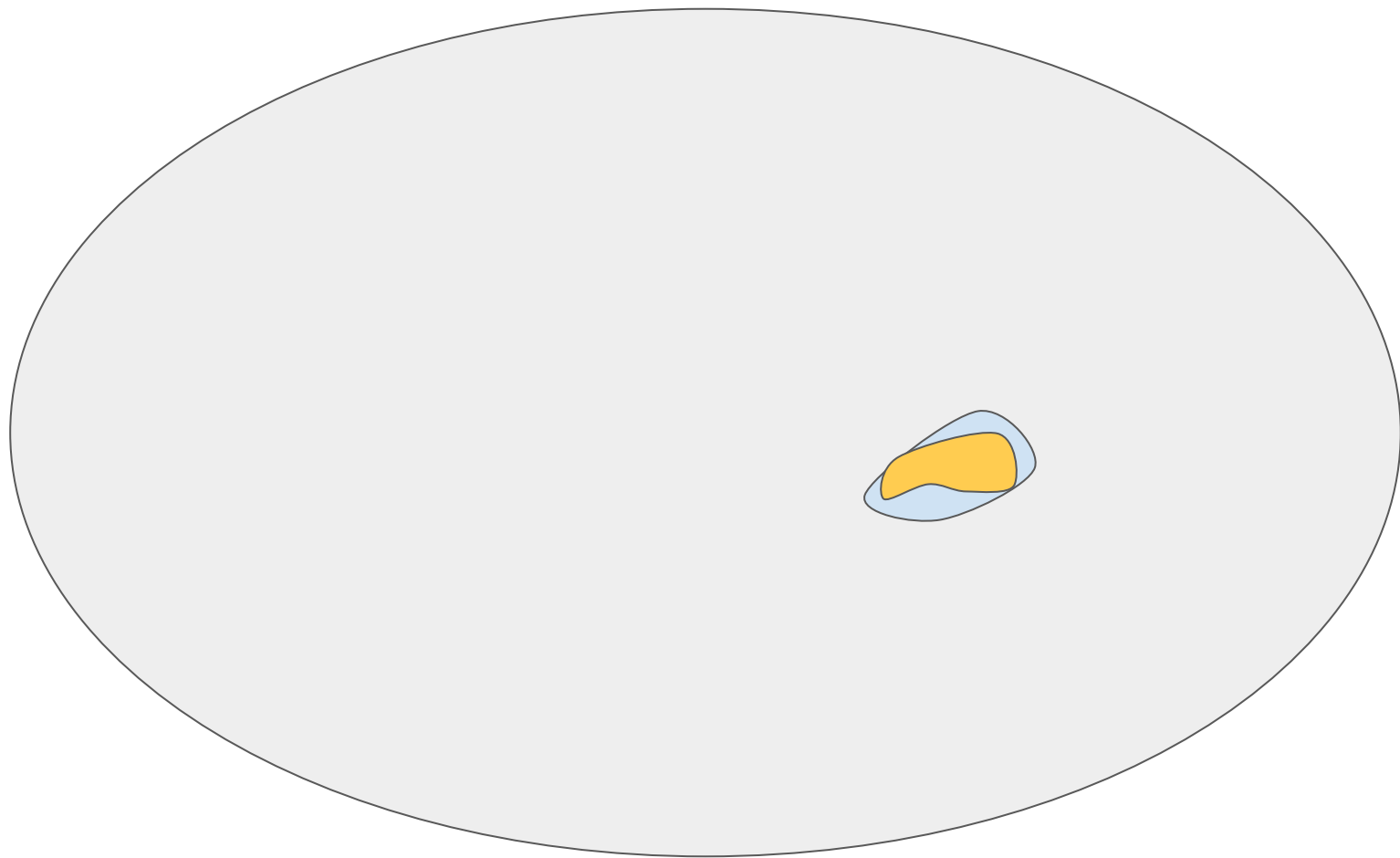
Noise is Cheap

$$E = k_B T \log 2$$

$$\frac{\text{energy to heat tea}}{k_B T \log 2} \sim 10^{25} \text{ bits}$$

1000 zetabytes,
approximately
1000x all human
generated data.

$$\frac{\text{solar flux on earth's surface}}{k_B T \log 2} \sim 10^{38} \text{ bits/sec}$$



Humans Do It

How Much Do People Remember?

Some Estimates of the Quantity of Learned Information in Long-term Memory

THOMAS K. LANDAUER

Bell Communications Research

How much information from experience does a normal adult remember? The "functional information content" of human memory was estimated in several ways. The methods depend on measured rates of input and loss from very long-term memory and on analyses of the informational demands of human memory-based performance. Estimates ranged around 10^9 bits. It is speculated that the flexible and creative retrieval of facts by humans is a function of a large ratio of "hardware" capacity to functional storage requirements.

TABLE 3
Estimates of information required for recognition memory for various kinds of materials
(in bits).

Pictures

Unselected pictures

- 10.0-14.6 (Standing, 1973; 20 to 10,000 tested after 2 days)
- 12.9 (Standing, 1973; 200 tested after 1 hr)
- 12.0 (Nickerson, 1968; 200 tested after 1 day)
- 13.1 (Shepard, 1967; 612 tested after 0.5 hr)

Vivid pictures

- 11.9-13.0 (Standing, 1973; 20 to 1000 tested after 2 days)

Words, visual

- 11.0 (Shepard, 1968; 540 tested after 0.5 hr)
- 6.5-11.0 (Standing, 1973; 20 to 1000 tested after 2 days)
- 11.5 (Standing, 1973; 200 tested after 1 hr)

Words, auditory

- 11.7 (Standing, 1973; 200 tested after 1 hr)

Sentences

- 11.3 (Shepard, 1968; 612 tested after 0.5 hr)

Nonsense syllables

- 10.2 (Standing, 1973; 200 tested after 1 hr)

Music segments

- 11.4 (Standing, 1973; 200 tested after 1 hr)
-

The Hunt

The Hunt

Find the best representation subject to constraints.

- Representation First
- Respect Stochasticity
- Embrace tradeoffs

Recipe for Creating Objectives

1. Describe the world.
2. Add what you plan to build.
3. Describe what you wish were the case.
4. Take the KL between the two.
5. Separate into pieces.
6. Remove redundant parts.
7. Add coefficients.

Relative Entropy as a Principle for Learning

A Principle For Learning

Make it hard to tell the difference between the real world and your model.

$$\min \mathcal{S}[\text{real}; \text{model}]$$

$$\min_q \int dx p(x) \log \frac{p(x)}{q(x)}$$

$$\min_q \int dx p(x) \log \frac{p(x)}{q(x)} = \int dx p(x) \log p(x) - \min_q \int dx p(x) \log q(x)$$

-Data Entropy

Cross Entropy

Maximum Likelihood

We don't have access to the true distribution of data, only samples.

$$\hat{p}(x) = \frac{1}{N} \sum_i \delta(x - x_i)$$

$$- \int dx \hat{p}(x) \log q(x) = - \int dx \frac{1}{N} \sum_i \delta(x - x_i) \log q(x) = - \frac{1}{N} \sum_i \log q(x_i)$$

Empirical Cross
Entropy

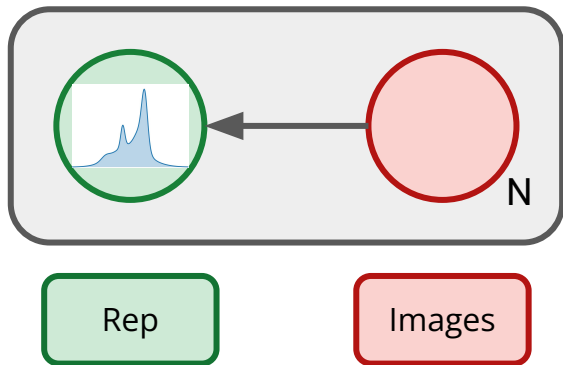
$$\min_q \mathcal{S}[\hat{p}; q]$$

Variational Autoencoders

World P

world we have

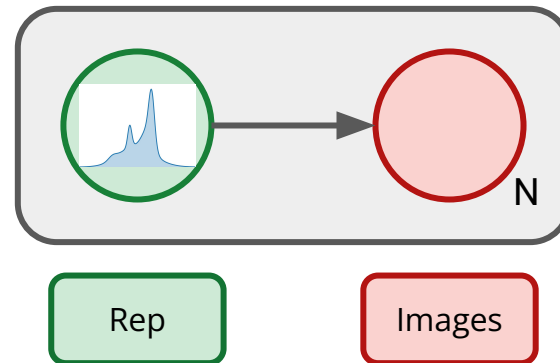
$$p(x, z) = p(z|x)p(x)$$



World Q

world we want

$$q(z, x) = q(z)q(x|z)$$



VAE

$$\begin{aligned}\text{KL}[p; q] &= \mathbb{E}_{p(x, z)} \left[\log \frac{p(z, x)}{q(z, x)} \right] \\ &= \left\langle \log \frac{p(x)p(z|x)}{q(x|z)q(z)} \right\rangle_p \\ &= \langle \log p(x) \rangle + \left\langle -\log q(x|z) + \log \frac{p(z|x)}{q(z)} \right\rangle \\ &= \mathbb{H}_p[x] + \left\langle -\log q(x|z) + \beta \log \frac{p(z|x)}{q(z)} \right\rangle\end{aligned}$$

Variational Autoencoder

$$\mathbb{E}_{p(x)p(z|x)} \left[-\log q(x|z) + \beta \log \frac{p(z|x)}{q(z)} \right]$$

Distortion

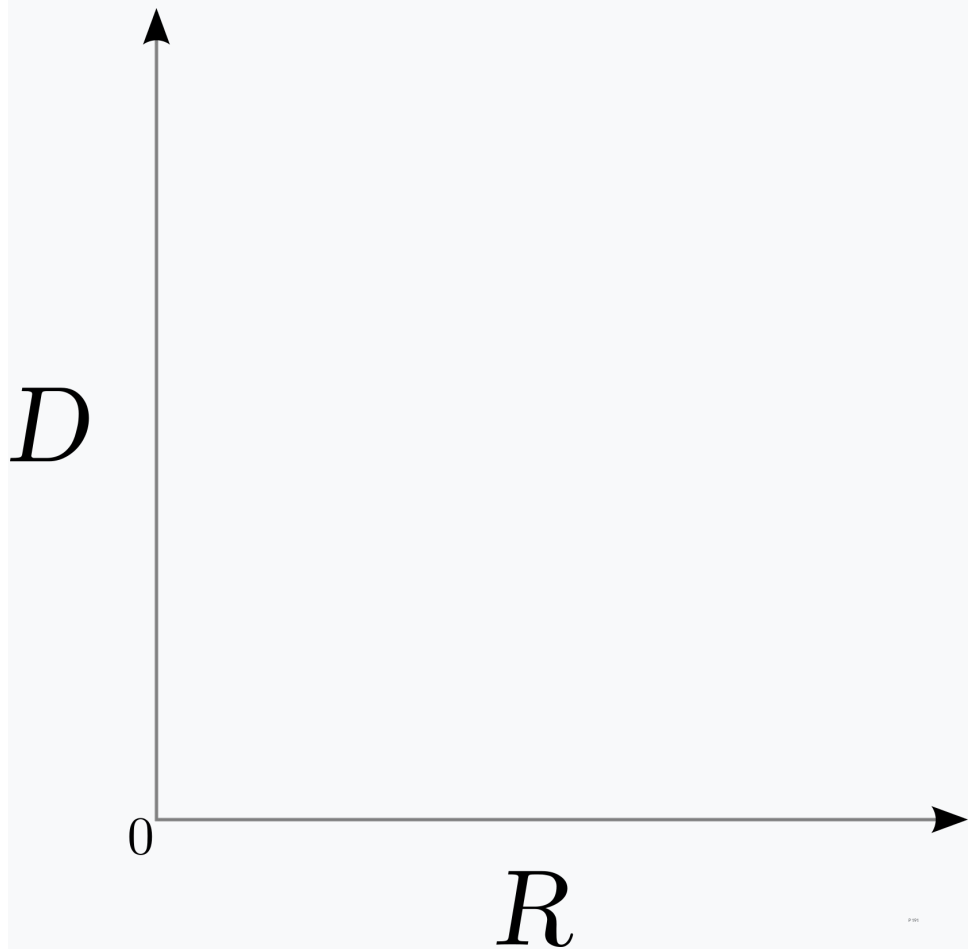
Rate

D

R

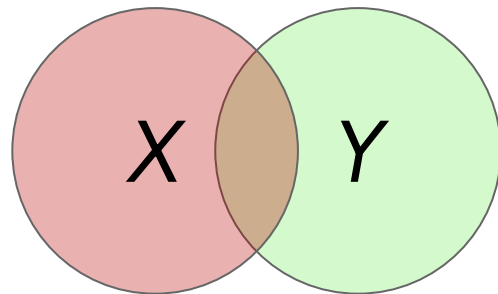
Phase Diagram

Rate and Distortion are both positive semidefinite.



Mutual Information

$$I(X; Z) = \mathbb{E}_{p(x,z)} \left[\log \frac{p(z, x)}{p(z)p(x)} \right]$$



Symmetric, positive, reparameterization independent measure of co-dependence.

$$I(X; Z) = \left\langle \log \frac{p(z|x)}{p(z)} \right\rangle = \left\langle \log \frac{p(x|z)}{p(x)} \right\rangle$$

$$I(X; Z) = H(X) - H(X|Z) = H(Z) - H(Z|X) = H(X, Z) - H(X|Z) - H(Z|X)$$

Bounds

$$I(X; Z) = \left\langle \log \frac{p(z|x)}{p(z)} \right\rangle = \left\langle \log \frac{p(z|x)}{q(z)} \frac{q(z)}{p(z)} \right\rangle = \left\langle \log \frac{p(z|x)}{q(z)} \right\rangle - \text{KL}[p(z); q(z)]$$

Replacing the thing downstairs gives an upper bound

$$I(X; Z) = \left\langle \log \frac{p(z|x)}{p(z)} \right\rangle = \left\langle \log \frac{q(x|z)}{p(x)} \frac{p(x|z)}{q(x|z)} \right\rangle = \langle \log q(x|z) \rangle - \mathbb{H}[p(x)] + \text{KL}[p(x|z); q(x|z)]$$

Replacing the thing upstairs gives a lower bound

$$H - D \leq I \leq R$$

I Mutual Information

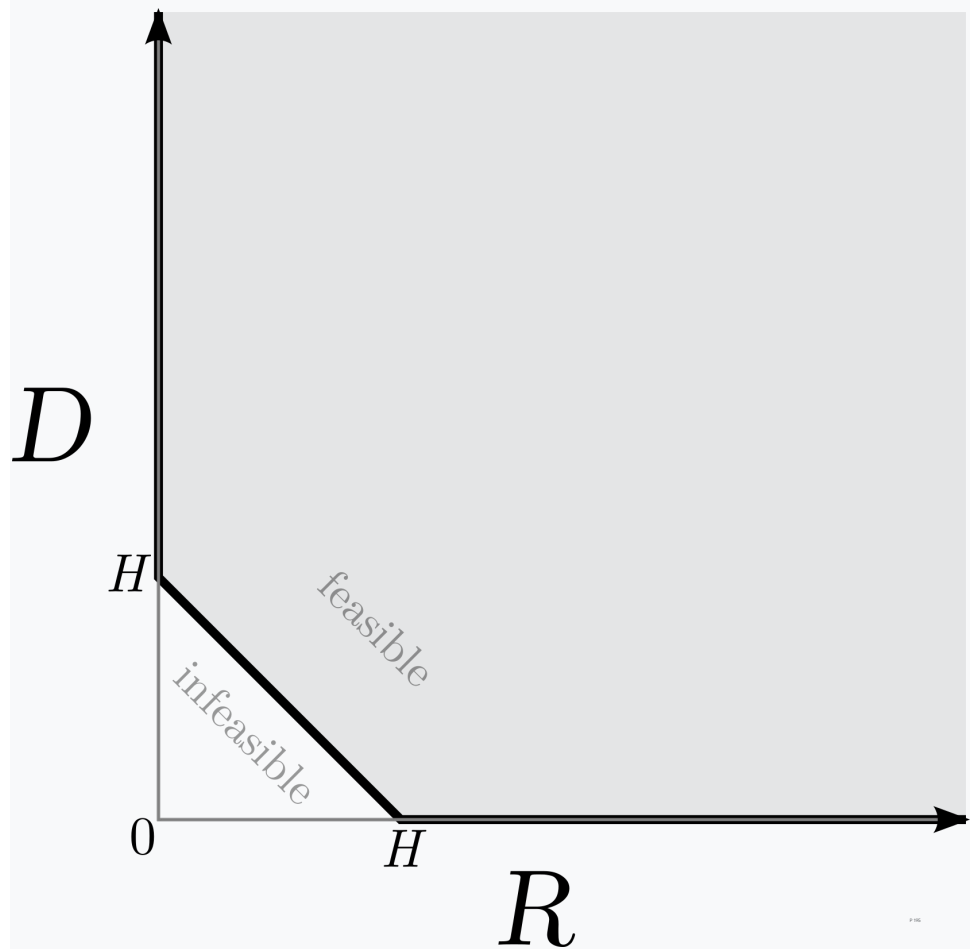
H Data Entropy

D Distortion

R Rate

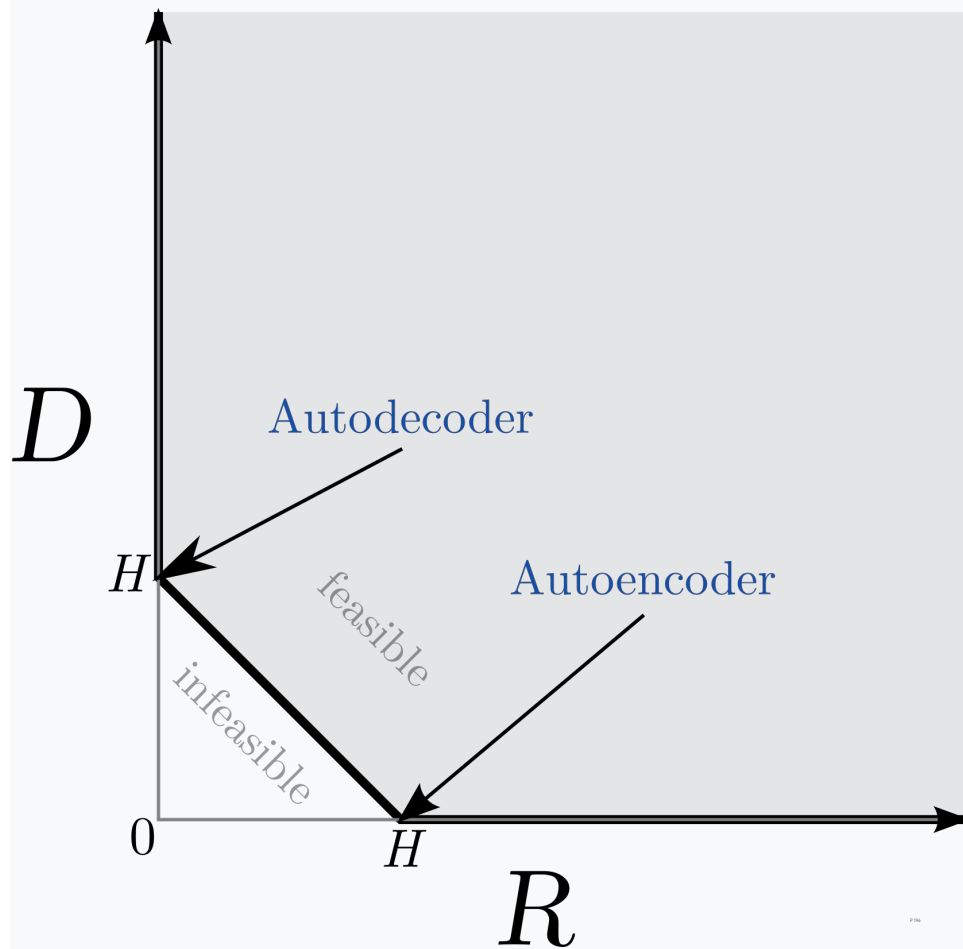
Feasible Region

$$H - D \leq I \leq R$$



Extremes

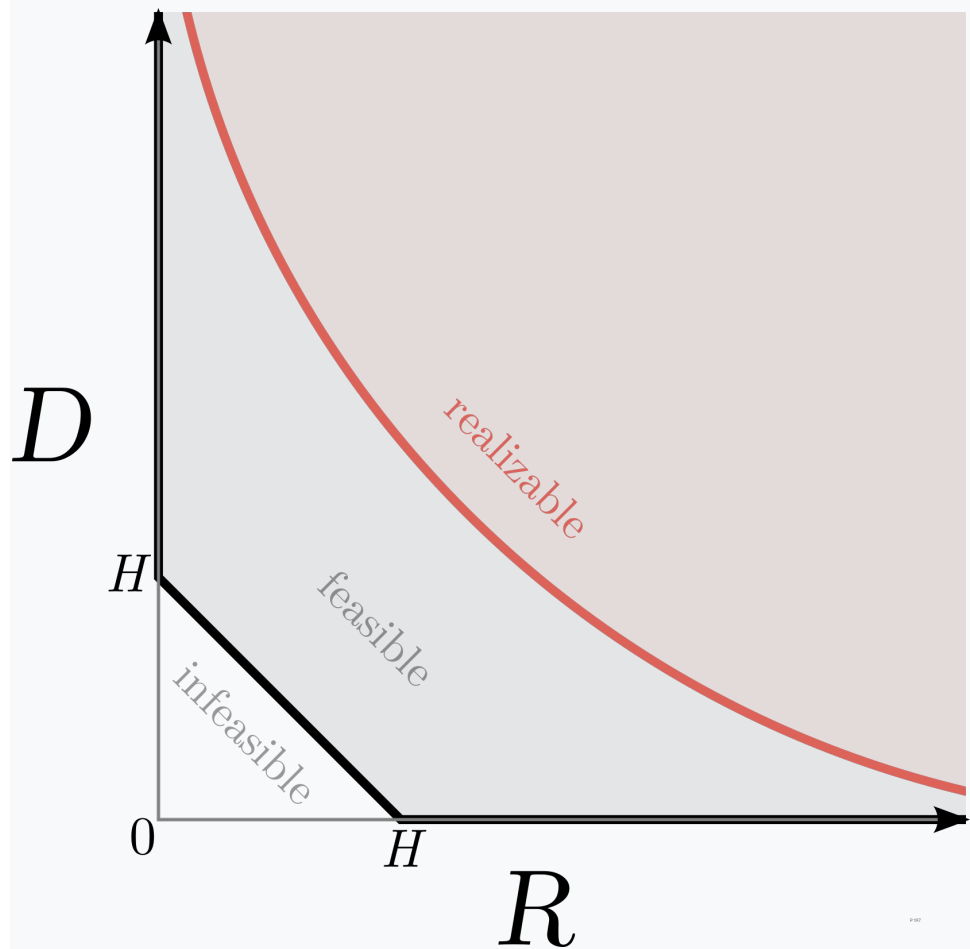
- Ideal autoencoders do lossless compression.
- Ideal "autodecoders" are perfect latent-less generative models.
- All points on diagonal line have equivalent ELBO.



Finite Family

Realizable frontier is a relaxation of the infinite model family limit.

$$\min D \text{ s.t. } R = R_0$$



β -VAE

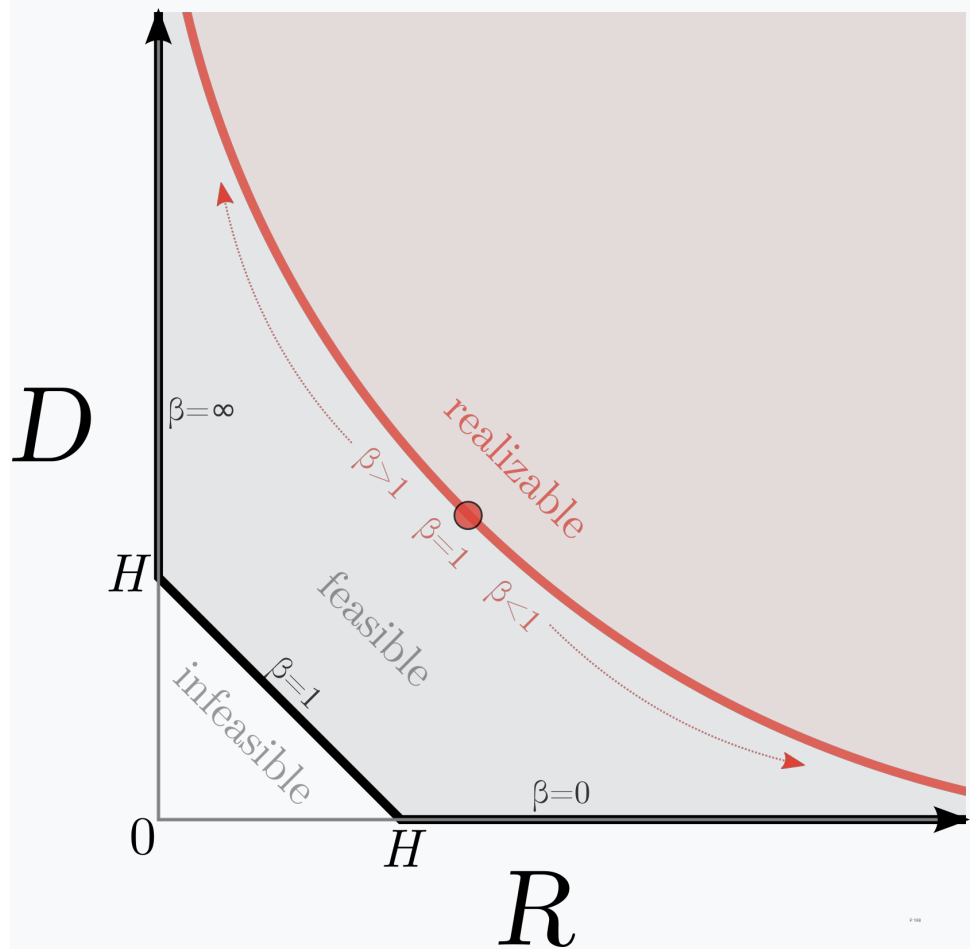
(beta-VAE... - Higgins et al. 2017)

Trace out curve.

$$\min D + \beta R$$

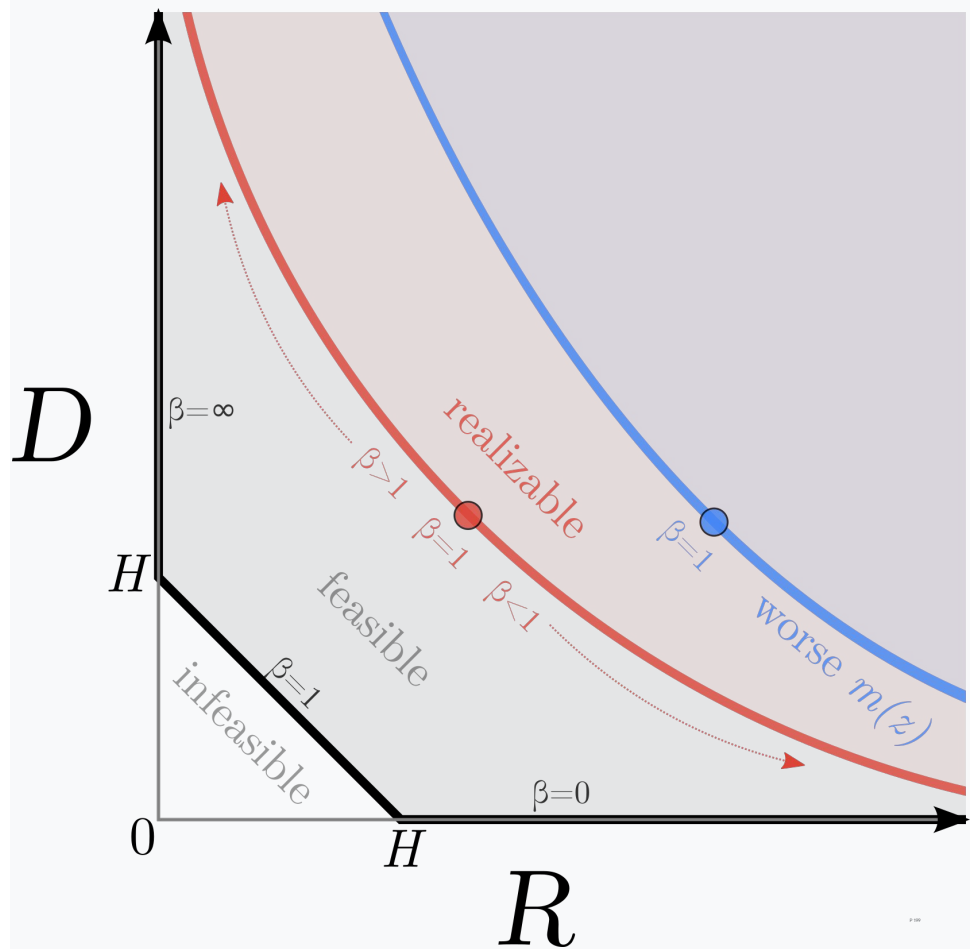
$$-\text{ELBO} = D + R$$

$$\beta = \frac{\partial D}{\partial R}$$



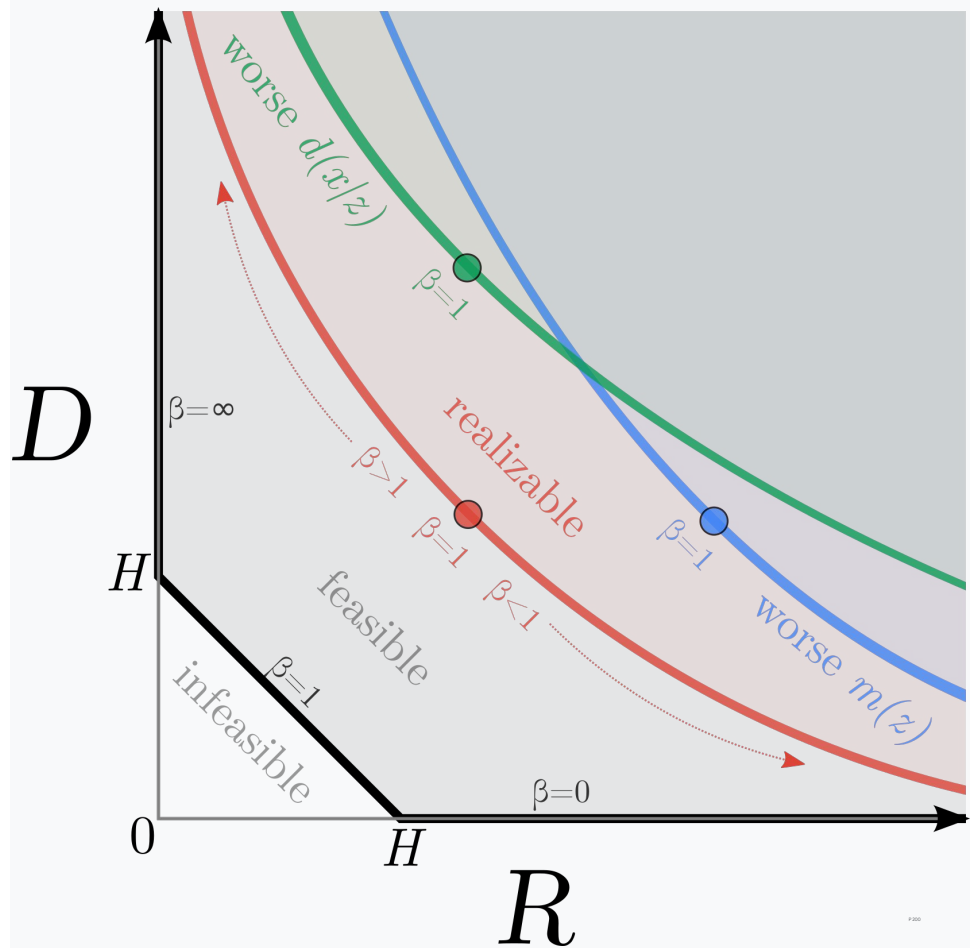
Architecture

The optimal ELBO point is a strong function of relative architectural power.



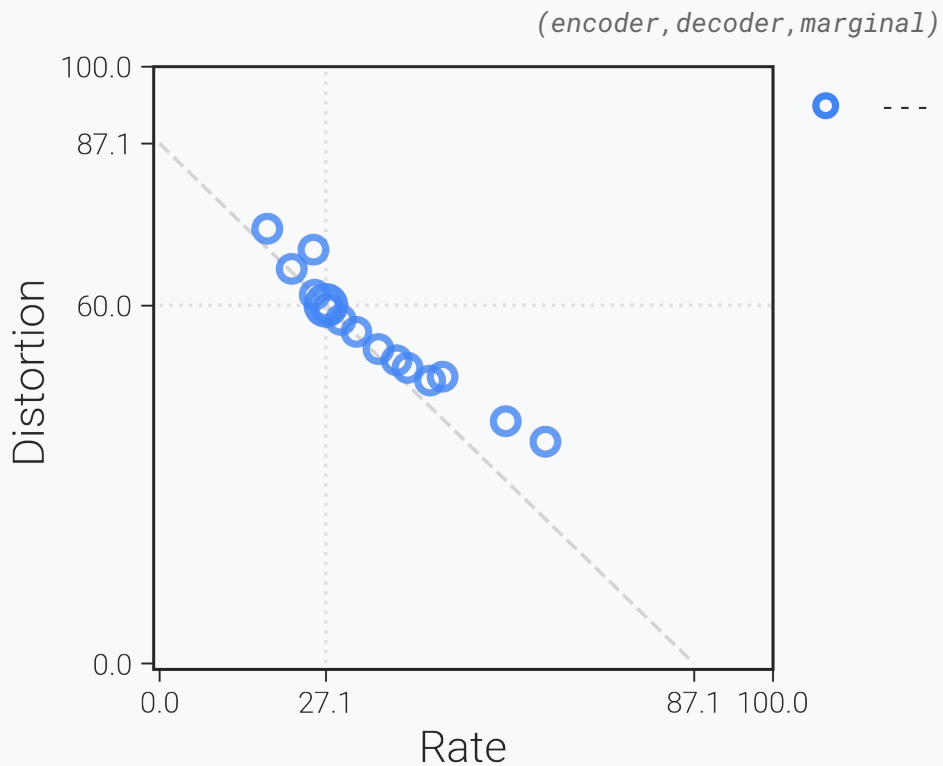
Phase Diagram

Instead of changing architecture,
explore frontier.



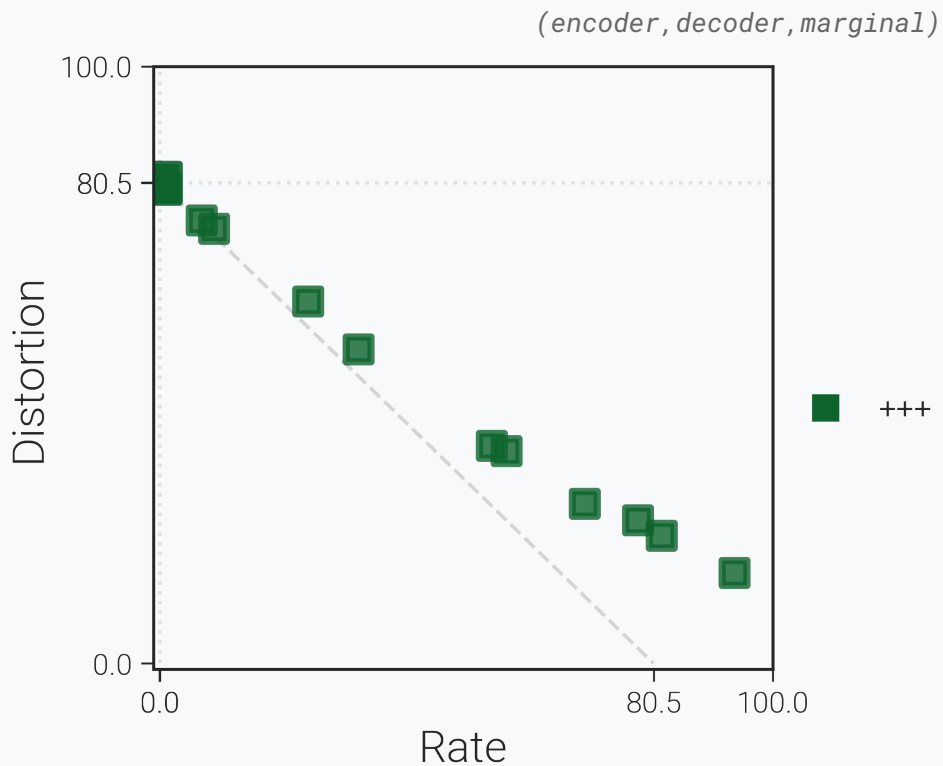
Baseline Sweep

Tracing out β traces out a curve.



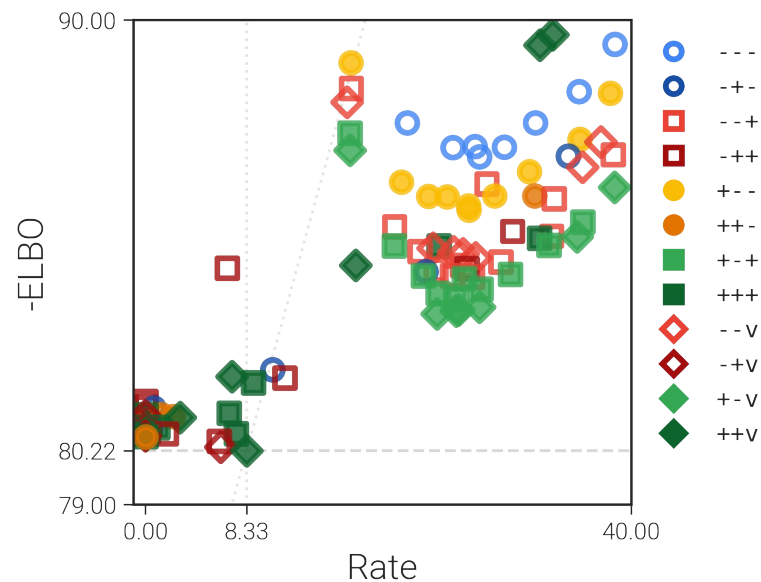
+++ Sweep

Tracing out β traces out a curve.

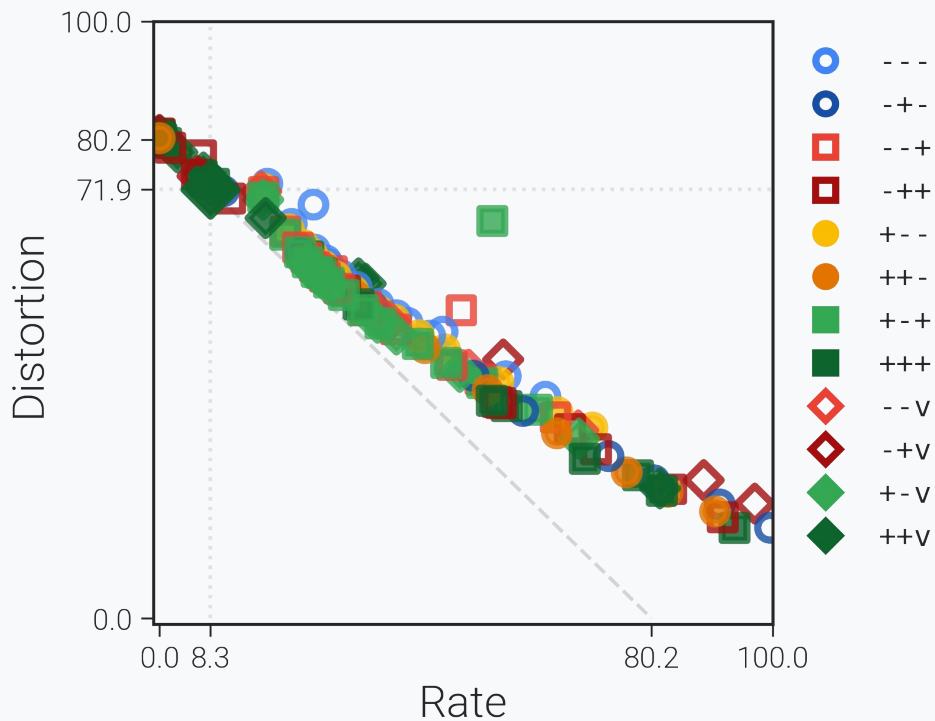


All Runs

Sweeping all model families.



(encoder, decoder, marginal)



Autodecoder

Model	-+v
Rate	0.0
Distortion	80.6
ELBO	-80.6

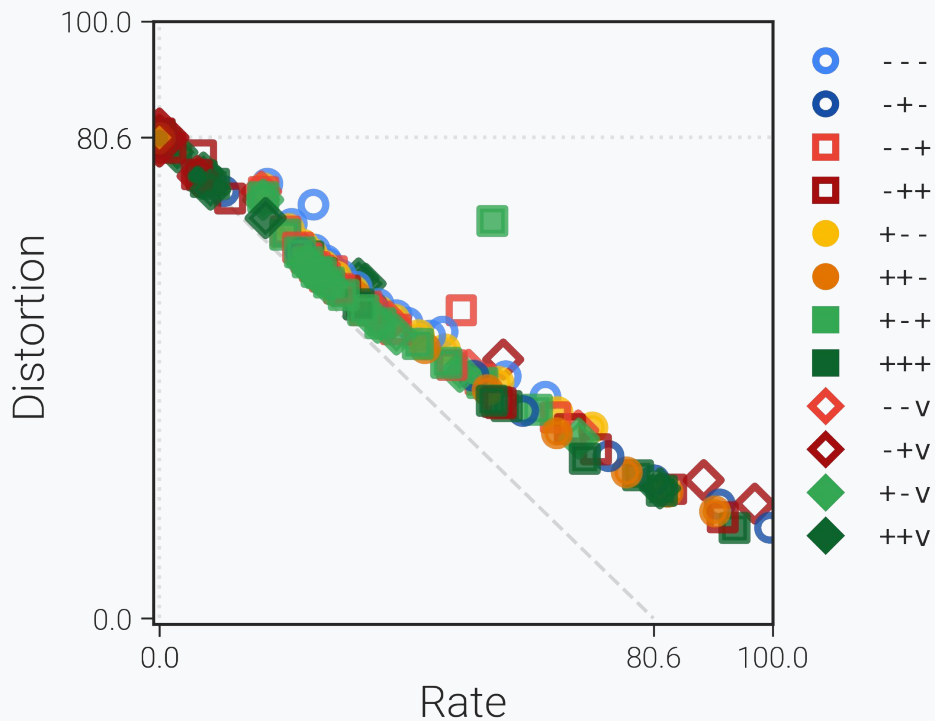
Generations



Reconstructions



(encoder, decoder, marginal)



"Autoencoder"

Model	+ - v
Rate	25.8
Distortion	57.7
ELBO	-83.4

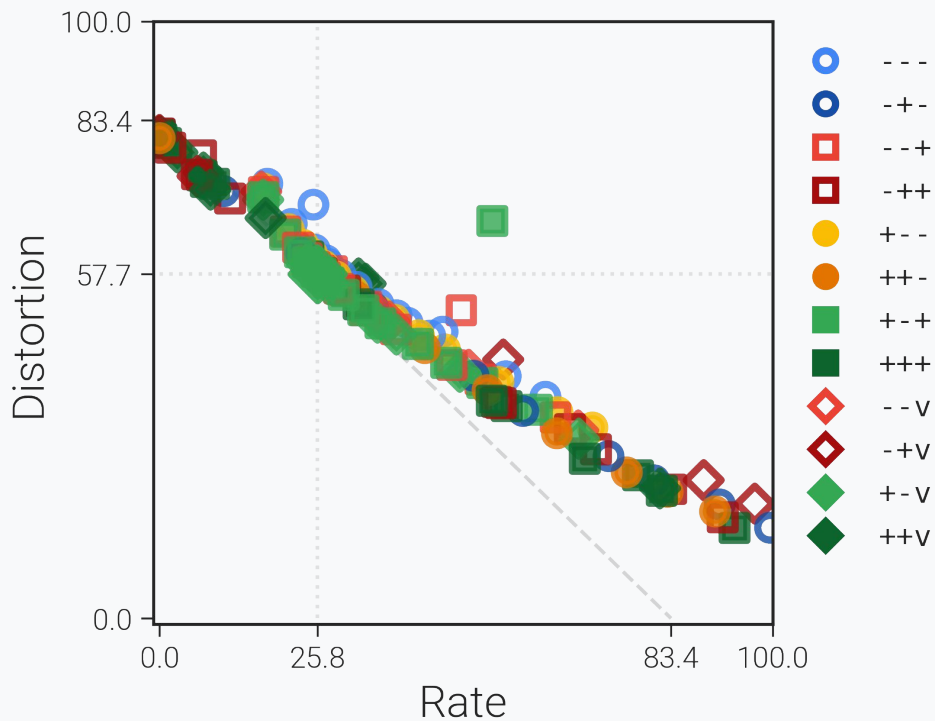
Generations



Reconstructions



(encoder, decoder, marginal)



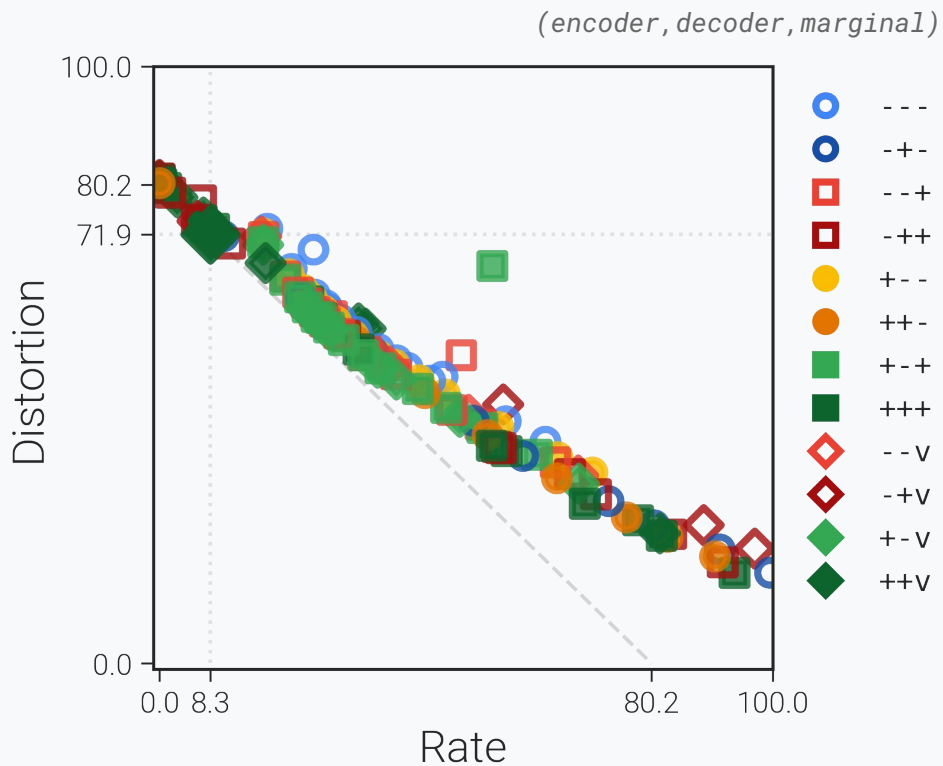
Semantic Encoder

Model	++v
Rate	8.3
Distortion	71.9
ELBO	-80.2

Generations



Reconstructions

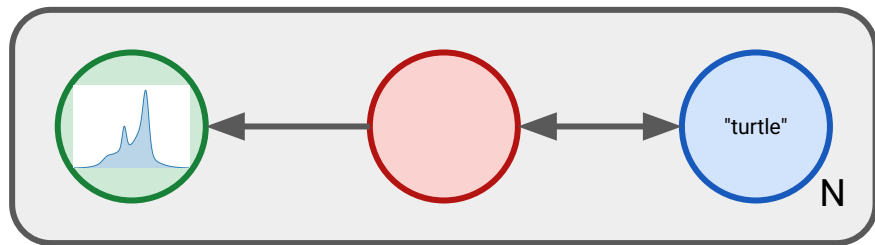


Variational Information Bottleneck

World P

world we have

$$p(x, y, z) = p(x, y)p(z|x)$$



Rep

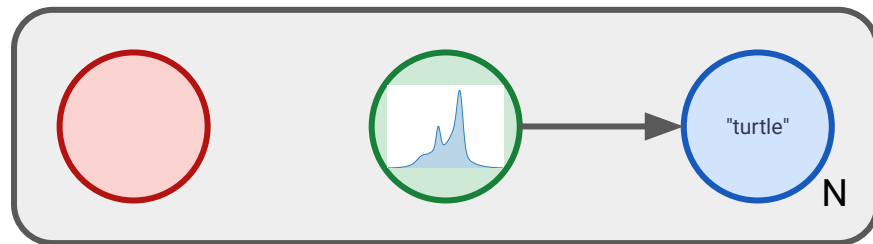
Images

Labels

World Q

world we want

$$q(x, y, z) = q(x)q(z)q(y|z)$$



Images

Rep

Labels

Variational Information Bottleneck

$$\begin{aligned}\text{KL}[p; q] &= \mathbb{E}_{p(x,y)p(z|x)} \left[\log \frac{p(x,y)p(z|x)}{q(x)q(z)q(y|z)} \right] \\ &= -\mathbb{H}[p(x,y)] - \langle \log q(x) \rangle + \left\langle \log q(y|z) + \beta \log \frac{p(z|x)}{q(z)} \right\rangle\end{aligned}$$

$$\mathbb{E}_{p(x)p(z|x)} \left[-\log q(y|z) + \beta \log \frac{p(z|x)}{q(z)} \right]$$

Classification
Error

Rate

Supervised Learning / Standard vs VIB

$$I(\textcolor{blue}{Z}; \textcolor{green}{Y}) - \beta I(\textcolor{blue}{Z}; \textcolor{red}{X})$$

Label

Representation

Image

Relevant sparse codes with variational information bottleneck - Chalk et al. / 1605.07332

Information Dropout: Learning Optimal Representations Through Noisy Computation - Achille et al. / 1611.01353

Nonlinear Information Bottleneck - Kolchinsky et al. / 1705.02436

Deep Variational Information Bottleneck - Alemi et al. 1706.02262

Extremes

Memorization requires $\log(N)$ nats.

Learning a perfect classifier requires $\log(C)$ nats.

This gap creates opportunity.

$$\log 60000 = 11$$

$$\log 10 = 2.3$$

Architecture

Encoder is a fully covariant Normal Distribution.

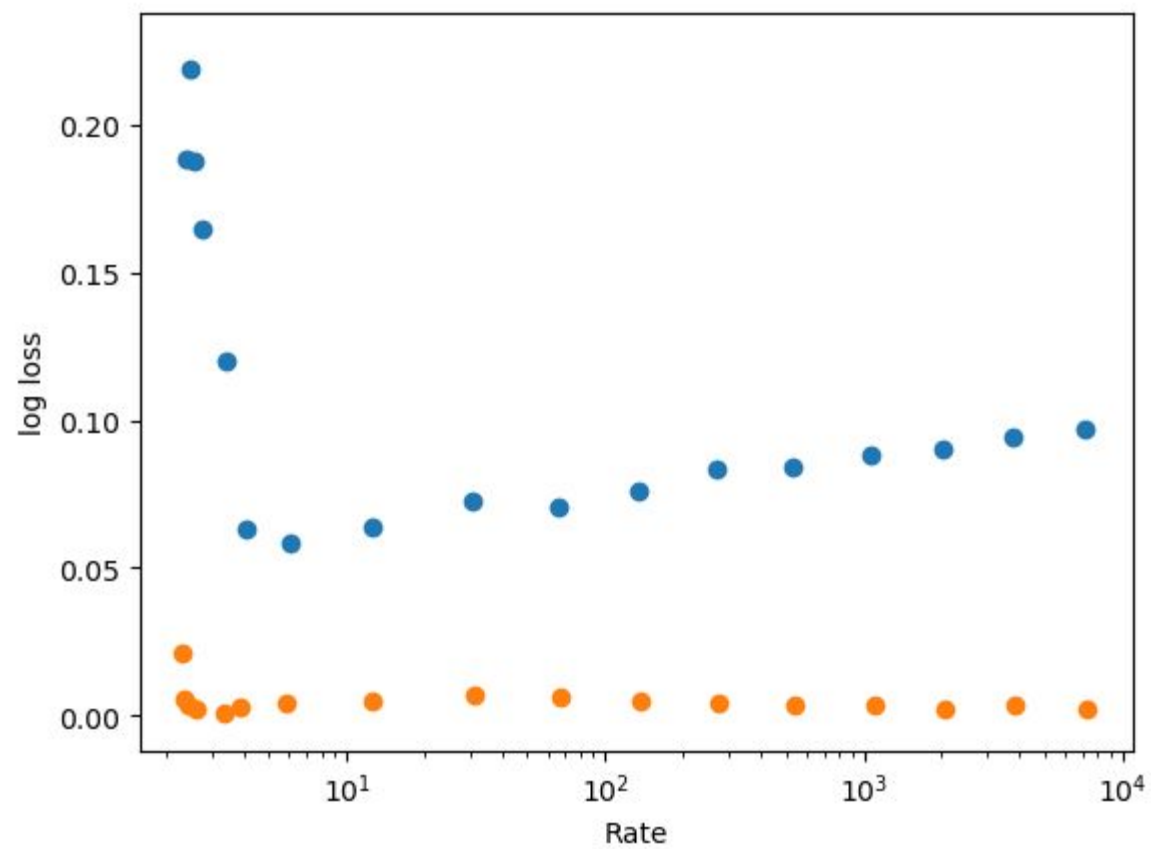
Use a clever energy model for both the classifier and the marginal distribution.

$$q(y, z) = \frac{1}{Z} e^{-\alpha \|z - c_y\|^2}$$

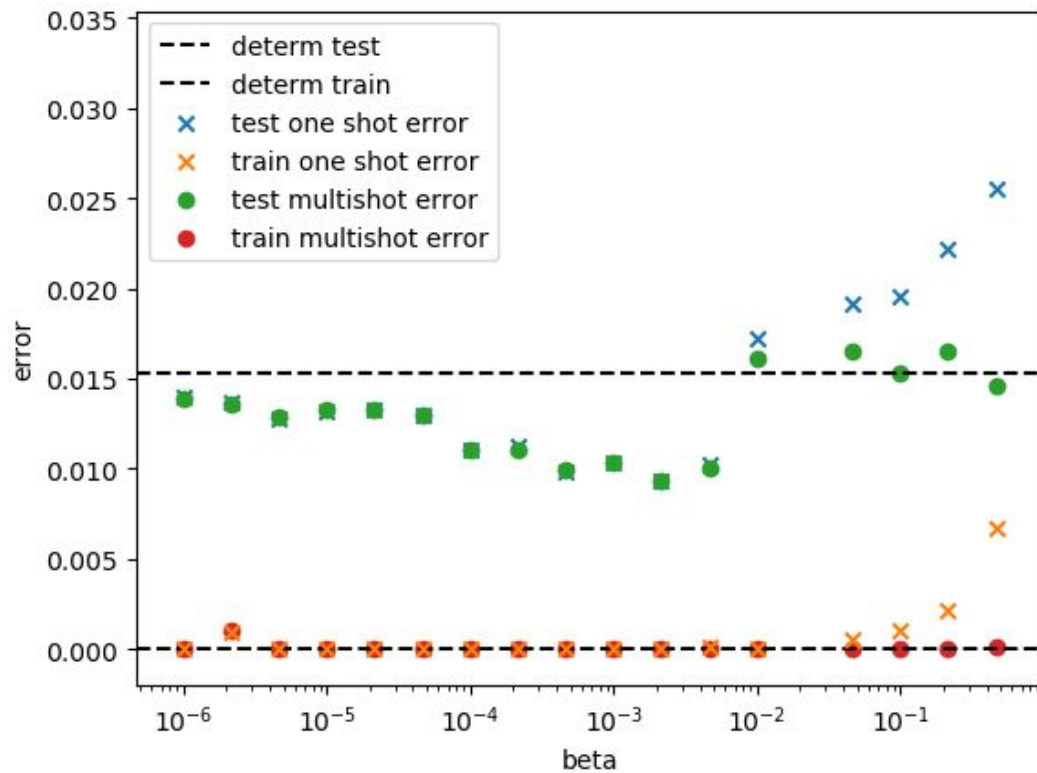
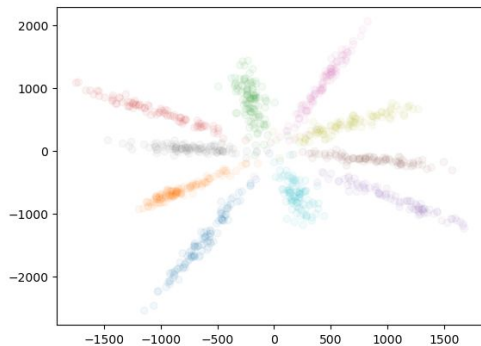
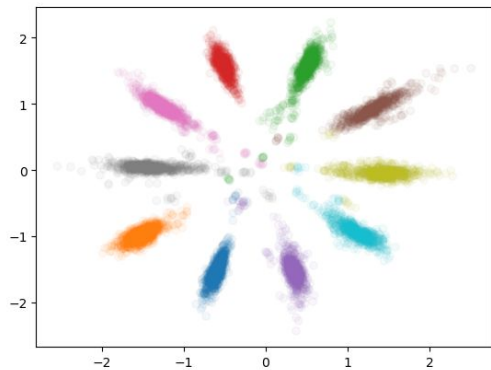
$$q(z) = \sum_y \frac{1}{Z} e^{-\alpha \|z - c_y\|^2}$$

$$q(y|z) = \frac{q(y, z)}{q(z)} = \frac{e^{-\alpha \|z - c_y\|^2}}{\sum_{y'} e^{-\alpha \|z - c_{y'}\|^2}}$$

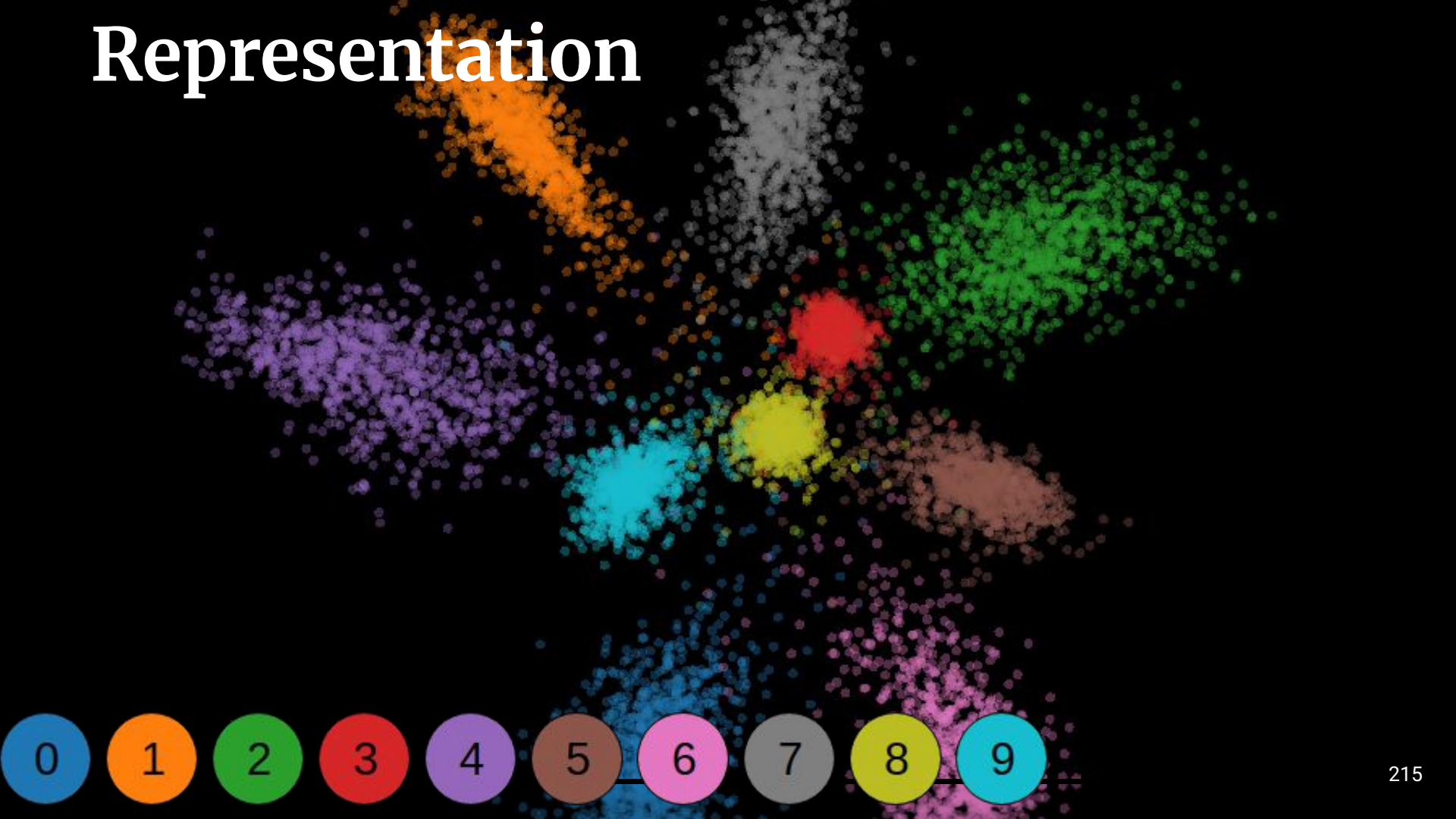
$$Z = \sum_y \int dz e^{-\alpha \|z - c_y\|^2} = \frac{K}{2} \left(\frac{\pi}{\alpha} \right)^{D/2}$$



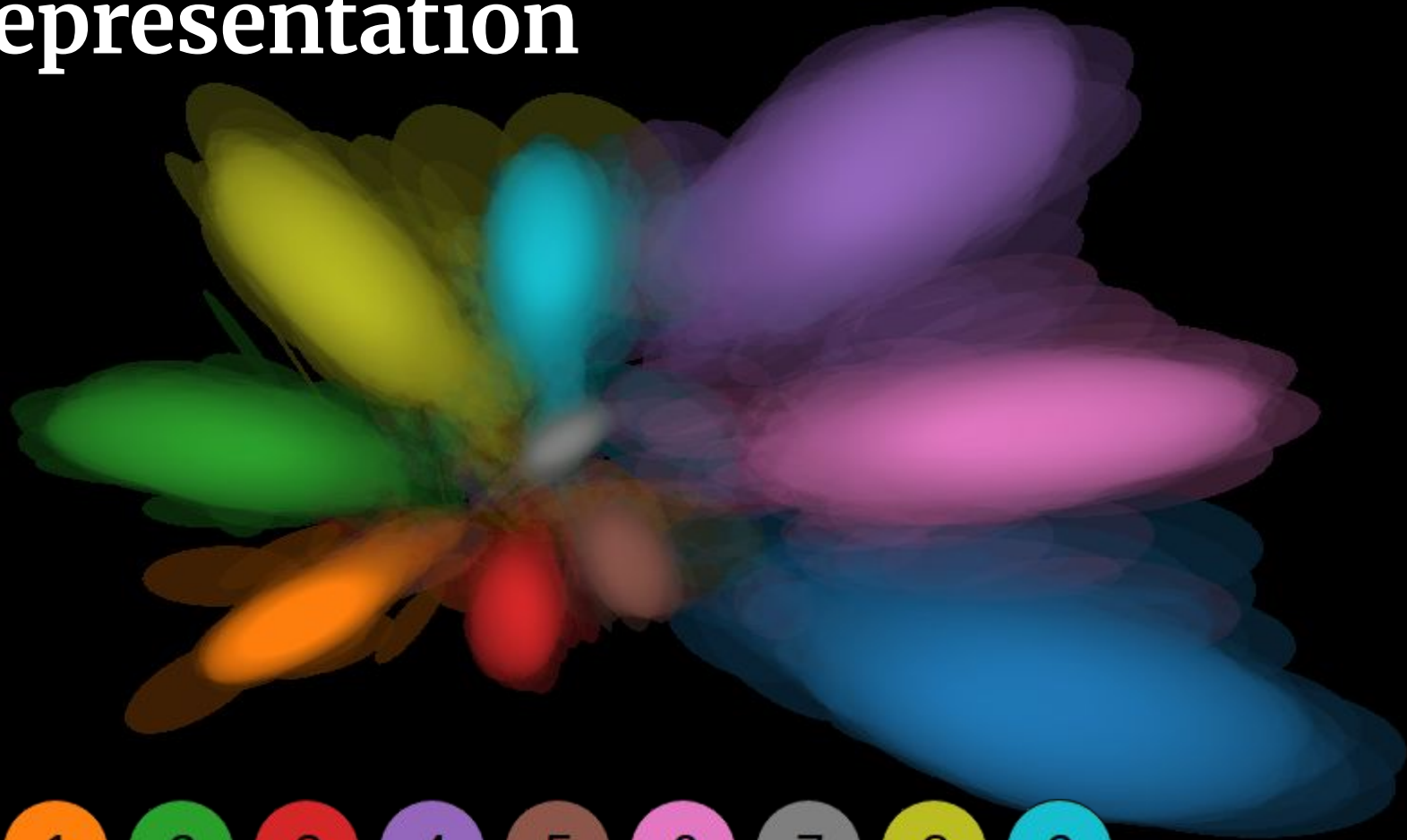
Performance as a function of beta

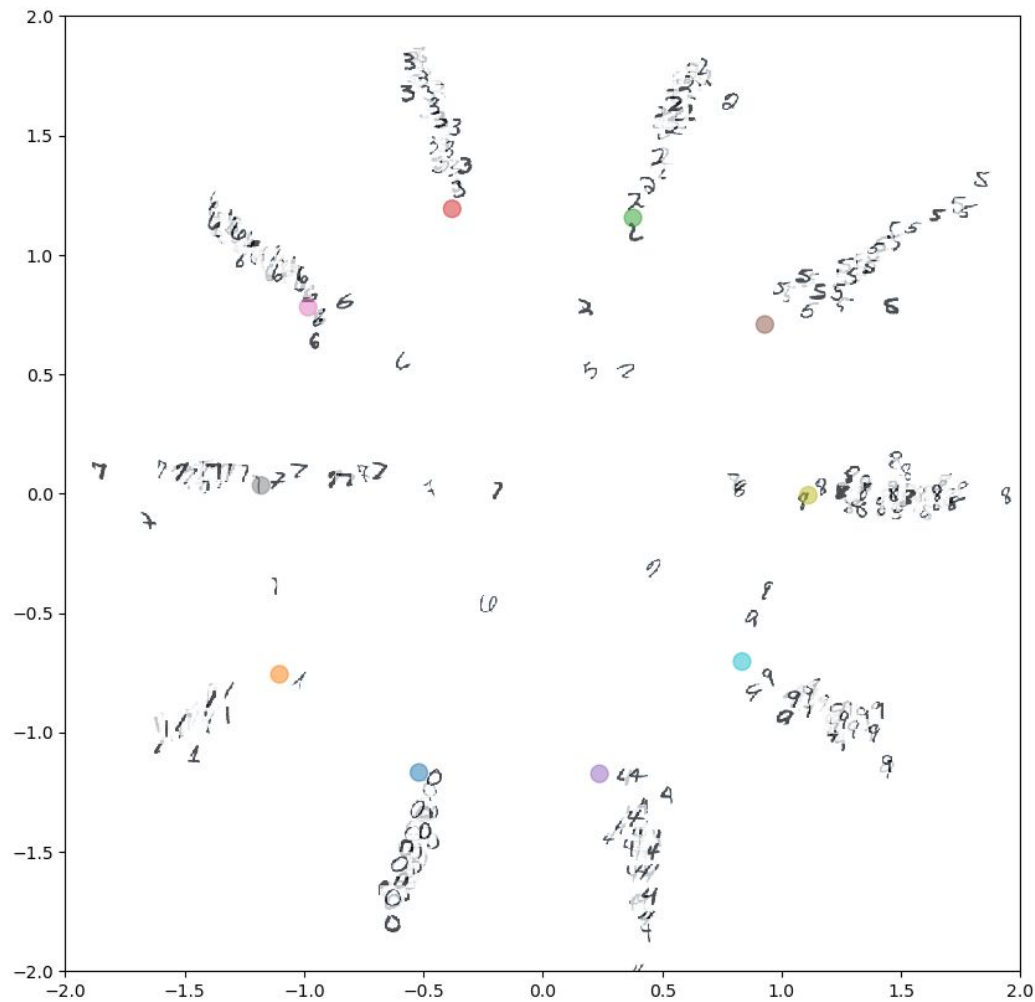
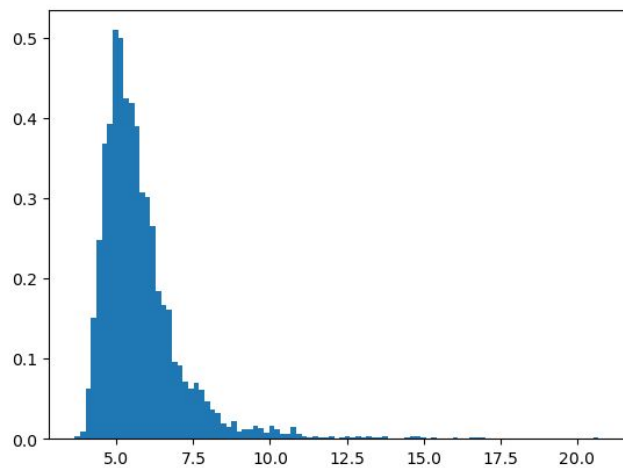


Representation

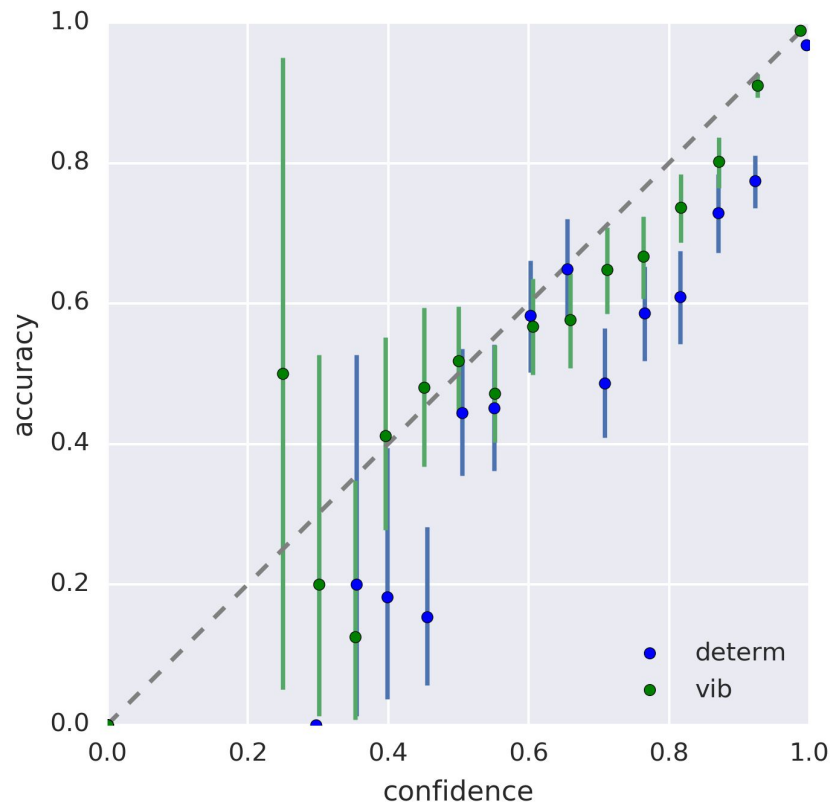


Representation





Calibration

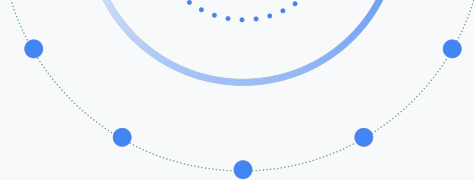


test likelihood

determ = 0.1921 nats

VIB = 0.0560 nats

~20 samples are
enough to strongly
select VIB over the
deterministic



Two Forms of Uncertainty

$$\max_{\theta, \phi, \psi} \mathbb{E}_{p(x, y) e_{\theta}(z|x)} \left[\underbrace{\log q_{\psi}(y|z)}_{\text{Confidence}} - \beta \log \underbrace{\frac{e_{\theta}(z|x)}{m_{\phi}(z)}}_{\text{Rate}} \right]$$

Fashion MNIST



T-shirt/top	Trouser	Pullover	Dress	Coat	Sandal	Shirt	Sneaker	Bag	Ankle boot
-------------	---------	----------	-------	------	--------	-------	---------	-----	------------

High Confidence Mislabellings



T-Shirt or Dress?



Boot or Sneaker?



Low Confidence



High Confidence



Low Confidence

High Confidence

Low Confidence, High Rate



Low Confidence

High Confidence



High Confidence, High Rate

Low Confidence

High Confidence



Low Confidence, Low Rate

Low Confidence

High Confidence



High Confidence, Low Rate

FIN

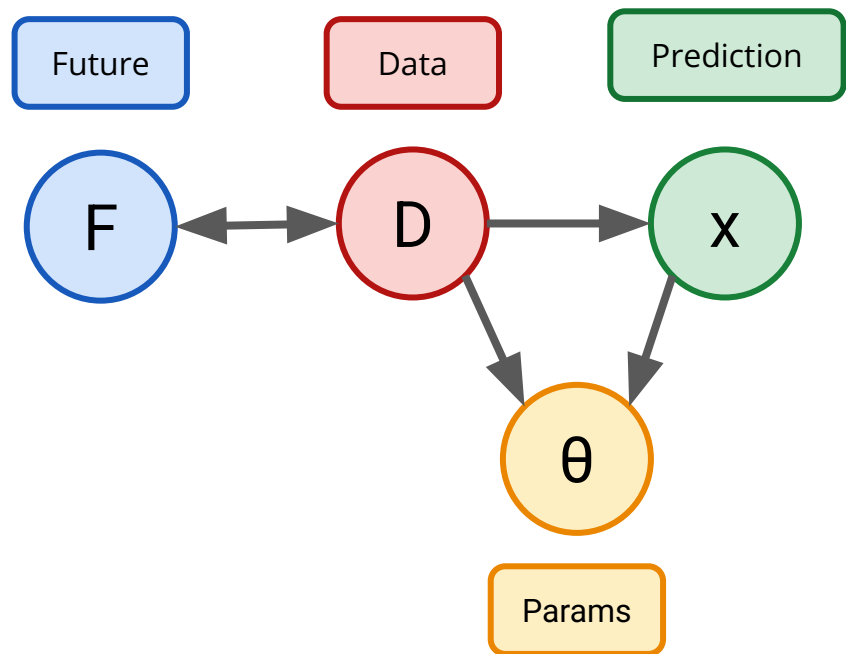
(THE END)

*Please check out the Problem Set [here](#)
Solutions [here](#)*

Variational Prediction

World P

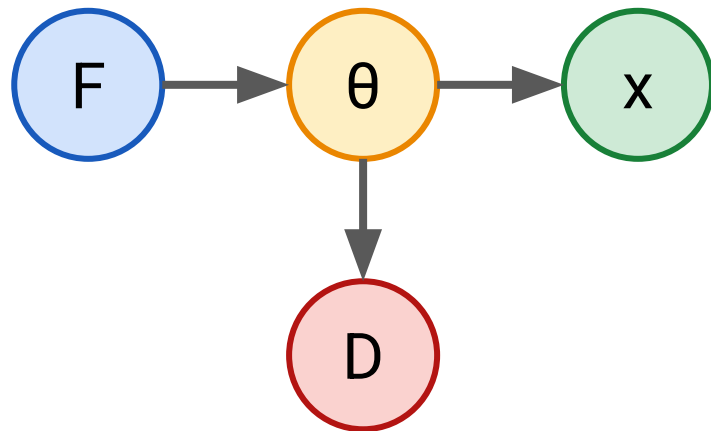
world we have



$$p(F, D, x, \theta) = p(D, F)p(x|D)p(\theta|x, D)$$

World Q

world we want

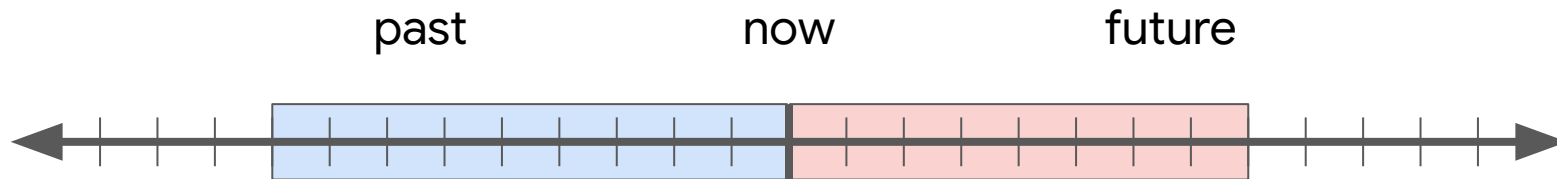


$$q(F, D, x, \theta) = q(\theta)q(x|\theta)q(F|\theta)q(D|\theta)^{\text{231}}$$

$$\begin{aligned}\text{KL}[p; q] &= \mathbb{E}_{p(D, F)p(x|D)p(\theta|x, D)} \left[\log \frac{p(D, F)p(x|D)p(\theta|x, D)}{q(F|\theta)q(D|\theta)q(x|\theta)q(\theta)} \right] \\ &= \left\langle \log \frac{p(x|D)p(\theta|x, D)}{q(x|\theta)q(\theta|F)q(D|\theta)} \right\rangle\end{aligned}$$

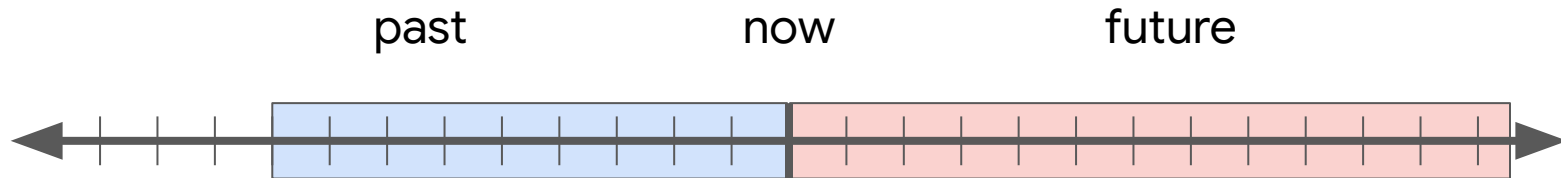
Predictive Information

Predictive Information



$$I(T; T')$$

Predictive Information



$$I_{\text{pred}}(T) \equiv \lim_{T' \rightarrow \infty} I(T; T')$$

Predictive Information

Positive

$$I_{\text{pred}}(T) \geq 0$$

Sublinear

$$\lim_{T \rightarrow \infty} \frac{I_{\text{pred}}(T)}{T} = 0$$

Vanishing Fraction

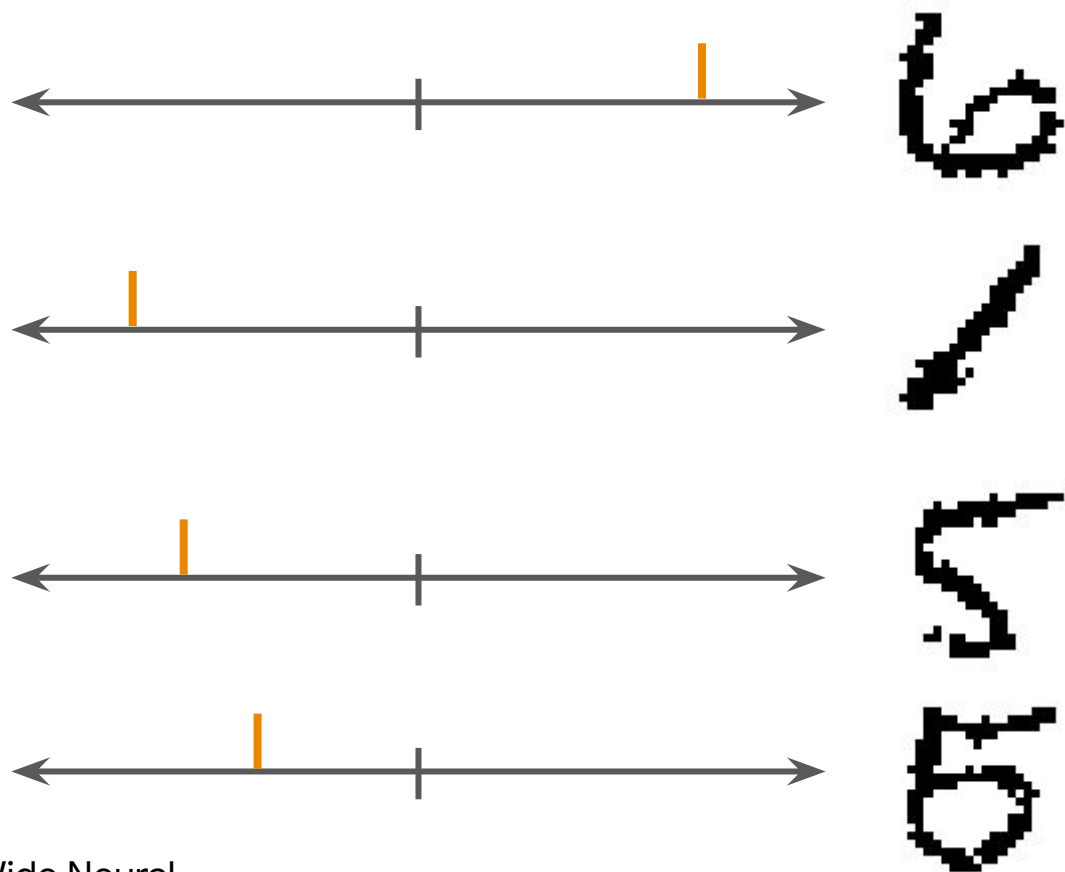
$$\lim_{T \rightarrow \infty} \frac{I_{\text{pred}}(T)}{S(T)} = 0$$

Parameter Uncertainty

How much
information is in a
deep neural networks
representation?

Individual Networks

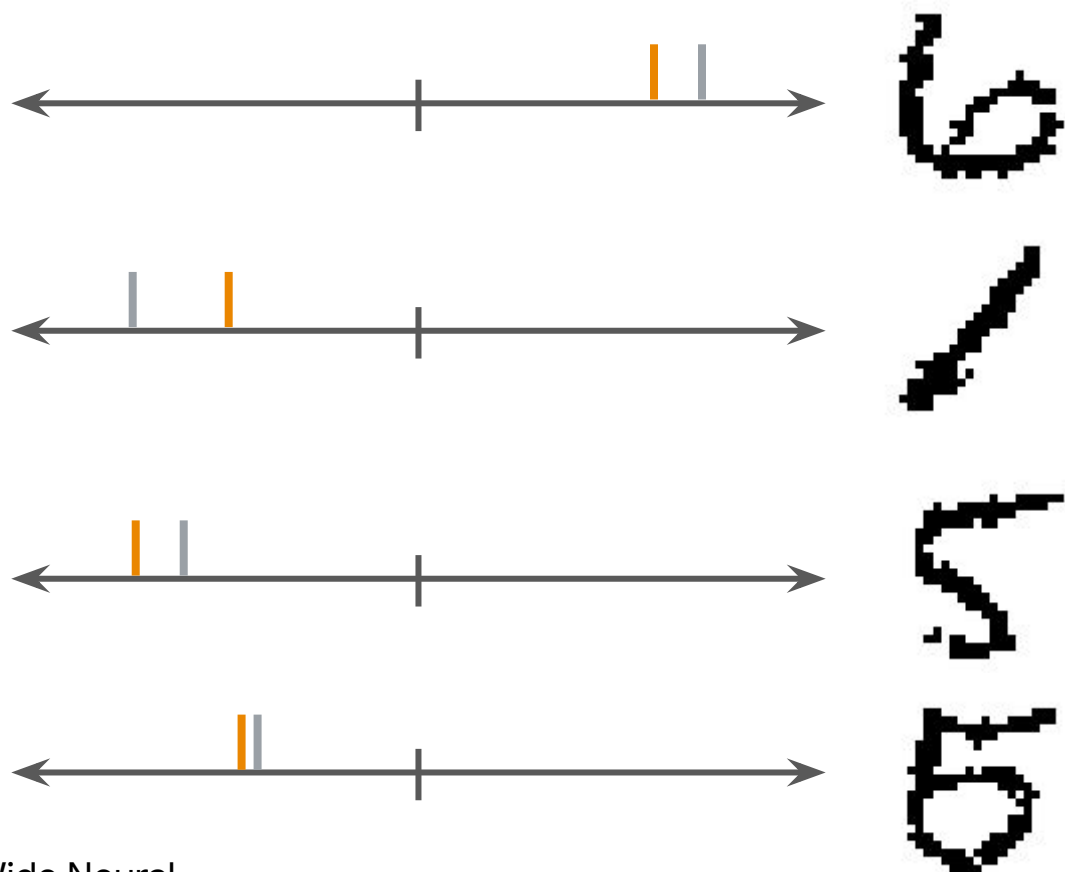
seed 0



Information in Infinite Ensembles of Infinitely-Wide Neural
Networks - Shwartz-Ziv & Alemi / 1911.09189

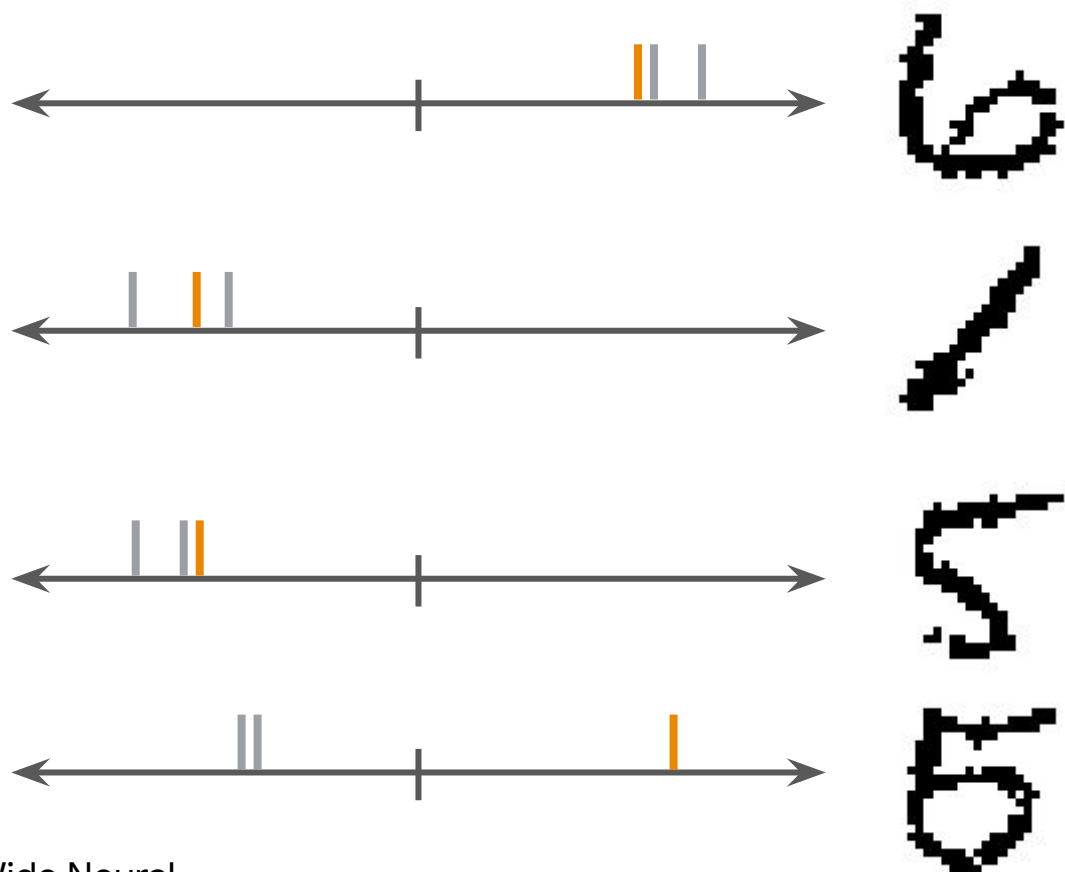
Individual Networks

seed 1



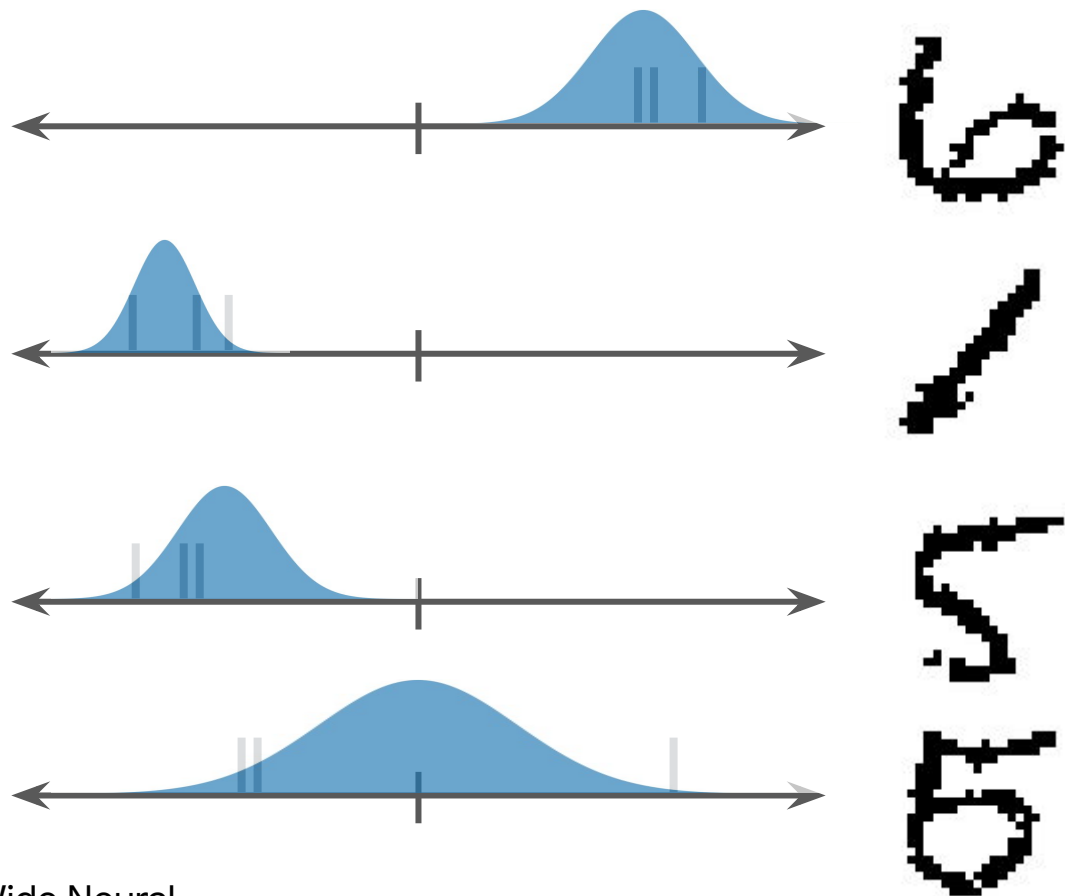
Individual Networks

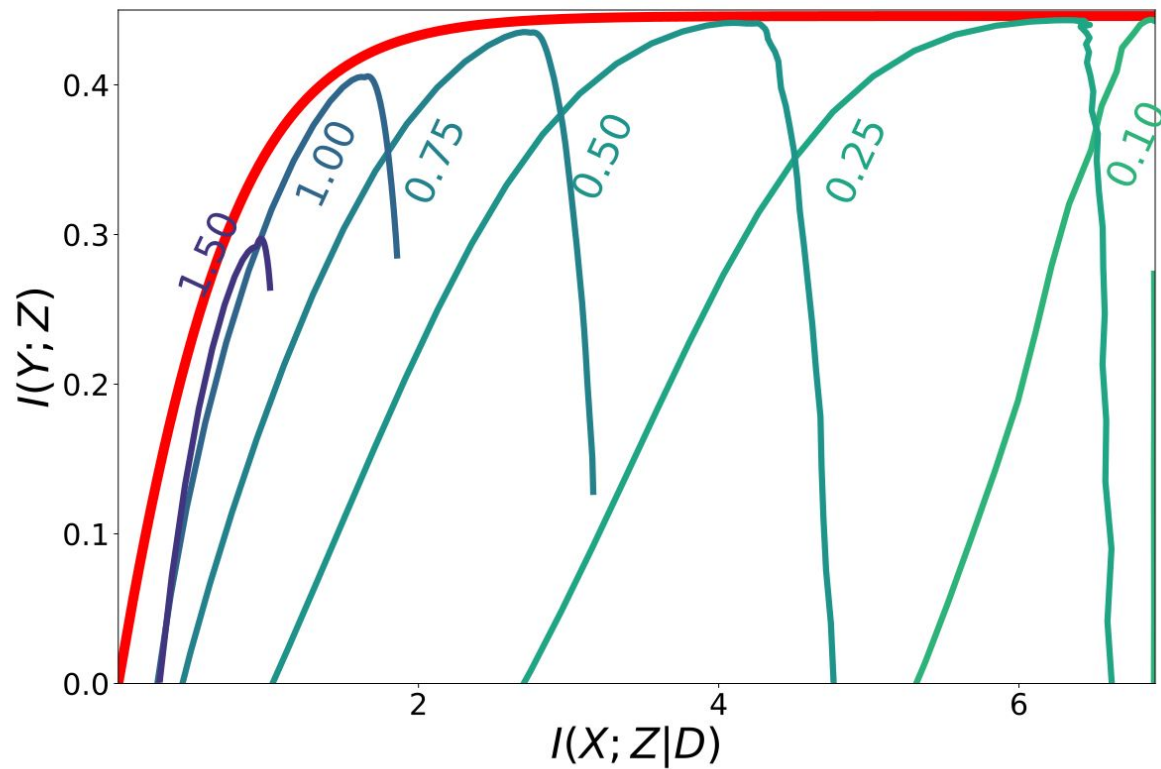
seed 2



Ensemble

marginalizing over all seeds
gives a stochastic
representation.

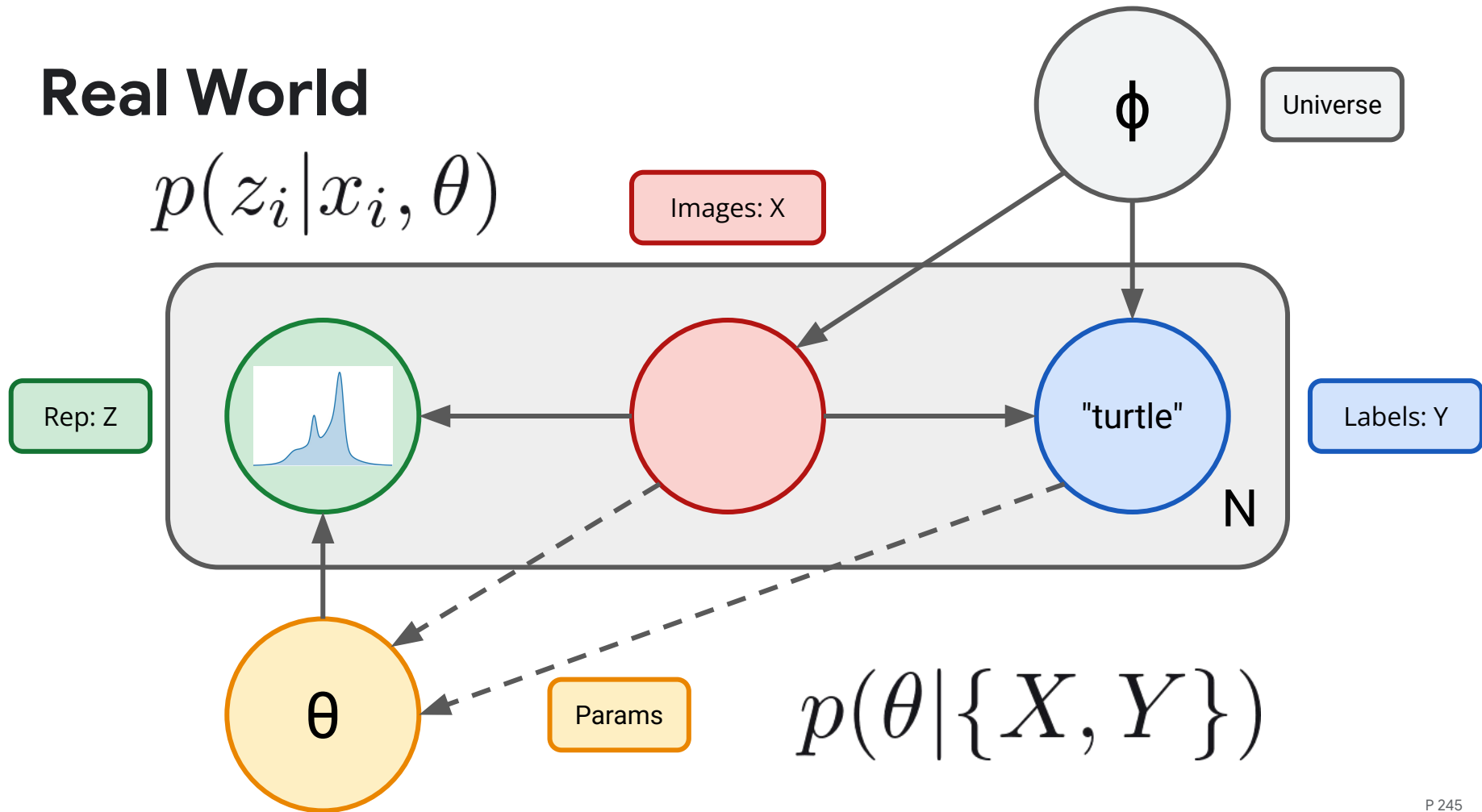




Information in Infinite Ensembles of Infinitely-Wide Neural
Networks - Schwartz-Ziv & Alemi / 1911.09189

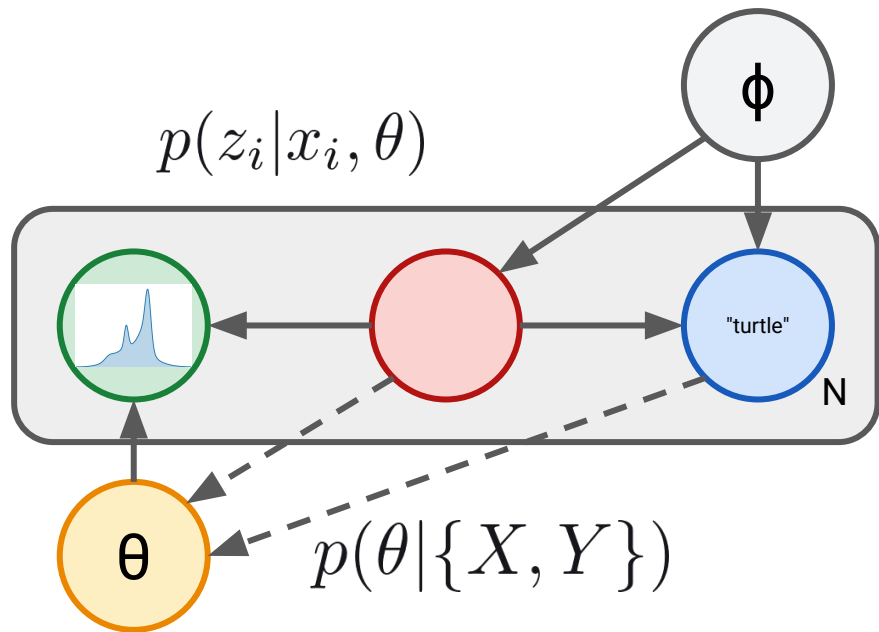
Semi Supervised Learning

Real World



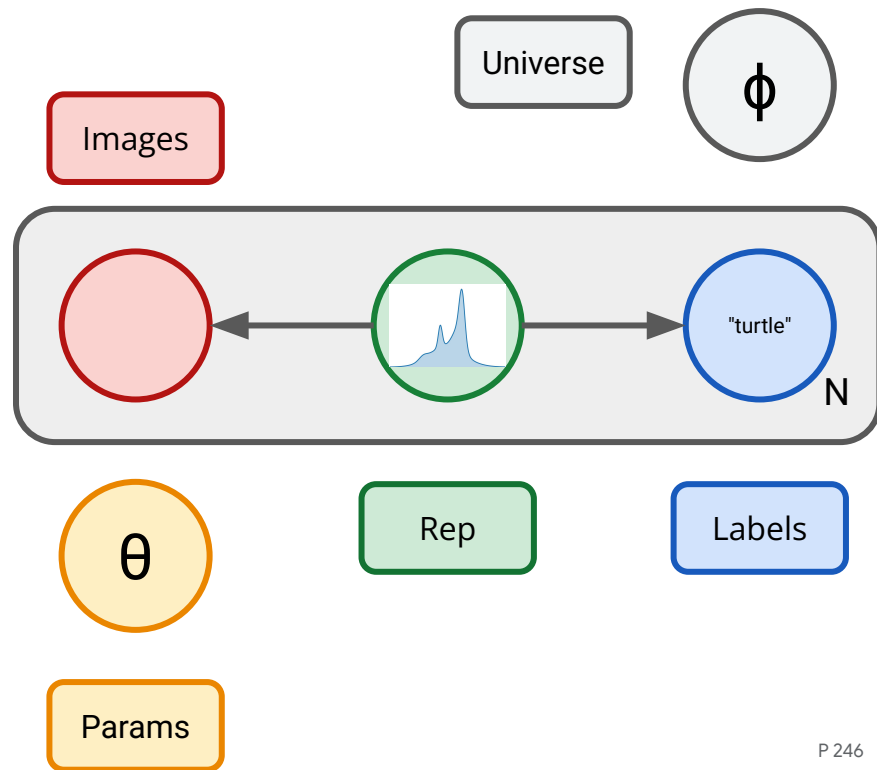
World P

world we have



World Q

world we want



Functionals

$$S \equiv \left\langle \log \frac{p(\theta | \{X, Y\})}{q(\theta)} \right\rangle$$

$$R \equiv \sum_i \left\langle \log \frac{p(z_i | x_i, \theta)}{q(z_i)} \right\rangle$$

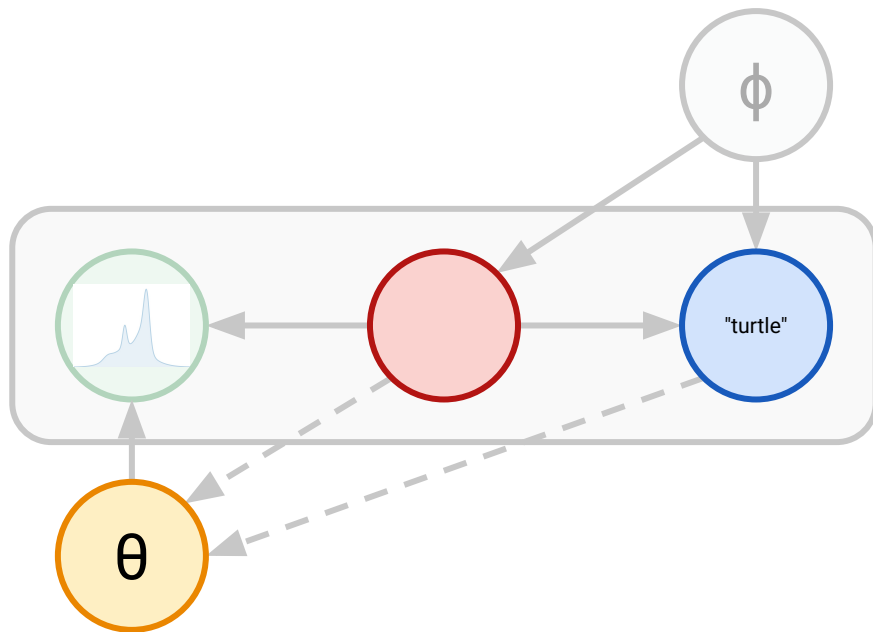
$$D \equiv - \sum_i \langle \log q(x_i | z_i) \rangle$$

$$C \equiv - \sum_i \langle \log q(y_i | z_i) \rangle$$

Entropy

The *entropy* is the relative entropy between our posterior over parameters and our *prior*.

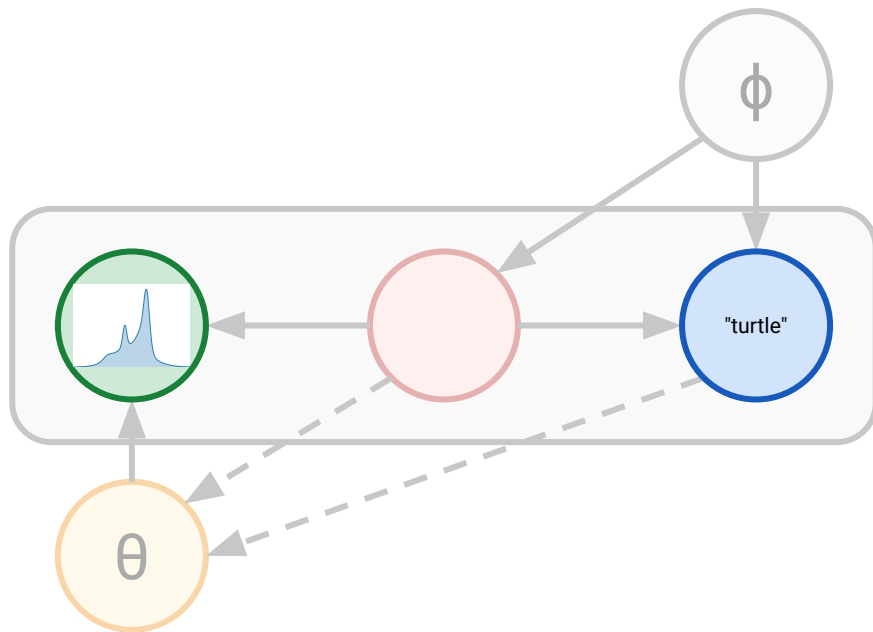
$$S \equiv \left\langle \log \frac{p(\theta | \{X, Y\})}{q(\theta)} \right\rangle$$



Classification Error

The *classification error* measures our supervised learning performance.

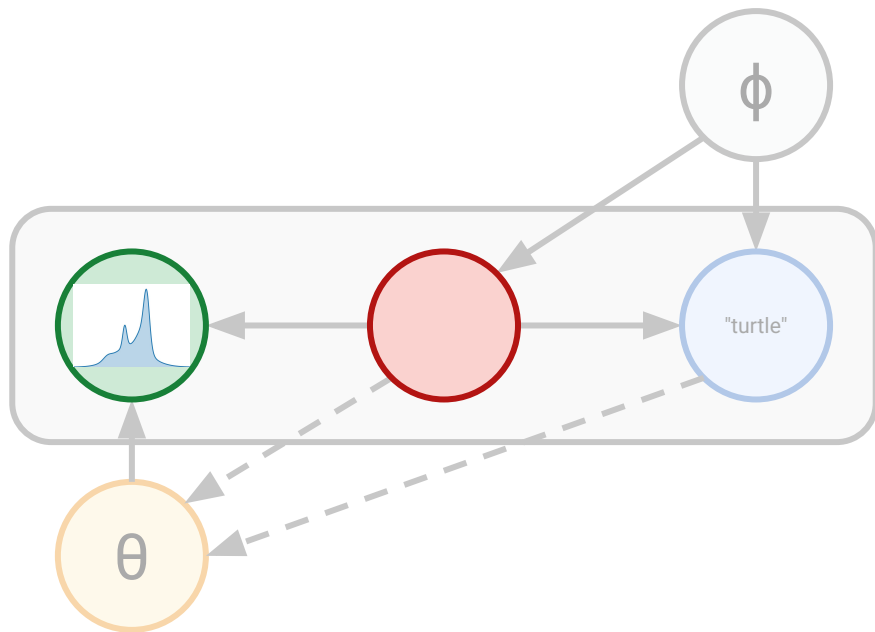
$$C \equiv - \sum_i \langle \log q(y_i | z_i) \rangle$$



Distortion

The *distortion* measures our unsupervised learning performance.

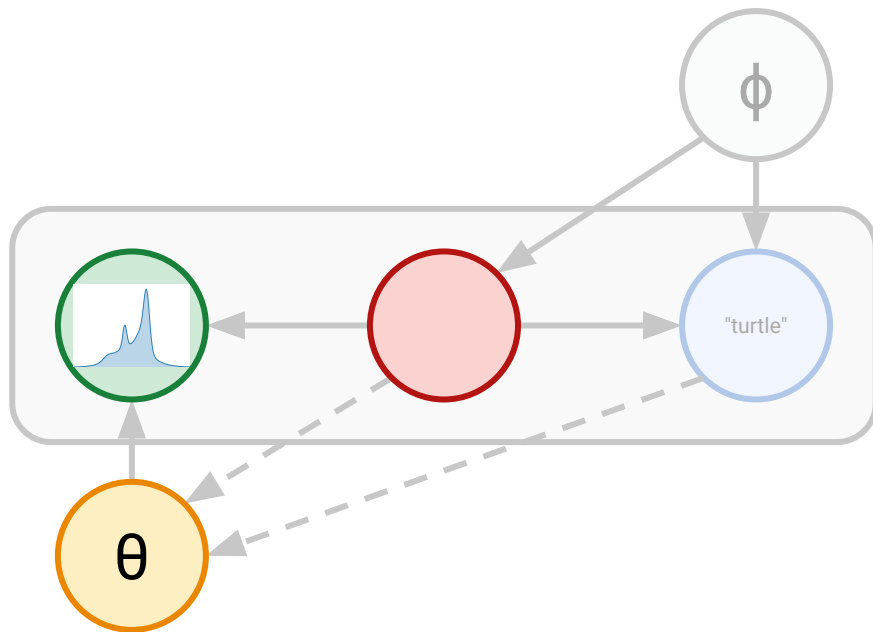
$$D \equiv - \sum_i \langle \log q(x_i | z_i) \rangle$$



Rate

The *rate* measures the cost of encoding samples into our representation.

$$R \equiv \sum_i \left\langle \log \frac{p(z_i | x_i, \theta)}{q(z_i)} \right\rangle$$



Functionals

$$S \equiv \left\langle \log \frac{p(\theta | \{X, Y\})}{q(\theta)} \right\rangle$$

$$R \equiv \sum_i \left\langle \log \frac{p(z_i | x_i, \theta)}{q(z_i)} \right\rangle$$

$$D \equiv - \sum_i \langle \log q(x_i | z_i) \rangle$$

$$C \equiv - \sum_i \langle \log q(y_i | z_i) \rangle$$

General Loss

$$\min \rho R + \delta D + \gamma C + \sigma S$$

Rate

Distortion

*Classification
Error*

Entropy

Classifier

$$\min \rho R + \delta D + \gamma \boxed{C} + \sigma S$$

Classification
Error

Normal Supervised Learning

VIB

$$\min \underbrace{\rho R}_{\text{Rate}} + \delta D + \gamma \underbrace{C}_{\text{Classification Error}} + \sigma S$$

Variational Information Bottleneck

Alemi et al. 2016

1612.00410

Autoencoder

$$\min \rho R + \delta \boxed{D} + \gamma C + \sigma S$$

Distortion

Autoencoder

VAE

$$\min \rho \boxed{R} + \delta \boxed{D} + \gamma C + \sigma S$$

Rate *Distortion*

Variational Autoencoder

Kingma / Rezende 2013

1312.6114/1401.4082

β -VAE

$$\min \underbrace{\rho R}_{\text{Rate}} + \underbrace{\delta D}_{\text{Distortion}} + \gamma C + \sigma S$$

β -VAE

Higgins et al. 2017

1606.05579

Fixing a Broken ELBO

Alemi et al. 2018

1711.00464

SS-VAE

$$\min \rho \boxed{R} + \delta \boxed{D} + \boxed{\gamma C} + \sigma S$$

Rate *Distortion* *Classification Error*

Semi Supervised VAE

Kingma et al. 2014

1406.5298

BBB

$$\min \rho R + \delta D + \gamma \boxed{C} + \sigma \boxed{S}$$

Classification Error *Entropy*

Bayes by Backprop
Blundell et al. 2015
1505.05424

Neural Process
Gernelo et al. 2018
1807.01622

IBL

$$\min \rho R + \delta D + \gamma \boxed{C} + \sigma \boxed{S}$$

Classification Error *Entropy*

Information Bottleneck Lagrangian

Achille & Soatto 2017

1706.01350

General Loss

$$\min \rho R + \delta D + \gamma C + \sigma S$$

Rate *Distortion* *Classification Error* *Entropy*

Profit?

Data Processing Inequality

What happens if we "process" our distribution?

$$x \rightarrow \boxed{t(y|x)} \rightarrow y$$

$$p(y) = \int dx \, t(y|x) p(x) \quad q(y) = \int dx \, t(y|x) q(x)$$

$$\mathcal{S}[p(x); q(x)] \geq \mathcal{S}[p(y); q(y)]$$

Any processing makes it harder to distinguish between the two distributions.

Monotonicity

Marginalization is a kind of processing.

$$(x, y) \rightarrow \boxed{} \rightarrow y$$

$$\mathcal{S}[p(x, y); q(x, y)] \geq \mathcal{S}[p(y); q(y)]$$

If you only look at a subset of variables, its harder to distinguish between two distributions.

Hypotheses

$$\Pr(X = x | P) \rightarrow p(x)$$

$$\Pr(X = x | Q) \rightarrow q(x)$$

The probability that some random variable X takes on a value x given that some model is true

Probability Densities

$$\Pr(x \leq X \leq x + dx \mid P) \rightarrow p(x) dx$$

The probability that some continuous random variable X takes on a value between x and $x + dx$ given that some model is true

Product Rule Again

$$p(x, y) = p(x)p(y|x)$$

Product Rule still applies

Maximum Likelihood

We don't have access to the true distribution of data, only samples.

$$\hat{p}(x) = \frac{1}{N} \sum_i \delta(x - x_i)$$

$$-\int dx \hat{p}(x) \log q(x) = -\int dx \frac{1}{N} \sum_i \delta(x - x_i) \log q(x) = -\frac{1}{N} \sum_i \log q(x_i)$$

Empirical Cross
Entropy

$$\min_q \mathcal{S}[\hat{p}; q]$$

Probabilities

$$\Pr(A)$$

The probability that A is true.

Conditional Probabilities

$$\Pr(A|\Omega)$$

The probability that X is true, given that Ω is true.

Joint Probabilities

$$\Pr(A, B)$$

The probability that A and B are true.

Product Rule

$$\Pr(A, B) = \Pr(A) \Pr(B|A)$$

The probability that A and B are true is the same as the probability that A is true times the probability that B is true, provided A is true

Independence

$$\Pr(B|A) = \Pr(B)$$

$$\Pr(A, B) = \Pr(A) \Pr(B)$$

Probabilities multiply for independent events.

Bayes Rule

$$\Pr(A, B) = \Pr(A) \Pr(B|A) = \Pr(B) \Pr(A|B)$$

$$\Pr(B|A) = \frac{\Pr(A|B) \Pr(B)}{\Pr(A)}$$

Bayes Rule = Joint probability sliced two ways + algebra.